BUET IPE · IPE 105

Cost & Management Accounting

Topic-wise Questions and Worked Answers

Term-final collection, 2009-2010 to 2024-2025

208 question entries · 15 topics · 14 available sessions

Each question is followed immediately by its model answer. Numerical cases include the data tables, formulas, computations and a decision or interpretation where requested.

The blue topic headings, green answer headings and clean mathematical typesetting follow your approved Chemistry and Math workbooks. This volume is designed for reading and revision without separate practice pages.

Coverage: All available questions in the supplied IPE 105 collection are retained under their exam tags. Repeated cases remain under each session. The topic map gives direct links to related questions.

Source accuracy: Tables were checked against the page images. Conditions, figures and requirements are preserved; nonessential company background is shortened. Missing or inconsistent data are identified, and assumptions are stated beside the answer.

Use the source PDF page references beneath each exam tag to compare the original scan. All money values are in the currency stated in the question; tables in thousands are explicitly labeled.

Topics and Navigation

Click the page number to open a topic. Original exam question numbers are retained in every entry.

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Questions and answers appear together. Read the calculation as well as the final result; costing conventions, activity bases and relevant-cost assumptions control the answer.

Accounting foundations, ethics and quality

Q1 · 2009-2010 · Exam Q1(b) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 1.

Lipex first quarter: inspection 0.5 day, process 2.8, preproduction wait 16, queue 4, move 0.7.

Required

1. Compute throughput, MCE, non-value-added percentage and delivery cycle time.

Model Answer

$$\text{Throughput} = 0.5 + 2.8 + 4 + 0.7 = 8 \text{ days}.$$
$$\text{MCE} = 2.8 / 8 = 35\%, \quad \text{Nonvalue} - \text{added} = 65\%.$$
$$\text{Delivery cycle} = 16 + 8 = 24 \text{ days}.$$

Q2 · 2011-2012 · Exam Q4(a) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 12.

Explain the changing business environment in a competitive market.

Model Answer

Competition, shorter product cycles, global supply chains, automation and customer demand for reliable quality make timely cost and performance information essential. Firms focus on lean operations, continuous improvement, quality management, responsive service and strategic pricing. Management accounting supports this through driver-based costs, budgets, relevant-cost decisions and financial/nonfinancial measures; relying only on historical totals can miss waste, idle capacity or customer-service problems.

Q3 · 2011-2012 · Exam Q5(a) [8 marks]

Source: Ipe 2009-25.pdf, PDF page 13.

What would an MCE greater than 1 mean? Interpret an MCE of 1.5.

Q3 · 2011-2012 · Exam Q5(a) [8 marks] (answer continued)

Model Answer

It is impossible under the standard definition: MCE = value-added process time / throughput, and throughput includes process plus nonnegative inspection, move and queue time. Therefore 0 <= MCE <= 1. A reported 1.5 would claim 150% of elapsed throughput is process time; it indicates a reversed formula, inconsistent scope, double-counted process time or measurement error. It does not indicate exceptional efficiency. An MCE of 1 means all throughput is value-added process time.

Q4 · 2012-2013 · Exam Q4(a) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 19.

Explain throughput time and manufacturing cycle efficiency.

Model Answer

$$\text{Throughput} = \text{Process} + \text{Inspection} + \text{Move} + \text{Queue time}.$$

$$\text{MCE} = \text{Value} - \text{added process time} \,/\, \text{Throughput time}.$$

Throughput is elapsed time from start of production to completion. Preproduction wait from customer order is excluded. In the usual model only process time adds value; MCE lies between 0 and 1. Delivery cycle time = preproduction wait + throughput; non-value-added proportion = 1 - MCE.

Q5 · 2018-2019 · Exam Q2(a) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 55.

Explain the balanced scorecard approach to organizational performance measurement.

Model Answer

A balanced scorecard translates strategy into objectives, measures, targets and initiatives across financial, customer, internal-process, and learning/growth perspectives. It balances financial outcomes with operational drivers of future outcomes.

Perspective	Example objective	Measure
Financial	Profitable growth	ROI and operating cash flow
Customer	Reliable service	Retention, satisfaction and delivery reliability

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Q5 · 2018-2019 · Exam Q2(a) [15 marks] (answer continued)

Perspective	Example objective	Measure
Internal process	Efficient high-quality operations	Defect rate, cycle time, yield
Learning/growth	Capable workforce and systems	Training outcomes, skill coverage, system availability

Connect the measures causally: employee skills can improve processes, which improve customer outcomes and eventually financial results. Select a small strategic set, rather than treating every available measure as equally useful.

Q6 · 2020-2021 · Exam Q3(b) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 71.

Mittel Rhein reports inspection 0.3 day; wait from order to start 14 days; process 2.7 days; move 1 day; queue 5 days.

Required

1. Compute manufacturing cycle efficiency (MCE) and the percentage of throughput spent in non-value-added activities.

Model Answer

$$\text{Throughput} = 0.3 + 2.7 + 1 + 5 = 9 \text{ days}.$$

$$\text{MCE} = 2.7 / 9 = 30\%, \quad \text{Nonvalue} - \text{added} = 70\%.$$

Only processing adds customer value in this model. The preproduction wait is excluded from throughput; delivery cycle time would include it: 14 + 9 = 23 days.

Q7 · 2023-2024 · Exam Q1(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF pages 98, 99.

Tombro is automating its plant and evaluating factory performance across four months.

Measure	Month 1	Month 2	Month 3	Month 4
Defects	185	163	124	91
Warranty claims	46	39	30	27

Q7 · 2023-2024 · Exam Q1(b) [15 marks] (question continued)

Measure	Month 1	Month 2	Month 3	Month 4
Complaints	102	96	79	58
Purchase lead time (days)	8	7	5	4
Scrap / total cost	1%	1%	2%	3%
Machine downtime	3%	4%	4%	6%
Machine use / availability	95%	92%	89%	85%
Setup hours	8	10	11	12
On-time deliveries	96%	95%	92%	89%

Time measure (same time unit)	Month 1	Month 2	Month 3	Month 4
Wait before production	9	11.50	12	14
Inspection	0.80	0.70	0.70	0.70
Process	2.10	2	1.90	1.80
Queue	2.80	4.40	6	7
Move	0.30	0.40	0.40	0.50

Required

1. Compute throughput time, manufacturing cycle efficiency (MCE), and delivery cycle time.

Model Answer

$$\text{Throughput} = \text{Process} + \text{Inspection} + \text{Move} + \text{Queue}.$$

$$\text{MCE} = \frac{\text{Process time}}{\text{Throughput time}}, \qquad \text{Delivery cycle} = \text{Wait} + \text{Throughput}.$$

Calculated performance	Month 1	Month 2	Month 3	Month 4
Throughput time	6.00	7.50	9.00	10.00
MCE	35.00%	26.67%	21.11%	18.00%
Delivery cycle time	15.00	19.00	21.00	24.00

Q7 · 2023-2024 · Exam Q1(b) [15 marks] (answer continued)

Only processing time is value-added in this measure. Throughput/delivery times rise and MCE falls, mainly as queues and waiting grow. Fewer defects and complaints do not by themselves establish better delivery performance; on-time delivery also declines.

Cost concepts and classifications

Q8 · 2009-2010 · Exam Q6(a) [9 marks]

Source: Ipe 2009-25.pdf, PDF page 6.

Classify costs by behavior, traceability and function with examples.

Model Answer

Basis	Classification	Examples
Behavior	Variable, fixed, mixed	Materials/unit; monthly rent; service charge + usage
Traceability	Direct or indirect relative to a named cost object	Patient prescription direct to patient; hospital heat indirect to patient
Function	Manufacturing, selling, administration	Factory wages; commissions; head-office salaries

These are separate dimensions. Factory rent can be fixed, indirect to a product, and manufacturing. The salary of a department head is direct to that department but indirect to a particular patient/product. State the cost object before deciding direct versus indirect.

Q9 · 2011-2012 · Exam Q3(b) [6 marks]

Source: Ipe 2009-25.pdf, PDF page 11.

How can costs be classified by function?

Model Answer

Function	Cost examples	Treatment
Manufacturing	DM, DL, factory overhead	Product costs under absorption costing
Selling/distribution	Advertising, commissions, delivery	Period expenses
Administration	Office salaries, office rent, accounting	Period expenses
Research/development and other support	Design and product-development work	Identify separately; financial treatment depends on applicable rules

Q9 · 2011-2012 · Exam Q3(b) [6 marks] (answer continued)

Function describes why the cost is incurred. It is independent of behavior or traceability: a factory cost can be fixed and indirect, while a commission is selling and variable.

Q10 · 2017-2018 · Exam Q6(b) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 52, 54.

Mark Hansen earns 20/hour; weekly hours above 40 earn time-and-a-half (30/hour). Ordinary overtime premiums and idle time are manufacturing overhead. Fringe benefits are 6 for every paid hour.

Required

1. 46 paid/worked hours: total wages, DL and OH.
2. 48 paid hours with 3 idle due to breakdown: wages, DL and OH.
3. 50 paid hours with 2 idle due to materials shortage: wages plus fringe; allocate if all fringe is OH.
4. Reallocate case 3 if fringe relating to direct labor is added to DL.

Model Answer

Case	Wages	Fringe	Total	Direct labor	MOH
1: 46 hours	980	0	980	920	60
2: 48 hours, 3 idle	1,040	0	1,040	900	140
3: 50 hours, all fringe in OH	1,100	300	1,400	960	440
4: productive fringe in DL	1,100	300	1,400	1,248	152

Case 1: 40 x 20 + 6 x 30 = 980; basic pay for 46 productive hours is direct, overtime premium 6 x 10 is OH. Case 2: productive time 45 hours x 20 = 900; OH = idle basic pay 60 + overtime premium 80. Case 3: productive basic wages 48 x 20 = 960; OH = premium 100 + idle wages 40 + all fringe 300. Case 4: productive fringe 48 x 6 = 288 moves to DL; idle fringe 2 x 6 = 12 remains OH. Premium attributable to a specific customer's rush request may instead be traced to that job; no such request is stated here.

Q11 · 2021-2022 · Exam Q5(a) [7 marks]

Source: Ipe 2009-25.pdf, PDF page 82.

What is an activity base? Give examples.

Model Answer

An activity base is a measurable level of work used to express or assign costs. For a variable cost, it should reflect the activity causing total cost to change. Examples: machine-hours for machine power; units produced for materials; kilometers for fuel; rental returns for car washing; direct labor-hours for labor-related overhead. Choose the causal measure, not simply the easiest number to count.

Q12 · 2022-2023 · Exam Q5(a) [6 marks]

Source: Ipe 2009-25.pdf, PDF page 90.

Explain the difference between product and period costs, with examples.

Model Answer

Product costs in a manufacturer	Period costs
Direct materials, direct labor, and manufacturing overhead	Selling and administrative costs
Recorded in inventory until the goods are sold	Expensed in the period incurred
Move RM -> WIP -> FG -> cost of goods sold	Do not pass through the production inventories
Factory rent, factory depreciation, indirect factory labor	Sales commissions, office rent, advertising

Product versus period is a financial-reporting classification. It is separate from variable versus fixed and direct versus indirect; a fixed factory cost can be a product cost.

Q13 · 2023-2024 · Exam Q5(b) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 104.

What are the three major elements of product costs in a manufacturing company? Explain with examples.

Q13 · 2023-2024 · Exam Q5(b) [10 marks] (answer continued)

Model Answer

Element	Meaning	Example
Direct materials	Materials economically traceable to the product	Wood in a cabinet
Direct labor	Production-worker effort economically traceable to the product	Assembly wages for that cabinet
Manufacturing overhead	All other factory production costs	Factory rent, indirect labor, factory depreciation and utilities

Selling and office costs are period costs, not manufacturing product costs. Prime cost = direct materials + direct labor; conversion cost = direct labor + manufacturing overhead.

Q14 · 2024-2025 · Exam Q5(b) [12 marks]

Source: Ipe 2009-25.pdf, PDF page 113.

Explain differential costs and revenues, opportunity costs, and sunk costs with suitable examples.

Model Answer

Term	Meaning	Example
Differential cost/revenue	The future difference in cost/revenue between alternatives	An order creates 12,000 revenue and 8,000 extra costs: 4,000 benefit
Opportunity cost	Benefit sacrificed by choosing one alternative	Using a room internally sacrifices rent obtainable from leasing it
Sunk cost	A past cost that cannot be changed by the current decision	The original purchase price of equipment already owned

Only future amounts that differ between alternatives affect the decision. An opportunity cost need not appear in the accounting ledger to be relevant; a sunk cost appears in the ledger but cannot be changed.

Cost behavior, high-low and regression

Q15 · 2009-2010 · Exam Q3(a) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 3.

Maxwell cost data	July	October
Units produced	9,000	12,000
COGM	285,000	390,000
Beginning WIP	14,000	22,000
Ending WIP	25,000	15,000
DM/unit	15	15
DL/unit	6	6

Required

1. Determine overhead added each month.
2. Estimate high-low formula.
3. Compute COGM at 95,000 units with opening/closing WIP 16,000 / 19,000.

Model Answer

July current manufacturing = 285,000 - 14,000 + 25,000 = 296,000; OH = 296,000 - 9,000 x 21 = 107,000. October current manufacturing = 383,000; OH = 383,000 - 12,000 x 21 = 131,000.

Choose the highest and lowest activity observations, not the highest and lowest cost observations.

$$b = \frac{131,000 - 107,000}{12,000 - 9,000} = 8.000000\,.$$

$$a = 131,000 - 8.000000(12,000) = 35,000.00\,.$$

$$Y = 35,000.00 + 8.000000X\,.$$

At 95,000 activity units, the high-low prediction is 795,000.00.

$$\text{COGM} = 95,000(15 + 6) + 795,000 + 16,000 - 19,000 = 2,787,000\,.$$

The model is monthly because its observations are monthly; fixed annual cost would be 35,000 x 12 = 420,000 if the same base applies all year. The printed forecast is 95,000 units, far outside observed 9,000-12,000; the result is conditional on extending the formula unchanged.

Q16 · 2011-2012 · Exam Q3(a) [6 marks]

Source: Ipe 2009-25.pdf, PDF page 11.

Briefly explain cost behavior.

Model Answer

Cost behavior	Total cost within the relevant range	Per-unit cost
Variable	Changes proportionately with activity	Approximately constant
Fixed	Approximately constant for the period	Falls as activity rises; rises as activity falls
Mixed	Fixed base plus a variable component	Neither entirely fixed nor entirely variable

$$Y = a + bX\,.$$

Here a is total fixed cost, b is variable cost per activity unit and X is activity. Examples: materials per unit are variable; factory rent is fixed; a utility bill with a service charge plus usage is mixed. The relevant range and the time horizon limit these approximations.

Q17 · 2011-2012 · Exam Q6(a) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 14.

Write a short note on semi-variable cost.

Model Answer

A semi-variable or mixed cost has a fixed base plus a variable component: Y = a + bX. A utility bill might include a 1,000 monthly connection charge and 0.05 per kWh. The fixed base remains within the relevant range even at zero activity; the variable component grows with usage. Separate the components using engineering analysis, scattergraphs, high-low or regression before using the cost in CVP or flexible budgets.

Q18 · 2011-2012 · Exam Q7(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 16.

Young Boys Club football-program cost data:

Leagues	Annual cost
5	130,000
2	70,000

Q18 · 2011-2012 · Exam Q7(b) [15 marks] (question continued)

Leagues	Annual cost
4	105,000
6	140,000
3	100,000

Required

1. Use least squares to estimate variable cost/league, fixed annual cost and Y = a + bX.
2. Predict cost for eight leagues.

Model Answer

Least-squares regression uses all observations to minimize squared residuals.

Regression totals	Value
n	5
Sum X	20.00
Sum Y	545,000.00
Sum X squared	90.00
Sum XY	2,350,000.00

$$b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2} = 17,000.000000,$$

$$a = \bar{Y} - b\bar{X} = 41,000.00, \quad Y = 41,000.00 + 17,000.000000X\,.$$

At 8 activity units, the regression prediction is 177,000.00.

Q19 · 2012-2013 · Exam Q8(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 23.

Activity/cost	March	June
Units produced	6,000	9,000
COGM	168,000	257,000

Q19 · 2012-2013 · Exam Q8(a) [20 marks] (question continued)

Activity/cost	March	June
Beginning WIP	9,000	32,000
Ending WIP	15,000	21,000
DM/unit	6	6
DL/unit	10	10

Required

1. Determine overhead added in each month, with no over/underapplication.
2. Estimate high-low overhead formula.
3. Estimate COGM at 7,000 units.

> **Source clarification:** The last part omits forecast WIP. Total manufacturing cost is calculable; COGM requires a stated inventory assumption.

Model Answer

Current manufacturing cost = COGM - opening WIP + ending WIP. March = 174,000; June = 246,000. Subtract materials/labor of 16 per produced unit: overhead = 78,000 and 102,000.

Choose the highest and lowest activity observations, not the highest and lowest cost observations.

$$b = \frac{102,000 - 78,000}{9,000 - 6,000} = 8.000000\,.$$

$$a = 102,000 - 8.000000(9,000) = 30,000.00\,.$$

$$Y = 30,000.00 + 8.000000X\,.$$

At 7,000 activity units, the high-low prediction is 86,000.00.

At 7,000 output, DM/DL = 112,000 and OH = 86,000, total current manufacturing cost = 198,000. COGM = 198,000 + beginning WIP - ending WIP. It is 198,000 only if WIP is unchanged; no WIP values for that forecast month are given.

Q20 · 2013-2014 · Exam Q2(a) [4 marks]

Source: Ipe 2009-25.pdf, PDF page 25.

Define true variable and step-variable costs with examples.

Model Answer

A true variable cost changes proportionately with activity within its relevant range: materials of 6/unit produce total cost 6Q. A step-variable cost changes in small blocks when another resource unit is needed: one hourly assistant can serve ten customers at a time, so staffing rises at each ten-customer threshold. A small-step cost may approximate variable behavior over a broad activity range; it is constant within each step.

Q21 · 2013-2014 · Exam Q2(b) [6 marks]

Source: Ipe 2009-25.pdf, PDF page 25.

Explain fixed cost and relevant range with examples.

Model Answer

A fixed cost is constant in total for the defined period over a relevant activity range, such as factory rent of 40,000/month for the existing building. Its average per-unit amount falls when output rises. The relevant range is the interval within which the assumed cost behavior is valid. Beyond capacity, renting another building or adding a supervisor can increase total fixed cost. Fixed means fixed within the stated range/time, not permanently unavoidable.

Q22 · 2013-2014 · Exam Q2(c) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 25.

A retailer records auto expenses and mileage (thousands of miles).

Month	Mileage (000)	Cost
Jan	4	5,600
Feb	8	11,000
Mar	7	10,400
Apr	12	14,800
May	6	6,800
Jun	11	13,900

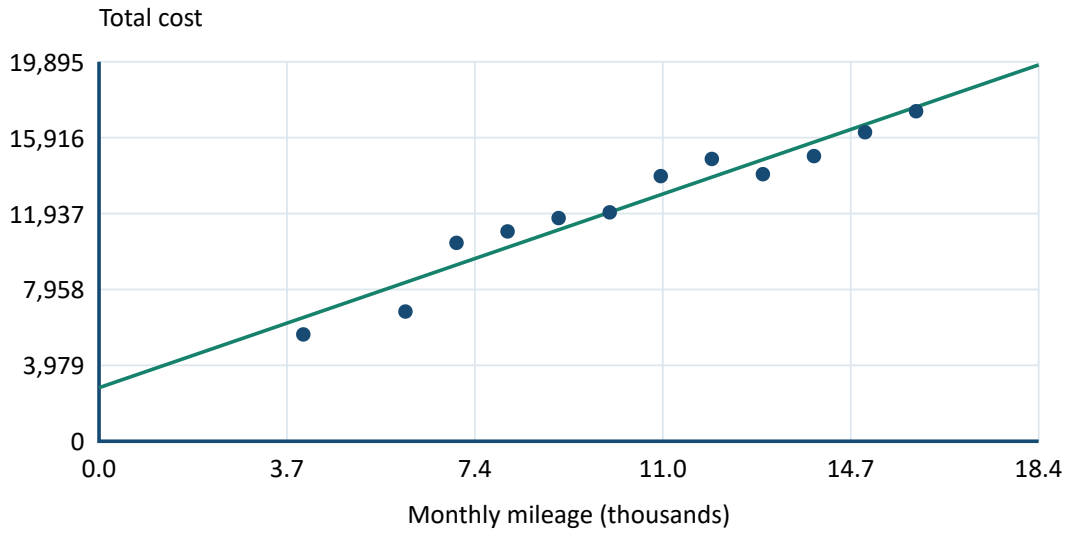
Q22 · 2013-2014 · Exam Q2(c) [25 marks] (question continued)

Month	Mileage (000)	Cost
Jul	14	14,950
Aug	10	12,000
Sep	13	14,000
Oct	15	16,200
Nov	9	11,700
Dec	16	17,300

Required

1. Draw a scattergraph and estimate fixed/variable cost.
2. Use least-squares regression and forecast at 20,000 miles.

Model Answer



The scatter shows an approximately linear mixed cost. A reasonable visual line is near Y = 3,000 + 900X, giving about 3,000 fixed per month and 0.90 per actual mile. Visual fits are approximate and may differ; the green line shows the precise least-squares fit below.

Least-squares regression uses all observations to minimize squared residuals.

Regression totals	Value
n	12
Sum X	125.00

Q22 · 2013-2014 · Exam Q2(c) [25 marks] (answer continued)

Regression totals	Value
Sum Y	148,650.00
Sum X squared	1,457.00
Sum XY	1,690,900.00

$$b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2} = 919.607316,$$

$$a = \bar{Y} - b\bar{X} = 2,808.26, \quad Y = 2,808.26 + 919.607316X\,.$$

At 20 activity units, the regression prediction is 21,200.40.

X is thousands of miles. Divide the regression slope by 1,000 to express cost per actual mile. A 20,000-mile prediction extrapolates beyond the observed maximum 16,000; validate the same relevant range.

Q23 · 2013-2014 · Exam Q8(c) [7 marks]

Source: Ipe 2009-25.pdf, PDF page 30.

Explain the trend toward fixed costs in business organizations.

Model Answer

Automation, information systems and specialized facilities often replace variable manual effort with committed depreciation, leases, technical staff and infrastructure. Unit variable cost can fall while fixed cost rises. This creates higher operating leverage and idle-capacity risk, so managers must plan utilization and evaluate investments over an appropriate long horizon. Outsourcing/cloud usage can reverse this tendency; it is a common pattern rather than a universal rule.

Q24 · 2014-2015 · Exam Q8(a) [23 marks]

Source: Ipe 2009-25.pdf, PDF pages 36, 38.

Zerbel shipping costs vary by month.

Month	Units shipped	Cost
Jan	4	18,000

Q24 · 2014-2015 · Exam Q8(a) [23 marks] (question continued)

Month	Units shipped	Cost
Feb	5	21,000
Mar	6	24,000
Apr	10	33,000
May	12	35,000
Jun	11	33,000
Jul	9	30,000
Aug	8	27,000
Sep	7	26,000

Required

1. Estimate high-low and least-squares formulas.
2. Predict cost at 14 units using each.

Model Answer

Choose the highest and lowest activity observations, not the highest and lowest cost observations.

$$b = \frac{35,000 - 18,000}{12 - 4} = 2,125.000000\,.$$

$$a = 35,000 - 2,125.000000(12) = 9,500.00\,.$$

$$Y = 9,500.00 + 2,125.000000X\,.$$

At 14 activity units, the high-low prediction is 39,250.00.

Least-squares regression uses all observations to minimize squared residuals.

Regression totals	Value
n	9
Sum X	72.00
Sum Y	247,000.00
Sum X squared	636.00
Sum XY	2,102,000.00

Q24 · 2014-2015 · Exam Q8(a) [23 marks] (answer continued)

$$b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2} = 2,100.000000,$$

$$a = \bar{Y} - b\bar{X} = 10,644.44, \quad Y = 10,644.44 + 2,100.000000X\,.$$

At 14 activity units, the regression prediction is 40,044.44.

Q25 · 2016-2017 · Exam Q1(b) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 39.

Hoi Chong truck average annual operating cost: 0.114/km at 105,000 km; 0.134/km at 70,000 km.

Required

1. Use high-low to find fixed/variable costs and Y = a + bX.
2. Predict cost at 80,000 km.

Model Answer

Convert average unit costs to totals first: 105,000 x 0.114 = 11,970; 70,000 x 0.134 = 9,380.

Choose the highest and lowest activity observations, not the highest and lowest cost observations.

$$b = \frac{11,970 - 9,380}{105,000 - 70,000} = 0.074000\,.$$

$$a = 11,970 - 0.074000(105,000) = 4,200.00\,.$$

$$Y = 4,200.00 + 0.074000X\,.$$

At 80,000 activity units, the high-low prediction is 10,120.00.

Q26 · 2017-2018 · Exam Q1(a) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 47.

Write a linear cost function using Y for cost and X for activity: (i) direct labor 10/hour; (ii) materials 9.20/cubic yard; (iii) utilities minimum 1,000 plus 0.05/kWh; (iv) annual machine depreciation 200,000 plus 75 utility cost per operating day.

Q26 · 2017-2018 · Exam Q1(a) [10 marks] (answer continued)

Model Answer

Case	Activity X	Cost function
i	Labor-hours	Y = 10X
ii	Cubic yards	Y = 9.20X
iii	Kilowatt-hours	Y = 1,000 + 0.05X
iv	Operating days per year	Y = 200,000 + 75X

Use an annual period in (iv); the fixed depreciation and the activity count must refer to the same time interval.

Q27 · 2021-2022 · Exam Q5(b) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 82.

Bargain Rental cleans cars after each rental return. Estimate the fixed monthly washing cost and variable cost per return using least-squares regression; round fixed cost to whole dollars and variable cost to two decimals.

Month	Rental returns	Cost
January	2,380	10,825
February	2,421	11,865
March	2,586	11,332
April	2,725	12,422
May	2,968	13,850
June	3,281	14,419
July	3,353	14,935
August	3,489	15,738
September	3,057	13,563
October	2,876	11,889
November	2,735	12,683
December	2,983	13,796

Q27 · 2021-2022 · Exam Q5(b) [20 marks] (answer continued)

Model Answer

Least-squares regression uses all observations to minimize squared residuals.

Regression totals	Value
n	12
Sum X	34,854.00
Sum Y	157,317.00
Sum X squared	102,623,516.00
Sum XY	462,541,971.00

$$b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2} = 4.039172,$$

$$a = \bar{Y} - b\bar{X} = 1,377.97, \quad Y = 1,377.97 + 4.039172X\,.$$

The requested rounded equation is Y = 1,378 + 4.04X: fixed washing cost 1,378 per month and variable cost 4.04 per rental return. Keep full precision for intermediate regression calculations.

Q28 · 2022-2023 · Exam Q5(b) [14 marks]

Source: Ipe 2009-25.pdf, PDF pages 90, 91.

Lakeshore Hotel has the following guest-day and custodial-supply costs.

Month	Guest-days	Expense
March	4,000	7,500
April	6,500	8,250
May	8,000	10,500
June	10,500	12,000
July	12,000	13,500
August	9,000	10,750
September	7,500	9,750

Q28 · 2022-2023 · Exam Q5(b) [14 marks] (question continued)

Required

1. Estimate the high-low cost formula.
2. Predict expense at 11,000 guest-days.

Model Answer

Choose the highest and lowest activity observations, not the highest and lowest cost observations.

$$b = \frac{13,500 - 7,500}{12,000 - 4,000} = 0.750000\,.$$

$$a = 13,500 - 0.750000(12,000) = 4,500.00\,.$$

$$Y = 4,500.00 + 0.750000X\,.$$

At 11,000 activity units, the high-low prediction is 12,750.00.

Q29 · 2022-2023 · Exam Q5(c) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 91.

Finance 101 has variable and fixed teaching costs; Professor Morton wants a cost formula based on sections offered.

Term	Sections	Total cost
Fall last year	4	10,000
Winter last year	6	14,000
Summer last year	2	7,000
Fall this year	5	13,000
Winter this year	3	9,500

Required

1. Use least-squares regression to estimate variable cost per section and fixed cost per term. Express the formula Y = a + bX.

Model Answer

Least-squares regression uses all observations to minimize squared residuals.

Q29 · 2022-2023 · Exam Q5(c) [15 marks] (answer continued)

Regression totals	Value
n	5
Sum X	20.00
Sum Y	53,500.00
Sum X squared	90.00
Sum XY	231,500.00

$$b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2} = 1,750.000000,$$

$$a = \bar{Y} - b\bar{X} = 3,700.00, \quad Y = 3,700.00 + 1,750.000000X\,.$$

Manufacturing schedules and cost flows

Q30 · 2009-2010 · Exam Q5(a) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 5.

Cost data: advertising 100,000; DL 90,000; RM purchases 132,000; factory rent 80,000; indirect labor 56,300; commissions 35,000; factory utilities 9,000; factory maintenance 24,000; office depreciation 8,000; factory depreciation 40,000; factory supplies 700. Opening/closing RM 8,000/10,000; WIP 5,000/20,000; FG 70,000/25,000.

Required

1. Prepare COGM schedule.

Model Answer

DM used = 8,000 + 132,000 - 10,000 = 130,000.

Manufacturing overhead	Amount
Factory rent	80,000
Indirect labor	56,300
Utilities	9,000
Maintenance	24,000
Factory depreciation	40,000
Factory supplies	700
Total MOH	210,000

Cost of goods manufactured	Amount
Direct materials used	130,000
Direct labor	90,000
Manufacturing overhead	210,000
Total current manufacturing costs	430,000
Add beginning WIP	5,000
Less ending WIP	20,000
Cost of goods manufactured	415,000

Q30 · 2009-2010 · Exam Q5(a) [15 marks] (answer continued)

COGM = 415,000. Advertising, commissions and office depreciation are excluded from manufacturing costs.

Q31 · 2009-2010 · Exam Q5(b) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 6.

Selected balances: advertising 215,000; factory insurance 8,000; sales-equipment depreciation 40,000; factory rent 90,000; utilities 52,000; commissions 35,000; factory supplies 6,000; factory depreciation 110,000; S&A salaries 85,000; factory maintenance 74,000; DL unknown; RM purchases 260,000. Opening/closing RM 50,000/40,000; WIP unknown/33,000; FG 30,000/unknown. Total current manufacturing costs 675,000; goods available for sale 720,000; COGS 635,000.

Required

1. Prepare COGS schedule.

Model Answer

DM = 270,000; MOH = 340,000; missing DL = 65,000. COGM = 720,000 - 30,000 = 690,000. Beginning WIP = 690,000 + 33,000 - 675,000 = 48,000. Ending FG = 720,000 - 635,000 = 85,000.

Manufacturing overhead	Amount
Insurance	8,000
Rent	90,000
Utilities	52,000
Supplies	6,000
Factory depreciation	110,000
Maintenance	74,000
Total MOH	340,000

Cost of goods manufactured	Amount
Direct materials used	270,000
Direct labor	65,000
Manufacturing overhead	340,000

Q31 · 2009-2010 · Exam Q5(b) [20 marks] (answer continued)

Cost of goods manufactured	Amount
Total current manufacturing costs	675,000
Add beginning WIP	48,000
Less ending WIP	33,000
Cost of goods manufactured	690,000

Cost of goods sold	Amount
Beginning finished goods	30,000
COGM	690,000
Goods available	720,000
Less ending finished goods	85,000
Unadjusted COGS	635,000

Q32 · 2011-2012 · Exam Q3(c) [23 marks]

Source: Ipe 2009-25.pdf, PDF pages 11, 12.

Art Factory: opening RM 40,000, WIP 20,000, FG 70,000; DL 110,000; accumulated depreciation 40,000; indirect-material purchases 15,000, direct-material purchases 80,000; indirect material used 10,000; sales commissions 100,000; ending RM 10,000, WIP 60,000, FG 30,000. Sales-office rent 35,000; factory rent 50,000; indirect labor 20,000; corporate rent 50,000; factory depreciation 10,000; factory utilities 10,000; corporate and sales utilities 10,000 each; prepaid insurance 38,000.

Required

1. Prepare COGM and COGS schedules.

> **Source clarification:** The coverage of the single RM inventory balance is not specified. The normal combined-inventory interpretation and the direct-only alternative are distinguished.

Model Answer

Treat the one raw-materials inventory as including direct and indirect materials: total material used = 40,000 + 80,000 + 15,000 - 10,000 = 125,000. Deduct indirect usage 10,000 to get DM used 115,000.

Q32 · 2011-2012 · Exam Q3(c) [23 marks] (answer continued)

Manufacturing overhead	Amount
Indirect material used	10,000
Factory rent	50,000
Indirect labor	20,000
Factory depreciation	10,000
Factory utilities	10,000
Total MOH	100,000

Cost of goods manufactured	Amount
Direct materials used	115,000
Direct labor	110,000
Manufacturing overhead	100,000
Total current manufacturing costs	325,000
Add beginning WIP	20,000
Less ending WIP	60,000
Cost of goods manufactured	285,000

Cost of goods sold	Amount
Beginning finished goods	70,000
COGM	285,000
Goods available	355,000
Less ending finished goods	30,000
Unadjusted COGS	325,000

Accumulated depreciation is a balance, not current expense; use factory depreciation expense 10,000. Prepaid insurance is an asset with no stated expiration, so it is not expensed. If the reported RM balances refer only to direct materials, DM used would instead be 110,000 and COGM/COGS 280,000/320,000; the main solution uses the usual combined RM account.

Q33 · 2012-2013 · Exam Q5 [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 20, 21.

Superior: S&A salaries 110,000; factory insurance 8,000, utilities 45,000, indirect labor 60,000, supplies 7,000, rent 120,000, maintenance 30,000. RM purchases 290,000; DL unknown; advertising 80,000; commissions 50,000. Opening/closing RM 40,000/10,000, WIP unknown/35,000, FG 50,000/unknown. Total manufacturing costs 683,000; goods available 750,000; COGS 660,000. Costs relate to 40,000 produced units.

Required

1. Prepare COGM/COGS and fill missing values.
2. Compute DM/unit.
3. At 50,000 units next year, estimate DM/unit and total.
4. Explain the difference between the two unit costs.

Model Answer

DM used = 320,000; OH = 270,000; DL = 683,000 - 320,000 - 270,000 = 93,000. COGM = 750,000 - 50,000 = 700,000; beginning WIP = 700,000 + 35,000 - 683,000 = 52,000. Ending FG = 750,000 - 660,000 = 90,000.

Manufacturing overhead	Amount
Insurance	8,000
Utilities	45,000
Indirect labor	60,000
Supplies	7,000
Rent	120,000
Maintenance	30,000
Total MOH	270,000

Cost of goods manufactured	Amount
Direct materials used	320,000
Direct labor	93,000
Manufacturing overhead	270,000
Total current manufacturing costs	683,000

Q33 · 2012-2013 · Exam Q5 [35 marks] (answer continued)

Cost of goods manufactured	Amount
Add beginning WIP	52,000
Less ending WIP	35,000
Cost of goods manufactured	700,000

Cost of goods sold	Amount
Beginning finished goods	50,000
COGM	700,000
Goods available	750,000
Less ending finished goods	90,000
Unadjusted COGS	660,000

DM/unit = 320,000/40,000 = 8. At 50,000 units, DM/unit remains 8 and total is 400,000, assuming unchanged prices, usage and relevant range. There is no per-unit difference: direct materials are variable in this model; only their total changes.

Q34 · 2013-2014 · Exam Q5 [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 27, 28.

A new manufacturer reports August sales 450,000 and expensed all the following: indirect labor 12,000; utilities 15,000; DL 70,000; factory depreciation 21,000; RM purchases 165,000; sales equipment depreciation 18,000; insurance 4,000; rent 50,000; S&A salaries 32,000; advertising 75,000. Reported loss 12,000 leads management to consider selling. Factory shares: utilities 60%, insurance 75%, rent 80%.

Inventory	August 1	August 31
RM	8,000	13,000
WIP	16,000	21,000
FG	40,000	60,000

Required

1. Prepare COGM and corrected income statement.
2. Recommend whether to sell the company.

Q34 · 2013-2014 · Exam Q5 [35 marks] (answer continued)

Model Answer

DM used = 8,000 + 165,000 - 13,000 = 160,000. Factory utilities 9,000; insurance 3,000; rent 40,000. S&A = 6,000 utilities + 1,000 insurance + 10,000 rent + 18,000 depreciation + 32,000 salaries + 75,000 ads = 142,000.

Manufacturing overhead	Amount
Indirect labor	12,000
Utilities	9,000
Depreciation	21,000
Insurance	3,000
Rent	40,000
Total MOH	85,000

Cost of goods manufactured	Amount
Direct materials used	160,000
Direct labor	70,000
Manufacturing overhead	85,000
Total current manufacturing costs	315,000
Add beginning WIP	16,000
Less ending WIP	21,000
Cost of goods manufactured	310,000

Cost of goods sold	Amount
Beginning finished goods	40,000
COGM	310,000
Goods available	350,000
Less ending finished goods	60,000
Unadjusted COGS	290,000

Q34 · 2013-2014 · Exam Q5 [35 marks] (answer continued)

Income statement	Amount
Sales	450,000
COGS	290,000
Gross margin	160,000
Selling/admin	142,000
Operating income	18,000

Correct NOI is 18,000 profit, not a 12,000 loss. The 30,000 difference represents net manufacturing costs retained in inventories (RM 5,000 + WIP 5,000 + FG 20,000). Do not sell solely because of the faulty statement; evaluate future profitability, cash needs and business value before deciding.

Q35 · 2014-2015 · Exam Q5(a) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 35.

Mason: DL 70,000; RM purchases 118,000; indirect labor 30,000; factory maintenance 6,000; advertising 90,000; factory insurance 800; sales salaries 50,000; factory rent 20,000; supplies 4,200; office depreciation 3,000; factory depreciation 19,000.

Inventory	Beginning	Ending
RM	7,000	15,000
WIP	10,000	5,000
FG	20,000	35,000

Required

1. Prepare COGM and the COGS section.

Model Answer

$$\text{DM used} = 7,000 + 118,000 - 15,000 = 110,000\,.$$

Manufacturing overhead	Amount
Indirect labor	30,000
Factory maintenance	6,000

Q35 · 2014-2015 · Exam Q5(a) [15 marks] (answer continued)

Manufacturing overhead	Amount
Factory insurance	800
Factory rent	20,000
Supplies (factory)	4,200
Factory depreciation	19,000
Total MOH	80,000

Cost of goods manufactured	Amount
Direct materials used	110,000
Direct labor	70,000
Manufacturing overhead	80,000
Total current manufacturing costs	260,000
Add beginning WIP	10,000
Less ending WIP	5,000
Cost of goods manufactured	265,000

Cost of goods sold	Amount
Beginning finished goods	20,000
COGM	265,000
Goods available	285,000
Less ending finished goods	35,000
Unadjusted COGS	250,000

Advertising, sales salaries and office depreciation are period expenses, so they are excluded from COGM. Supplies are treated as factory supplies in the manufacturing-cost context.

Q36 · 2014-2015 · Exam Q5(b) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 35.

XYZ: FG opening/closing 20,000/40,000; factory depreciation 27,000; admin expense 110,000; factory utilities 8,000, maintenance 40,000, supplies 11,000, insurance 4,000; RM purchases 125,000, opening 9,000, closing 6,000; DL 70,000, indirect labor 15,000; WIP opening 17,000, closing 30,000; sales 500,000; selling expense 80,000.

Required

1. Prepare income statement for the year.

Model Answer

Manufacturing overhead	Amount
Factory depreciation	27,000
Utilities	8,000
Maintenance	40,000
Supplies	11,000
Insurance	4,000
Indirect labor	15,000
Total MOH	105,000

Cost of goods manufactured	Amount
Direct materials used	128,000
Direct labor	70,000
Manufacturing overhead	105,000
Total current manufacturing costs	303,000
Add beginning WIP	17,000
Less ending WIP	30,000
Cost of goods manufactured	290,000

Cost of goods sold	Amount
Beginning finished goods	20,000
COGM	290,000

Q36 · 2014-2015 · Exam Q5(b) [20 marks] (answer continued)

Cost of goods sold	Amount
Goods available	310,000
Less ending finished goods	40,000
Unadjusted COGS	270,000

Income statement	Amount
Sales	500,000
COGS	270,000
Gross margin	230,000
Selling/admin	190,000
Operating income	40,000

Q37 · 2016-2017 · Exam Q1(a) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 39.

Mason: DL 70,000; RM purchases 118,000; indirect labor 30,000; factory maintenance 6,000; advertising 90,000; factory insurance 800; sales salaries 50,000; factory rent 20,000; supplies 4,200; office depreciation 3,000; factory depreciation 19,000.

Inventory	Beginning	Ending
RM	7,000	15,000
WIP	10,000	5,000
FG	20,000	35,000

Required

1. Prepare COGM and the COGS section.

Model Answer

$$\text{DM used} = 7,000 + 118,000 - 15,000 = 110,000\,.$$

Manufacturing overhead	Amount
Indirect labor	30,000

Q37 · 2016-2017 · Exam Q1(a) [25 marks] (answer continued)

Manufacturing overhead	Amount
Factory maintenance	6,000
Factory insurance	800
Factory rent	20,000
Supplies (factory)	4,200
Factory depreciation	19,000
Total MOH	80,000

Cost of goods manufactured	Amount
Direct materials used	110,000
Direct labor	70,000
Manufacturing overhead	80,000
Total current manufacturing costs	260,000
Add beginning WIP	10,000
Less ending WIP	5,000
Cost of goods manufactured	265,000

Cost of goods sold	Amount
Beginning finished goods	20,000
COGM	265,000
Goods available	285,000
Less ending finished goods	35,000
Unadjusted COGS	250,000

Advertising, sales salaries and office depreciation are period expenses, so they are excluded from COGM. Supplies are treated as factory supplies in the manufacturing-cost context.

Q38 · 2017-2018 · Exam Q6(a) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 52.

Mason: DL 70,000; RM purchases 118,000; indirect labor 30,000; factory maintenance 6,000; advertising 90,000; factory insurance 800; sales salaries 50,000; factory rent 20,000; supplies 4,200; office depreciation 3,000; factory depreciation 19,000.

Inventory	Beginning	Ending
RM	7,000	15,000
WIP	10,000	5,000
FG	20,000	35,000

Required

1. Prepare COGM and the COGS section.

Model Answer

$$\text{DM used} = 7,000 + 118,000 - 15,000 = 110,000\,.$$

Manufacturing overhead	Amount
Indirect labor	30,000
Factory maintenance	6,000
Factory insurance	800
Factory rent	20,000
Supplies (factory)	4,200
Factory depreciation	19,000
Total MOH	80,000

Cost of goods manufactured	Amount
Direct materials used	110,000
Direct labor	70,000
Manufacturing overhead	80,000
Total current manufacturing costs	260,000
Add beginning WIP	10,000

Q38 · 2017-2018 · Exam Q6(a) [15 marks] (answer continued)

Cost of goods manufactured	Amount
Less ending WIP	5,000
Cost of goods manufactured	265,000

Cost of goods sold	Amount
Beginning finished goods	20,000
COGM	265,000
Goods available	285,000
Less ending finished goods	35,000
Unadjusted COGS	250,000

Advertising, sales salaries and office depreciation are period expenses, so they are excluded from COGM. Supplies are treated as factory supplies in the manufacturing-cost context.

Q39 · 2018-2019 · Exam Q8(b) [8 marks]

Source: Ipe 2009-25.pdf, PDF page 63.

RM beginning/end 40,000 / 50,000; purchases 400,000; DL 150,000; MOH 800,000; WIP beginning/end 200,000 / 180,000; FG beginning/end 250,000 / 300,000; sales 1,800,000; S&A 190,000.

Required

1. Compute COGM, COGS and operating income.

Model Answer

Manufacturing overhead	Amount
MOH	800,000
Total MOH	800,000

Cost of goods manufactured	Amount
Direct materials used	390,000
Direct labor	150,000

Q39 · 2018-2019 · Exam Q8(b) [8 marks] (answer continued)

Cost of goods manufactured	Amount
Manufacturing overhead	800,000
Total current manufacturing costs	1,340,000
Add beginning WIP	200,000
Less ending WIP	180,000
Cost of goods manufactured	1,360,000

Cost of goods sold	Amount
Beginning finished goods	250,000
COGM	1,360,000
Goods available	1,610,000
Less ending finished goods	300,000
Unadjusted COGS	1,310,000

Income statement	Amount
Sales	1,800,000
COGS	1,310,000
Gross margin	490,000
Selling/admin	190,000
Operating income	300,000

Q40 · January 2020 Term · Exam Q4(a) [45 marks]

Source: Ipe 2009-25.pdf, PDF page 66.

Visic produces 29,000 units; price 50; sales 1,300,000. Costs: advertising 105,000; entertainment/travel 40,000; DL 90,000; indirect labor 85,000; purchases printed 480,00; rent 40,000 (80% factory); factory utilities 108,000; patent royalty 1.50/unit produced; maintenance 9,000; special equipment rent 7,000 + 0.30/unit; S&A salaries 210,000; other factory overhead 6,800; other S&A 17,000.

Q40 · January 2020 Term · Exam Q4(a) [45 marks] (question continued)

Inventory	Beginning	Ending
Raw materials	20,000	30,000
WIP	50,000	40,000
Finished goods	?	?

Finished goods are carried at this year's average unit production cost.

Required

1. Prepare COGM.
2. Compute ending FG units and cost.
3. Prepare an income statement.

> **Source clarification:** The purchases amount is printed 480,00 and interpreted as 480,000. Beginning FG is a question mark, not explicitly zero; the zero-beginning solution is labeled and the general result is shown.

Model Answer

Manufacturing overhead	Amount
Indirect labor	85,000
Factory rent: 80% of 40,000	32,000
Factory utilities	108,000
Royalty: 29,000 x 1.50	43,500
Factory maintenance	9,000
Equipment rent: 7,000 + 0.30 x 29,000	15,700
Other factory OH	6,800
Total MOH	300,000

Cost of goods manufactured	Amount
Direct materials used	470,000
Direct labor	90,000
Manufacturing overhead	300,000

Q40 · January 2020 Term · Exam Q4(a) [45 marks] (answer continued)

Cost of goods manufactured	Amount
Total current manufacturing costs	860,000
Add beginning WIP	50,000
Less ending WIP	40,000
Cost of goods manufactured	870,000

COGM = 870,000, so unit completed cost = 870,000 / 29,000 = 30. Sales units = 1,300,000 / 50 = 26,000. If beginning FG is zero, ending FG = 3,000 units costing 90,000; COGS = 780,000.

Conditional income statement: no beginning FG	Amount
Sales	1,300,000
COGS	780,000
Gross margin	520,000
S&A: 105,000 + 40,000 + 8,000 + 210,000 + 17,000	380,000
Operating income	140,000

More generally, if beginning FG contains B units also valued at 30, ending FG = B + 3,000 units costing 30B + 90,000. COGS still equals 780,000. If beginning FG has a different cost, its amount/units are needed for an unconditional statement.

Job-order costing, overhead and journal entries

Q41 · 2009-2010 · Exam Q6(b) [26 marks]

Source: Ipe 2009-25.pdf, PDF pages 6, 7.

Manufacturer opening RM/WIP/FG 16,000 / 10,000 / 30,000. Estimated MHs 36,000 and OH 153,000. Actual: RM purchases 200,000; issued 190,000 (80% direct, 20% indirect); DL 160,000; indirect labor 27,000; sales commissions printed 36,0000; admin salaries 80,000; factory utilities 42,000; insurance expiration 10,000 (90% factory); ads 50,000; depreciation 60,000 (85% factory). Actual MHs 40,000. Completed jobs 480,000; credit sales 700,000; printed job cost sold 745,000.

Required

1. Find over/underapplied OH and closing journal.
2. Prepare income statement.

> **Source clarification:** Printed sales commissions have an extra zero, and sold job costs exceed available FG. Neither is silently corrected. Insurance is interpreted as expired insurance, consistent with its manufacturing/S&A split.

Model Answer

$$\text{POHR} = 153,000 / 36,000 = 4.25; \quad \text{Applied} = 40,000(4.25) = 170,000.$$

Actual MOH	Amount
Indirect materials: 20% x 190,000	38,000
Indirect labor	27,000
Utilities	42,000
Insurance: 90% x 10,000	9,000
Depreciation: 85% x 60,000	51,000
Total	167,000

Overapplied OH = 3,000. Close: Dr MOH 3,000; Cr COGS 3,000.

Income using printed COGS, commission interpreted 36,000	Amount
Sales	700,000

Q41 · 2009-2010 · Exam Q6(b) [26 marks] (answer continued)

Income using printed COGS, commission interpreted 36,000	Amount
Unadjusted job-cost COGS	745,000
Less overapplication	3,000
Adjusted COGS	742,000
Gross loss	-42,000
S&A: 36,000 + 80,000 + 1,000 + 50,000 + 9,000	176,000
Operating loss	-218,000

The data are internally inconsistent: only 30,000 opening FG + 480,000 completions = 510,000 available, below the printed 745,000 sold cost. The statement above follows the explicit COGS figure, but an inventory reconciliation cannot be made without correcting it. If commissions are literally 360,000, the operating loss is 542,000 instead; the customary intended reading is 36,000.

Q42 · 2011-2012 · Exam Q4(b) [30 marks]

Source: Ipe 2009-25.pdf, PDF page 12.

High Desert Pottery applies Molding OH on machine-hours and Painting OH on direct labor cost.

Annual estimate	Molding	Painting
DLHs	12,000	60,000
MHs	70,000	8,000
DM	510,000	650,000
DL cost	130,000	420,000
OH	602,000	735,000

Job 205	Molding	Painting
DLHs	30	85
MHs	110	20
DM	470	332

Q42 · 2011-2012 · Exam Q4(b) [30 marks] (question continued)

Job 205	Molding	Painting
DL	290	680

Annual actual	Molding	Painting
DLHs	10,000	62,000
MHs	65,000	9,000
DM	430,000	680,000
DL cost	108,000	436,000
OH	570,000	750,000

Required

1. Compute departmental rates, Job 205 OH and total cost/unit for 50 units, and annual over/underapplication.

Model Answer

Molding rate = 602,000/70,000 = 8.60/MH; Painting rate = 735,000/420,000 = 175% of DL cost.

Job 205	Molding	Painting	Total
DM	470	332	802
DL	290	680	970
Applied OH	946	1,190	2,136
Total	1,706	2,202	3,908

Unit cost = 3,908/50 = 78.16. Annual Molding applied = 65,000 x 8.60 = 559,000; underapplied 11,000. Painting applied = 436,000 x 175% = 763,000; overapplied 13,000. Do not use the labor-hour basis from later papers: this question explicitly uses labor dollars in Painting.

Q43 · 2012-2013 · Exam Q6 [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 21, 22.

Teledex applies overhead on direct labor cost. Departments Fabricating, Machining and Assembly:

Q43 · 2012-2013 · Exam Q6 [35 marks] (question continued)

Annual estimate	Fabricating	Machining	Assembly	Total
DL	200,000	100,000	300,000	600,000
OH	350,000	400,000	90,000	840,000

Customer job	Fabricating	Machining	Assembly	Total
DL	2,800	500	6,200	9,500
DM	3,000	200	1,400	4,600

Annual actual	Fabricating	Machining	Assembly	Total
DL	210,000	108,000	262,000	580,000
DM	190,000	16,000	114,000	320,000
OH	360,000	420,000	84,000	864,000

Required

1. Compute plantwide rate and job OH.
2. Compute departmental rates and job OH.
3. Find annual over/underapplication under both approaches.

Model Answer

Plantwide rate = 840,000/600,000 = 140% of DL; job OH = 9,500 x 140% = 13,300.
Departmental rates = 175%, 400%, 30%.

Department	Job OH	Annual applied OH	Actual - applied
Fabricating	4,900	367,500	7,500 over
Machining	2,000	432,000	12,000 over
Assembly	1,860	78,600	5,400 under
Total	8,760	878,100	14,100 over

Plantwide annual applied = 580,000 x 1.40 = 812,000; underapplied = 864,000 - 812,000 = 52,000. Job totals including DM/DL are 27,400 plantwide vs 22,860 departmental. Differences reflect the actual mix of work, not a difference in cash overhead incurred.

Q44 · 2013-2014 · Exam Q6 [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 28, 29.

A law firm has Research/Documents and Litigation departments. Research overhead uses research-hours; Litigation uses attorney cost.

Annual estimate	Research	Litigation
Research-hours	24,000	-
Attorney-hours	9,000	18,000
Forms/supplies	16,000	5,000
Attorney cost	450,000	900,000
OH	840,000	360,000

Client case	Research	Litigation
Research-hours	26	-
Attorney-hours	7	114
Forms/supplies	80	30
Attorney cost	350	5,700

Annual actual	Research	Litigation
Research-hours	26,000	-
Attorney-hours	8,000	15,000
Forms/supplies	19,000	6,000
Attorney cost	400,000	750,000
OH	870,000	315,000

Required

1. Compute departmental rates, case OH, total case cost by department, and annual over/underapplied OH.

Model Answer

$$\text{Research rate} = 840,000 / 24,000 = 35 \text{ per research hour}.$$
$$\text{Litigation rate} = 360,000 / 900,000 = 40\% \text{ of attorney cost}.$$

Q44 · 2013-2014 · Exam Q6 [35 marks] (answer continued)

Case cost	Research	Litigation	Total
Forms	80	30	110
Attorney	350	5,700	6,050
Applied OH	910	2,280	3,190
Total cost	1,340	8,010	9,350

Annual OH	Research	Litigation
Actual	870,000	315,000
Applied	910,000	300,000
Difference	40,000 overapplied	15,000 underapplied

Use research-hours for Research and attorney dollars for Litigation; attorney-hours do not directly enter these overhead formulas.

Q45 · 2014-2015 · Exam Q6(a) [35 marks]

Source: Ipe 2009-25.pdf, PDF page 37.

A recording studio has all-fixed OH 160,000/year and capacity 1,600 hours. Estimates for 1998/1999 are 1,000/800 hours; actual hours 750/500. Actual OH equals estimate. A particular CD requires 40 hours.

Required

1. Using estimated annual activity, find applied OH for the CD and annual under/overapplication.
2. Repeat using capacity hours as denominator.
3. Interpret capacity-based under/overapplication.

Model Answer

Calculation	1998	1999
Estimate-based rate/hour	160	200
CD applied OH	6,400	8,000
Annual applied OH	120,000	100,000

Q45 · 2014-2015 · Exam Q6(a) [35 marks] (answer continued)

Model Answer

Calculation	1998	1999
Annual underapplied OH	40,000	60,000
Capacity rate/hour	100	100
CD capacity-based OH	4,000	4,000
Annual capacity-based applied	75,000	50,000
Unused-capacity cost / underapplied OH	85,000	110,000

At capacity rate 100/hour, unused hours are 850 and 1,100. Their costs are idle capacity, not higher resource consumption by a 40-hour CD. Using shrinking demand as the denominator increases cost/job, which can force higher prices and further customer losses. Capacity-based rates keep job costs stable and reveal idle-resource costs separately.

Q46 · 2016-2017 · Exam Q2 [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 40, 41.

Gold Nest birdcages: all transactions with suppliers, employees and customers are cash. Currency Rmb. Estimated DL cost 200,000; MOH 330,000. Beginning RM/WIP/FG: 25,000 / 10,000 / 40,000.

Required

1. a. RM purchased for cash 275,000.
2. b. Materials issued 280,000: direct 220,000, indirect 60,000.
3. c. DL 180,000; indirect labor 72,000; commissions 63,000; admin salaries 90,000.
4. d. Rent 18,000, factory portion 13,000.
5. e. Factory utilities 57,000.
6. f. Advertising 140,000.
7. g. Depreciation 100,000, factory portion 88,000.
8. h. Apply MOH on actual DL cost.
9. i. Complete jobs costing 675,000.
10. j. Cash sales 1,250,000, job cost sold 700,000.

Q46 · 2016-2017 · Exam Q2 [35 marks] (question continued)

Required

1. Prepare journals and inventory/MOH/COGS T-accounts.
2. Find and close over/underapplied MOH.
3. Prepare income statement.

Model Answer

$$\text{POHR} = 330,000 / 200,000 = 165\% \text{ of DL}; \quad \text{Applied} = 297,000.$$

Actual MOH = 60,000 + 72,000 + 13,000 + 57,000 + 88,000 = 290,000; overapplied = 7,000.

Debit is the left side and credit the right side. Inventory increases are debits; inventory transfers credit the sending account and debit the receiving account. Actual overhead is debited to MOH; applied overhead is credited to MOH and debited to WIP.

Entry	Account	Debit	Credit
a	RM	275,000.00	
a	Cash		275,000.00
b	WIP	220,000.00	
b	MOH	60,000.00	
b	RM		280,000.00
c	WIP	180,000.00	
c	MOH	72,000.00	
c	Sales commissions	63,000.00	
c	Admin salaries	90,000.00	
c	Cash		405,000.00
d	MOH	13,000.00	
d	S&A rent	5,000.00	
d	Cash		18,000.00
e	MOH	57,000.00	
e	Cash		57,000.00
f	Advertising	140,000.00	

Q46 · 2016-2017 · Exam Q2 [35 marks] (answer continued)

Entry	Account	Debit	Credit
f	Cash		140,000.00
g	MOH	88,000.00	
g	S&A depreciation	12,000.00	
g	Accumulated depreciation		100,000.00
h	WIP	297,000.00	
h	MOH		297,000.00
i	FG	675,000.00	
i	WIP		675,000.00
j-sale	Cash	1,250,000.00	
j-sale	Sales		1,250,000.00
j-cost	COGS	700,000.00	
j-cost	FG		700,000.00
OH-close	MOH	7,000.00	
OH-close	COGS		7,000.00

RM

Debit	Credit
Beginning: 25,000.00	b: 280,000.00
a: 275,000.00	
Ending balance: 20,000.00	

WIP

Debit	Credit
Beginning: 10,000.00	i: 675,000.00
b: 220,000.00	
c: 180,000.00	
h: 297,000.00	
Ending balance: 32,000.00	

Q46 · 2016-2017 · Exam Q2 [35 marks] (answer continued)

FG

Debit	Credit
Beginning: 40,000.00	j-cost: 700,000.00
i: 675,000.00	
Ending balance: 15,000.00	

Cash

Debit	Credit
j-sale: 1,250,000.00	a: 275,000.00
Ending balance: 355,000.00	c: 405,000.00
	d: 18,000.00
	e: 57,000.00
	f: 140,000.00

MOH

Debit	Credit
b: 60,000.00	h: 297,000.00
c: 72,000.00	
d: 13,000.00	
e: 57,000.00	
g: 88,000.00	
OH-close: 7,000.00	
Ending balance: 0.00	

Sales commissions

Debit	Credit
c: 63,000.00	
Ending balance: 63,000.00	

Admin salaries

Debit	Credit
c: 90,000.00	
Ending balance: 90,000.00	

S&A rent

Debit	Credit
d: 5,000.00	
Ending balance: 5,000.00	

Advertising

Debit	Credit
f: 140,000.00	
Ending balance: 140,000.00	

S&A depreciation

Debit	Credit
g: 12,000.00	
Ending balance: 12,000.00	

Q46 · 2016-2017 · Exam Q2 [35 marks] (answer continued)

Accumulated depreciation

Debit	Credit
	g: 100,000.00
	Ending balance: 100,000.00

Sales

Debit	Credit
	j-sale: 1,250,000.00
	Ending balance: 1,250,000.00

COGS

Debit	Credit
j-cost: 700,000.00	OH-close: 7,000.00
Ending balance: 693,000.00	

Ending balance after overhead adjustment	Debit	Credit
RM	20,000.00	
WIP	32,000.00	
FG	15,000.00	
Cash	355,000.00	
MOH	0.00	
Sales commissions	63,000.00	
Admin salaries	90,000.00	
S&A rent	5,000.00	
Advertising	140,000.00	
S&A depreciation	12,000.00	
Accumulated depreciation		100,000.00
Sales		1,250,000.00
COGS	693,000.00	

Every journal entry balances. Closing a debit (underapplied) overhead balance debits COGS and credits overhead; closing a credit (overapplied) balance does the reverse. The T-account balances shown precede general year-end closing of revenue/expense accounts.

Q46 · 2016-2017 · Exam Q2 [35 marks] (answer continued)

Income statement	Amount
Sales	1,250,000
Adjusted COGS	693,000
Gross margin	557,000
S&A: 63,000 + 90,000 + 5,000 + 140,000 + 12,000	310,000
Operating income	247,000

Cash and accumulated-depreciation columns show movements because their opening balances are not supplied. Required inventories end at RM 20,000, WIP 32,000 and FG 15,000.

Q47 · 2017-2018 · Exam Q5(a) [18 marks]

Source: Ipe 2009-25.pdf, PDF page 51.

Polaris October: purchase RM on account 210,000; issue 190,000 (direct 178,000, indirect 12,000); DL 90,000, indirect labor 110,000; factory depreciation 40,000; other OH 70,000 credited AP. Apply OH 8/MH on 30,000 MHs. Complete jobs 520,000. Sell jobs costing 480,000 on credit at 25% above cost. Beginning WIP 42,000.

Required

1. Prepare journal entries.
2. Post MOH and WIP T-accounts and determine ending balances.

Model Answer

Debit is the left side and credit the right side. Inventory increases are debits; inventory transfers credit the sending account and debit the receiving account. Actual overhead is debited to MOH; applied overhead is credited to MOH and debited to WIP.

Entry	Account	Debit	Credit
a	Raw materials	210,000.00	
a	Accounts payable		210,000.00
b	WIP	178,000.00	
b	MOH	12,000.00	
b	Raw materials		190,000.00

Q47 · 2017-2018 · Exam Q5(a) [18 marks] (answer continued)

Entry	Account	Debit	Credit
c	WIP	90,000.00	
c	MOH	110,000.00	
c	Wages payable		200,000.00
d	MOH	40,000.00	
d	Accumulated depreciation		40,000.00
e	MOH	70,000.00	
e	Accounts payable		70,000.00
f	WIP	240,000.00	
f	MOH		240,000.00
g	Finished goods	520,000.00	
g	WIP		520,000.00
h-sale	Accounts receivable	600,000.00	
h-sale	Sales		600,000.00
h-cost	COGS	480,000.00	
h-cost	Finished goods		480,000.00

WIP

Debit	Credit
Beginning: 42,000.00	g: 520,000.00
b: 178,000.00	
c: 90,000.00	
f: 240,000.00	
Ending balance: 30,000.00	

Raw materials

Debit	Credit
a: 210,000.00	b: 190,000.00
Ending balance: 20,000.00	

Q47 · 2017-2018 · Exam Q5(a) [18 marks] (answer continued)

Accounts payable

Debit	Credit
	a: 210,000.00
	e: 70,000.00
	Ending balance: 280,000.00

MOH

Debit	Credit
b: 12,000.00	f: 240,000.00
c: 110,000.00	Ending balance: 8,000.00
d: 40,000.00	
e: 70,000.00	

Wages payable

Debit	Credit
	c: 200,000.00
	Ending balance: 200,000.00

Accumulated depreciation

Debit	Credit
	d: 40,000.00
	Ending balance: 40,000.00

Finished goods

Debit	Credit
g: 520,000.00	h-cost: 480,000.00
Ending balance: 40,000.00	

Accounts receivable

Debit	Credit
h-sale: 600,000.00	
Ending balance: 600,000.00	

Sales

Debit	Credit
	h-sale: 600,000.00
	Ending balance: 600,000.00

COGS

Debit	Credit
h-cost: 480,000.00	
Ending balance: 480,000.00	

Ending balance after overhead adjustment	Debit	Credit
WIP	30,000.00	
Raw materials	20,000.00	
Accounts payable		280,000.00
MOH		8,000.00
Wages payable		200,000.00
Accumulated depreciation		40,000.00
Finished goods	40,000.00	

Q47 · 2017-2018 · Exam Q5(a) [18 marks] (answer continued)

Ending balance after overhead adjustment	Debit	Credit
Accounts receivable	600,000.00	
Sales		600,000.00
COGS	480,000.00	

Every journal entry balances. Closing a debit (underapplied) overhead balance debits COGS and credits overhead; closing a credit (overapplied) balance does the reverse. The T-account balances shown precede general year-end closing of revenue/expense accounts.

WIP ends at 30,000 debit. Actual OH = 232,000; applied = 240,000; MOH ends at 8,000 credit (overapplied). Other accounts with unspecified opening balances show current movements, not a claim that every opening balance was zero. No overhead-closing entry is requested.

Q48 · 2017-2018 · Exam Q5(b) [17 marks]

Source: Ipe 2009-25.pdf, PDF pages 51, 52.

Kingsport makes custom steel drums and finds quarterly unit costs confusing.

Forecast	Q1	Q2	Q3	Q4
Materials	240,000	120,000	60,000	180,000
Labor	128,000	64,000	32,000	96,000
OH	300,000	220,000	180,000	260,000
Total cost	668,000	404,000	272,000	536,000
Units	80,000	40,000	20,000	60,000
Reported unit cost	8.35	10.10	13.60	8.93

OH is mostly fixed and shows little sensitivity to production.

Required

1. Recommend a specific overhead-assignment method and recompute unit product costs.

Model Answer

$$\text{Annual POHR} = (300,000 + 220,000 + 180,000 + 260,000) / 200,000 = 4.80 \text{ pe}$$

Q48 · 2017-2018 · Exam Q5(b) [17 marks] (answer continued)

Unit product cost	Q1	Q2	Q3	Q4
Materials/unit	3	3	3	3
Labor/unit	1.60	1.60	1.60	1.60
Applied annual OH/unit	4.80	4.80	4.80	4.80
Total normal product cost/unit	9.40	9.40	9.40	9.40

Use an annual predetermined rate rather than dividing actual fixed overhead by each quarter's fluctuating output. Annual applied OH reconciles to 960,000 at expected production. If jobs differ in processing effort, use a more causal annual driver such as labor- or machine-hours rather than assuming identical drums.

Q49 · 2018-2019 · Exam Q5 [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 59, 60.

Smithson estimates Molding/Fabrication MHs 20,000 / 30,000; fixed OH 800,000 / 300,000; variable OH 5/MH in both. No opening/closing inventory or over/underapplied OH. Only two jobs are completed and sold.

Job/dept	DM	DL	MH
D-75 Molding	375,000	200,000	15,000
D-75 Fabrication	325,000	160,000	5,000
C-100 Molding	305,000	175,000	5,000
C-100 Fabrication	250,000	225,000	25,000

Required

1. Under plantwide and departmental MH rates, find rates, job costs, bids at 150% of costs and total COGS.
2. Explain the managerial insights and why COGS and bids do or do not differ.

Model Answer

The estimated total overhead includes both the fixed estimate and variable rate x estimated driver quantity.

Q49 · 2018-2019 · Exam Q5 [35 marks] (answer continued)

Department	Estimated OH	Driver quantity	Departmental rate
Molding	900,000.00	20,000	45.00
Fabrication	450,000.00	30,000	15.00

$$\text{Plantwide rate} = \frac{1,350,000}{50,000} = 27.00\,.$$

Job	DM	DL	Plantwide OH	Plantwide total	Departmental OH	Departmental total
D-75	700,000.00	360,000.00	540,000.00	1,600,000.00	750,000.00	1,810,000.00
C-100	555,000.00	400,000.00	810,000.00	1,765,000.00	600,000.00	1,555,000.00

Job	Plantwide bid	Departmental bid
D-75	2,400,000.00	2,715,000.00
C-100	2,647,500.00	2,332,500.00

Total manufacturing costs / COGS	Plantwide	Departmental
All completed and sold jobs	3,365,000.00	3,365,000.00

Total COGS is identical because the same aggregate overhead is allocated and every job is sold, with no inventory or over/underapplied balance. Individual jobs and bids change because they use departments in different proportions. A plantwide average can overcost jobs that use the cheaper department heavily and undercost jobs that use the more expensive department heavily.

Q50 · 2018-2019 · Exam Q8(a) [27 marks]

Source: Ipe 2009-25.pdf, PDF pages 62, 63.

Nature's Way Blending opening WIP 32,800 consists of materials 8,000, labor 4,000 and OH 20,800. March added Blending materials 147,600, labor 73,200, applied OH 481,000; Bottling materials 45,000, labor 17,000, applied OH 108,000. Actual whole-factory MOH 596,000 is credited to AP. Transfers: Blending to Bottling 722,000 for 760,000 ounces; Bottling to FG 920,000. Credit sales 1,400,000; job cost sold 890,000. Opening RM 198,600; Bottling WIP 49,000; FG 20,000.

Q50 · 2018-2019 · Exam Q8(a) [27 marks] (question continued)

Required

1. Record entries (i)-(vii), post to T-accounts, and determine ending inventory and MOH balances.

Model Answer

Debit is the left side and credit the right side. Inventory increases are debits; inventory transfers credit the sending account and debit the receiving account. Actual overhead is debited to MOH; applied overhead is credited to MOH and debited to WIP.

Entry	Account	Debit	Credit
i	WIP-Blending	147,600.00	
i	WIP-Bottling	45,000.00	
i	Raw materials		192,600.00
ii	WIP-Blending	73,200.00	
ii	WIP-Bottling	17,000.00	
ii	Wages payable		90,200.00
iii	MOH	596,000.00	
iii	Accounts payable		596,000.00
iv	WIP-Blending	481,000.00	
iv	WIP-Bottling	108,000.00	
iv	MOH		589,000.00
v	WIP-Bottling	722,000.00	
v	WIP-Blending		722,000.00
vi	Finished goods	920,000.00	
vi	WIP-Bottling		920,000.00
vii-sale	Accounts receivable	1,400,000.00	
vii-sale	Sales		1,400,000.00
vii-cost	COGS	890,000.00	
vii-cost	Finished goods		890,000.00

Q50 · 2018-2019 · Exam Q8(a) [27 marks] (answer continued)

Raw materials

Debit	Credit
Beginning: 198,600.00	i: 192,600.00
Ending balance: 6,000.00	

WIP-Blending

Debit	Credit
Beginning: 32,800.00	v: 722,000.00
i: 147,600.00	
ii: 73,200.00	
iv: 481,000.00	
Ending balance: 12,600.00	

WIP-Bottling

Debit	Credit
Beginning: 49,000.00	vi: 920,000.00
i: 45,000.00	
ii: 17,000.00	
iv: 108,000.00	
v: 722,000.00	
Ending balance: 21,000.00	

Finished goods

Debit	Credit
Beginning: 20,000.00	vii-cost: 890,000.00
vi: 920,000.00	
Ending balance: 50,000.00	

Wages payable

Debit	Credit
	ii: 90,200.00
	Ending balance: 90,200.00

MOH

Debit	Credit
iii: 596,000.00	iv: 589,000.00
Ending balance: 7,000.00	

Accounts payable

Debit	Credit
	iii: 596,000.00
	Ending balance: 596,000.00

Accounts receivable

Debit	Credit
vii-sale: 1,400,000.00	
Ending balance: 1,400,000.00	

Q50 · 2018-2019 · Exam Q8(a) [27 marks] (answer continued)

Sales		COGS	
Debit	**Credit**	**Debit**	**Credit**
	vii-sale: 1,400,000.00	vii-cost: 890,000.00	
	Ending balance: 1,400,000.00	Ending balance: 890,000.00	

Ending balance after overhead adjustment	Debit	Credit
Raw materials	6,000.00	
WIP-Blending	12,600.00	
WIP-Bottling	21,000.00	
Finished goods	50,000.00	
Wages payable		90,200.00
MOH	7,000.00	
Accounts payable		596,000.00
Accounts receivable	1,400,000.00	
Sales		1,400,000.00
COGS	890,000.00	

Every journal entry balances. Closing a debit (underapplied) overhead balance debits COGS and credits overhead; closing a credit (overapplied) balance does the reverse. The T-account balances shown precede general year-end closing of revenue/expense accounts.

Ending balances: RM 6,000; Blending WIP 12,600; Bottling WIP 21,000; FG 50,000; MOH debit 7,000 (underapplied). No overhead closing entry is requested here.

Q51 · January 2020 Term · Exam Q5(a) [45 marks]

Source: Ipe 2009-25.pdf, PDF pages 66, 67.

Film Specialties applies overhead using camera-hours; estimated OH 1,350,000 and hours 15,000. Opening trial balance follows.

Account	Opening debit	Opening credit
Cash	60,000	

Q51 · January 2020 Term · Exam Q5(a) [45 marks] (question continued)

Account	Opening debit	Opening credit
Accounts receivable	210,000	
Materials/supplies	130,000	
Films in process	75,000	
Finished films	860,000	
Prepaid insurance	90,000	
Studio/equipment	5,200,000	
Accumulated depreciation		1,990,000
Accounts payable		700,000
Wages payable		35,000
Capital stock		2,500,000
Retained earnings		1,400,000

Required

1. a. Purchase materials/supplies on account 690,000.
2. b. Issue 700,000, 80% direct and 20% indirect.
3. c. Studio utilities incurred 90,000.
4. d. Incur direct actors/directors/crew wages 1,300,000; support labor 230,000; marketing/admin salaries 650,000.
5. e. Advertising incurred 800,000.
6. f. Insurance expires 70,000: studio 60,000, S&A 10,000.
7. g. Depreciation 650,000, 80% studio and 20% S&A.
8. h. Rent incurred: production 360,000, S&A 40,000.
9. i. Apply overhead on 16,500 camera-hours.
10. j. Complete films costing 3,400,000.
11. k. Sell films on account 6,000,000; job cost 4,000,000.
12. l. Collect 5,400,000 from customers.
13. m. Pay creditors 2,500,000 and employees 2,200,000.

Q51 · January 2020 Term · Exam Q5(a) [45 marks] (question continued)

Required

1. Prepare journals, all T-accounts and ending balances.
2. Close production overhead to cost of films sold.
3. Prepare the income statement.

> **Source clarification:** Costs described as incurred, without payment, are credited to payables. Materials purchases and advertising/rent/utilities use accounts payable; employee costs use wages payable.

Model Answer

$$\text{Rate} = 1,350,000 / 15,000 = 90; \quad \text{Applied} = 16,500(90) = 1,485,000.$$

Actual production OH = 140,000 + 90,000 + 230,000 + 60,000 + 520,000 + 360,000 = 1,400,000. It is overapplied by 85,000.

Debit is the left side and credit the right side. Inventory increases are debits; inventory transfers credit the sending account and debit the receiving account. Actual overhead is debited to MOH; applied overhead is credited to MOH and debited to WIP.

Entry	Account	Debit	Credit
a	Materials/supplies	690,000.00	
a	Accounts payable		690,000.00
b	Films in process	560,000.00	
b	Production overhead	140,000.00	
b	Materials/supplies		700,000.00
c	Production overhead	90,000.00	
c	Accounts payable		90,000.00
d	Films in process	1,300,000.00	
d	Production overhead	230,000.00	
d	S&A salaries	650,000.00	
d	Wages payable		2,180,000.00
e	Advertising expense	800,000.00	
e	Accounts payable		800,000.00

Q51 · January 2020 Term · Exam Q5(a) [45 marks] (answer continued)

Entry	Account	Debit	Credit
f	Production overhead	60,000.00	
f	S&A insurance	10,000.00	
f	Prepaid insurance		70,000.00
g	Production overhead	520,000.00	
g	S&A depreciation	130,000.00	
g	Accumulated depreciation		650,000.00
h	Production overhead	360,000.00	
h	S&A rent	40,000.00	
h	Accounts payable		400,000.00
i	Films in process	1,485,000.00	
i	Production overhead		1,485,000.00
j	Finished films	3,400,000.00	
j	Films in process		3,400,000.00
k-sale	Accounts receivable	6,000,000.00	
k-sale	Sales		6,000,000.00
k-cost	Cost of films sold	4,000,000.00	
k-cost	Finished films		4,000,000.00
l	Cash	5,400,000.00	
l	Accounts receivable		5,400,000.00
m-creditors	Accounts payable	2,500,000.00	
m-creditors	Cash		2,500,000.00
m-payroll	Wages payable	2,200,000.00	
m-payroll	Cash		2,200,000.00
OH-close	Production overhead	85,000.00	
OH-close	Cost of films sold		85,000.00

Q51 · January 2020 Term · Exam Q5(a) [45 marks] (answer continued)

Cash

Debit	Credit
Beginning: 60,000.00	m-creditors: 2,500,000.00
l: 5,400,000.00	m-payroll: 2,200,000.00
Ending balance: 760,000.00	

Accounts receivable

Debit	Credit
Beginning: 210,000.00	l: 5,400,000.00
k-sale: 6,000,000.00	
Ending balance: 810,000.00	

Materials/supplies

Debit	Credit
Beginning: 130,000.00	b: 700,000.00
a: 690,000.00	
Ending balance: 120,000.00	

Films in process

Debit	Credit
Beginning: 75,000.00	j: 3,400,000.00
b: 560,000.00	
d: 1,300,000.00	
i: 1,485,000.00	
Ending balance: 20,000.00	

Finished films

Debit	Credit
Beginning: 860,000.00	k-cost: 4,000,000.00
j: 3,400,000.00	
Ending balance: 260,000.00	

Prepaid insurance

Debit	Credit
Beginning: 90,000.00	f: 70,000.00
Ending balance: 20,000.00	

Studio/equipment

Debit	Credit
Beginning: 5,200,000.00	
Ending balance: 5,200,000.00	

Accumulated depreciation

Debit	Credit
	Beginning: 1,990,000.00
	g: 650,000.00
	Ending balance: 2,640,000.00

Q51 · January 2020 Term · Exam Q5(a) [45 marks] (answer continued)

Accounts payable

Debit	Credit
m-creditors: 2,500,000.00	Beginning: 700,000.00
	a: 690,000.00
	c: 90,000.00
	e: 800,000.00
	h: 400,000.00
	Ending balance: 180,000.00

Wages payable

Debit	Credit
m-payroll: 2,200,000.00	Beginning: 35,000.00
	d: 2,180,000.00
	Ending balance: 15,000.00

Capital stock

Debit	Credit
	Beginning: 2,500,000.00
	Ending balance: 2,500,000.00

Retained earnings

Debit	Credit
	Beginning: 1,400,000.00
	Ending balance: 1,400,000.00

Production overhead

Debit	Credit
b: 140,000.00	i: 1,485,000.00
c: 90,000.00	
d: 230,000.00	
f: 60,000.00	
g: 520,000.00	
h: 360,000.00	
OH-close: 85,000.00	
Ending balance: 0.00	

S&A salaries

Debit	Credit
d: 650,000.00	
Ending balance: 650,000.00	

Advertising expense

Debit	Credit
e: 800,000.00	
Ending balance: 800,000.00	

S&A insurance

Debit	Credit
f: 10,000.00	
Ending balance: 10,000.00	

Q51 · January 2020 Term · Exam Q5(a) [45 marks] (answer continued)

S&A depreciation

Debit	Credit
g: 130,000.00	
Ending balance: 130,000.00	

S&A rent

Debit	Credit
h: 40,000.00	
Ending balance: 40,000.00	

Sales

Debit	Credit
	k-sale: 6,000,000.00
	Ending balance: 6,000,000.00

Cost of films sold

Debit	Credit
k-cost: 4,000,000.00	OH-close: 85,000.00
Ending balance: 3,915,000.00	

Ending balance after overhead adjustment	Debit	Credit
Cash	760,000.00	
Accounts receivable	810,000.00	
Materials/supplies	120,000.00	
Films in process	20,000.00	
Finished films	260,000.00	
Prepaid insurance	20,000.00	
Studio/equipment	5,200,000.00	
Accumulated depreciation		2,640,000.00
Accounts payable		180,000.00
Wages payable		15,000.00
Capital stock		2,500,000.00
Retained earnings		1,400,000.00
Production overhead	0.00	
S&A salaries	650,000.00	
Advertising expense	800,000.00	
S&A insurance	10,000.00	
S&A depreciation	130,000.00	

Q51 · January 2020 Term · Exam Q5(a) [45 marks] (answer continued)

Ending balance after overhead adjustment	Debit	Credit
S&A rent	40,000.00	
Sales		6,000,000.00
Cost of films sold	3,915,000.00	

Every journal entry balances. Closing a debit (underapplied) overhead balance debits COGS and credits overhead; closing a credit (overapplied) balance does the reverse. The T-account balances shown precede general year-end closing of revenue/expense accounts.

Income statement	Amount
Sales	6,000,000
Adjusted cost of films sold	3,915,000
Gross margin	2,085,000
S&A: salaries + advertising + insurance + depreciation + rent	1,630,000
Operating income	455,000

Q52 · 2020-2021 · Exam Q5(a) [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 72, 73.

Luzadis estimates overhead 900,000 for 75,000 MHs. Actual MHs 60,000 and overhead 850,000. Ending RM 30,000; WIP 100,000 including 36,000 applied OH; FG 500,000 including 180,000 OH; COGS 1,400,000 including 504,000 OH.

Required

1. Find over/underapplied OH.
2. Close directly to COGS.
3. Alternatively allocate among WIP, FG and COGS.
4. Compare NOI under the two methods.

Model Answer

$$\text{Rate} = 900,000 / 75,000 = 12; \quad \text{Applied} = 60,000(12) = 720,000\,.$$

Q52 · 2020-2021 · Exam Q5(a) [35 marks] (answer continued)

Underapplied overhead = 850,000 - 720,000 = 130,000 (debit MOH balance). Direct closing: Dr COGS 130,000; Cr MOH 130,000. Adjusted COGS = 1,530,000.

Applied-OH basis	Weight	Underapplication allocated	Adjusted balance
WIP	36,000 / 720,000 = 5%	6,500	106,500
FG	180,000 / 720,000 = 25%	32,500	532,500
COGS	504,000 / 720,000 = 70%	91,000	1,491,000

Allocation entry: Dr WIP 6,500; Dr FG 32,500; Dr COGS 91,000; Cr MOH 130,000. RM receives none because no applied manufacturing overhead is in RM. Allocation leaves 39,000 in inventories, so NOI is 39,000 higher than direct closing, before income taxes.

Q53 · 2021-2022 · Exam Q5(c) [8 marks]

Source: Ipe 2009-25.pdf, PDF page 82.

Why use a predetermined overhead rate rather than actual overhead rates?

Model Answer

Actual annual overhead and total annual activity are not known while jobs are being produced and priced. A predetermined rate gives timely job costs, supports estimates and smooths seasonal overhead or activity fluctuations. It is based on estimated overhead / estimated activity before the period. At period end compare actual overhead with applied overhead and dispose of the difference; the estimate is not assumed to equal actual cost.

Q54 · 2021-2022 · Exam Q6(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 83.

White Company uses departmental overhead rates: Cutting uses machine-hours; Finishing uses direct labor-hours.

Annual estimate	Cutting	Finishing
DLHs	6,000	30,000
Machine-hours	48,000	5,000
Fixed MOH	264,000	366,000

Q54 · 2021-2022 · Exam Q6(a) [20 marks] (question continued)

Annual estimate	Cutting	Finishing
Variable MOH rate	2 per MH	4 per DLH

Job 203	Cutting	Finishing
DLHs	6	20
Machine-hours	80	4
Materials	500	310
Direct labor	70	150

Required

1. Compute each predetermined departmental rate.
2. Compute Job 203 total manufacturing cost.
3. Would a plantwide rate based on DLHs assign different costs to some jobs? Explain without calculations.

Model Answer

$$\text{Cutting rate} = 264,000 / 48,000 + 2 = 7.50 \text{ per MH}.$$
$$\text{Finishing rate} = 366,000 / 30,000 + 4 = 16.20 \text{ per DLH}.$$

Job 203 cost	Amount
Cutting OH: 80 x 7.50	600
Finishing OH: 20 x 16.20	324
Total OH	924
Direct materials	810
Direct labor	220
Total manufacturing cost	1,954

Yes. Cutting is machine-intensive, whereas Finishing uses labor-hours. A single labor-hour rate ignores differing departmental use and cost drivers. Jobs with different mixes of machine/labor use would receive different costs, even though aggregate overhead still reconciles.

Q55 · 2022-2023 · Exam Q6(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 91, 92.

High Desert Pottery applies Molding overhead using machine-hours and Painting overhead using direct labor-hours.

Annual estimates	Molding	Painting
DLHs	12,000	60,000
Machine-hours	70,000	8,000
Direct materials	510,000	650,000
Direct labor cost	130,000	420,000
Fixed MOH	497,000	615,000
Variable MOH per machine-hour	1.50	-
Variable MOH per DLH	-	2

Job 205	Molding	Painting
DLHs	30	84
Machine-hours	110	20
Direct materials	470	332
Direct labor cost	325	588

Actual year-end data	Molding	Painting
DLHs	10,000	62,000
Machine-hours	65,000	9,000
Direct materials	430,000	680,000
Direct labor cost	108,000	436,000
Actual MOH	570,000	750,000

Required

1. Compute each department's predetermined overhead rate.
2. Compute overhead applied to Job 205.
3. Compute total job cost and unit cost if the job contains 50 units.

Q55 · 2022-2023 · Exam Q6(a) [20 marks] (question continued)

4. Compute under- or overapplied overhead for each department.

Model Answer

$$\text{Molding rate} = \frac{497,000 + 1.50(70,000)}{70,000} = 8.60 \text{ per MH}.$$

$$\text{Painting rate} = \frac{615,000 + 2(60,000)}{60,000} = 12.25 \text{ per DLH}.$$

Job 205 cost	Molding	Painting	Total
Direct materials	470	332	802
Direct labor	325	588	913
Applied MOH	946	1,029	1,975
Total job cost	1,741	1,949	3,690

$$\text{Unit product cost} = 3,690 / 50 = 73.80.$$

Year-end overhead	Molding	Painting
Actual activity for allocation	65,000	62,000
Applied MOH	559,000	759,500
Actual MOH	570,000	750,000
Actual less applied	11,000 underapplied	9,500 overapplied

Actual MOH is recorded on the debit side of Manufacturing Overhead; applied MOH is recorded on its credit side. A remaining debit is underapplied and a remaining credit is overapplied.

Q56 · 2023-2024 · Exam Q6(b) [22 marks]

Source: Ipe 2009-25.pdf, PDF pages 104, 105.

Delph uses job-order costing in Molding and Fabrication. There are no beginning or ending inventories, only jobs D-70 and C-200 were completed and sold, and no overhead was under- or overapplied.

Q56 · 2023-2024 · Exam Q6(b) [22 marks] (question continued)

Annual estimates	Molding	Fabrication
Machine-hours	20,000	30,000
Fixed MOH	700,000	210,000
Variable MOH per MH	3	3

Job data	D-70 Molding	D-70 Fabrication	C-200 Molding	C-200 Fabrication
Direct materials	375,000	325,000	300,000	250,000
Direct labor	200,000	160,000	175,000	225,000
Machine-hours	14,000	6,000	6,000	24,000

Required

1. Using a plantwide machine-hour rate, compute the rate, each job's manufacturing cost, bids at 150% of cost, and annual COGS.
2. Repeat using departmental machine-hour rates.
3. Explain the managerial insights, including total COGS and job bid differences.

Model Answer

The estimated total overhead includes both the fixed estimate and variable rate x estimated driver quantity.

Department	Estimated OH	Driver quantity	Departmental rate
Molding	760,000.00	20,000	38.00
Fabrication	300,000.00	30,000	10.00

$$\text{Plantwide rate} = \frac{1,060,000}{50,000} = 21.20\,.$$

Job	DM	DL	Plantwide OH	Plantwide total	Departmental OH	Departmental total
D-70	700,000.00	360,000.00	424,000.00	1,484,000.00	592,000.00	1,652,000.00
C-200	550,000.00	400,000.00	636,000.00	1,586,000.00	468,000.00	1,418,000.00

Q56 · 2023-2024 · Exam Q6(b) [22 marks] (answer continued)

Job	Plantwide bid	Departmental bid
D-70	2,226,000.00	2,478,000.00
C-200	2,379,000.00	2,127,000.00

Total manufacturing costs / COGS	Plantwide	Departmental
All completed and sold jobs	3,070,000.00	3,070,000.00

Total COGS is identical because the same aggregate overhead is allocated and every job is sold, with no inventory or over/underapplied balance. Individual jobs and bids change because they use departments in different proportions. A plantwide average can overcost jobs that use the cheaper department heavily and undercost jobs that use the more expensive department heavily.

Q57 · 2024-2025 · Exam Q6(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 114.

High Desert Pottery applies Molding overhead using machine-hours and Painting overhead using direct labor-hours.

Annual estimates	Molding	Painting
DLHs	12,000	60,000
Machine-hours	70,000	8,000
Direct materials	510,000	650,000
Direct labor cost	130,000	420,000
Fixed MOH	497,000	615,000
Variable MOH per machine-hour	1.50	-
Variable MOH per DLH	-	2

Job 205	Molding	Painting
DLHs	30	84
Machine-hours	110	20
Direct materials	470	332

Q57 · 2024-2025 · Exam Q6(a) [20 marks] (question continued)

Job 205	Molding	Painting
Direct labor cost	325	588

Actual year-end data	Molding	Painting
DLHs	10,000	62,000
Machine-hours	65,000	9,000
Direct materials	430,000	680,000
Direct labor cost	108,000	436,000
Actual MOH	570,000	750,000

Required

1. Compute each department's predetermined overhead rate.
2. Compute overhead applied to Job 205.
3. Compute total job cost and unit cost if the job contains 50 units.
4. Compute under- or overapplied overhead for each department.

Model Answer

$$\text{Molding rate} = \frac{497,000 + 1.50(70,000)}{70,000} = 8.60 \text{ per MH}.$$

$$\text{Painting rate} = \frac{615,000 + 2(60,000)}{60,000} = 12.25 \text{ per DLH}.$$

Job 205 cost	Molding	Painting	Total
Direct materials	470	332	802
Direct labor	325	588	913
Applied MOH	946	1,029	1,975
Total job cost	1,741	1,949	3,690

$$\text{Unit product cost} = 3,690 / 50 = 73.80.$$

Q57 · 2024-2025 · Exam Q6(a) [20 marks] (answer continued)

Year-end overhead	Molding	Painting
Actual activity for allocation	65,000	62,000
Applied MOH	559,000	759,500
Actual MOH	570,000	750,000
Actual less applied	11,000 underapplied	9,500 overapplied

Actual MOH is recorded on the debit side of Manufacturing Overhead; applied MOH is recorded on its credit side. A remaining debit is underapplied and a remaining credit is overapplied.

Process costing: weighted average and FIFO

Q58 · 2009-2010 · Exam Q7(a) [8 marks]

Source: Ipe 2009-25.pdf, PDF page 7.

Distinguish process and job-order costing.

Model Answer

Feature	Job-order costing	Process costing
Production	Distinct jobs or customer orders	Continuous production of similar units
Cost accumulation	Each job cost sheet	Each department/process for a period
Unit cost	Job cost divided by units in the job	Department cost assigned through equivalent units
Work in process	Identified by unfinished jobs	Identified by units and completion percentages
Examples	Custom furniture, construction, printing orders	Cement, chemicals, petroleum, food processing

Both record direct materials, direct labor and manufacturing overhead. The difference is how production costs are accumulated and averaged, not whether overhead exists.

Q59 · 2009-2010 · Exam Q7(b) [27 marks]

Source: Ipe 2009-25.pdf, PDF pages 7, 8.

Paint Base Fab: beginning 30,000 gallons, materials 100%, conversion 60%; started 420,000; completed 370,000; ending 80,000, materials 50%, conversion 25%. Beginning costs materials 92,000, labor 21,000, OH 37,000. Added materials 851,000, labor 330,000, OH 665,000.

Required

1. Calculate equivalent material/labor/OH costs and reconcile.

> **Source clarification:** The method is not explicitly named; weighted average is used. Materials cost/EU 2.30 plus labor 0.90 plus OH 1.80 = 5.00.

Q59 · 2009-2010 · Exam Q7(b) [27 marks] (answer continued)

Model Answer

Weighted average merges beginning WIP costs and current costs. It does not deduct the work already done on beginning WIP from equivalent units.

Physical unit flow	Units
Beginning WIP	30,000
Started	420,000
Total to account for	450,000
Transferred out	370,000
Ending WIP	80,000
Total accounted for	450,000

Equivalent-unit calculation	Materials	Conversion
Transferred out (fully complete)	370,000	370,000
Ending WIP	40,000	20,000
Total equivalent units	410,000	390,000

Cost per equivalent unit	Materials	Conversion
Costs in numerator	943,000.00	1,053,000.00
Equivalent units	410,000	390,000
Cost per EU (unrounded in assignments)	2.300000	2.700000

Cost assignment	Amount
Transferred out	1,850,000.00
Ending WIP: materials	92,000.00
Ending WIP: conversion	54,000.00
Ending WIP total	146,000.00
Transferred out + ending WIP	1,996,000.00
Beginning costs + added costs	1,996,000.00

Q59 · 2009-2010 · Exam Q7(b) [27 marks] (answer continued)

Use unrounded rates when assigning costs. The physical units reconcile, and total costs assigned equal total costs to account for.

Separate conversion components	Labor	OH
Total cost	351,000	702,000
Equivalent units	390,000	390,000
Cost/EU	0.90	1.80
Ending WIP cost	18,000	36,000
Transferred cost	333,000	666,000

Q60 · 2011-2012 · Exam Q1(a) [8 marks]

Source: Ipe 2009-25.pdf, PDF page 10.

Distinguish job-order and process costing.

Model Answer

Feature	Job-order costing	Process costing
Production	Distinct jobs or customer orders	Continuous production of similar units
Cost accumulation	Each job cost sheet	Each department/process for a period
Unit cost	Job cost divided by units in the job	Department cost assigned through equivalent units
Work in process	Identified by unfinished jobs	Identified by units and completion percentages
Examples	Custom furniture, construction, printing orders	Cement, chemicals, petroleum, food processing

Both record direct materials, direct labor and manufacturing overhead. The difference is how production costs are accumulated and averaged, not whether overhead exists.

Q61 · 2011-2012 · Exam Q1(b) [27 marks]

Source: Ipe 2009-25.pdf, PDF page 10.

Lindex Cutting adds materials at the start and conversion uniformly. February beginning WIP 50,000, material 100%, conversion 40%; costs material 70,500 and conversion 34,050. Started 225,000; completed 200,000; ending WIP 75,000, material 100%, conversion 20%. Added material 342,000; conversion 352,950.

Required

1. Prepare FIFO quantity/EU schedule and cost reconciliation.

Model Answer

FIFO separates prior-period work from work performed this period. Divide only this period's added costs by this period's equivalent units.

Physical unit flow	Units
Beginning WIP	50,000
Started	225,000
Total to account for	275,000
Transferred out	200,000
Ending WIP	75,000
Total accounted for	275,000

Equivalent-unit calculation	Materials	Conversion
Complete beginning WIP	0	30,000
Started and completed	150,000	150,000
Ending WIP	75,000	15,000
Total equivalent units	225,000	195,000

Cost per equivalent unit	Materials	Conversion
Costs in numerator	342,000.00	352,950.00
Equivalent units	225,000	195,000
Cost per EU (unrounded in assignments)	1.520000	1.810000

Q61 · 2011-2012 · Exam Q1(b) [27 marks] (answer continued)

Cost assignment	Amount
Beginning WIP cost brought forward	104,550.00
Cost to finish beginning WIP	54,300.00
Started and completed this period	499,500.00
Total transferred out	658,350.00
Ending WIP: materials	114,000.00
Ending WIP: conversion	27,150.00
Ending WIP total	141,150.00
Transferred out + ending WIP	799,500.00
Beginning costs + added costs	799,500.00

Use unrounded rates when assigning costs. The physical units reconcile, and total costs assigned equal total costs to account for.

Q62 · 2012-2013 · Exam Q7 [35 marks]

Source: Ipe 2009-25.pdf, PDF page 22.

A company has ending WIP 300,000 units, 50% conversion complete, reported cost 660,960; ending FG 200,000 units, reported cost 1,009,800. Materials enter at the start; OH is 60% of DL. Beginning WIP 200,000 units, 80% conversion complete, materials 200,000, labor 315,000; no beginning FG. Started 1,000,000; completed 900,000. Added materials 1,300,000; labor 1,995,000.

Required

1. Compute EUs and material/labor/OH rates.
2. Verify ending WIP and FG.
3. Compute COGS if no over/underapplied OH.

> **Source clarification:** The method is unspecified and printed inventory values are inconsistent. Weighted average is the principal solution, with FIFO alternatives provided explicitly.

Model Answer

Using weighted average, beginning OH = 60% x 315,000 = 189,000; added OH = 1,197,000.

Q62 · 2012-2013 · Exam Q7 [35 marks] (answer continued)

Weighted-average calculation	Material	Labor	OH
Total cost	1,500,000	2,310,000	1,386,000
EUs: completed + ending work	1,200,000	1,050,000	1,050,000
Cost/EU	1.25	2.20	1.32

$$\text{Ending WIP} = 300,000[1.25 + 0.5(2.20 + 1.32)] = 903,000\,.$$

$$\text{Ending FG} = 200,000(1.25 + 2.20 + 1.32) = 954,000\,.$$

Total costs 5,196,000 = transferred cost 4,293,000 + ending WIP 903,000. If FIFO is intended instead, current material EU = 1,000,000, labor/OH EU = 890,000; rates = 1.30, 2.241573, 1.344944. FIFO ending WIP = 927,977.53; ending FG (from current units) = 977,303.37.

The two printed inventory amounts are inconsistent with the cost/unit data under either method. Indeed, their per-unit costs would imply a negative material rate if both used one weighted-average rate. They must not be accepted as checked answers.

Under the weighted-average solution, sold units = 900,000 completed - 200,000 ending FG = 700,000. COGS = 700,000 x 4.77 = 3,339,000. If FIFO is used, transferred cost 4,268,022.47 less ending FG 977,303.37 gives COGS 3,290,719.10. The beginning inventory row incorrectly says materials are 80% complete while also stating they enter at the start; apply the 80% to conversion.

Q63 · 2013-2014 · Exam Q7 [35 marks]

Source: Ipe 2009-25.pdf, PDF page 29.

A company has ending WIP 300,000 units, 50% conversion complete, reported cost 660,960; ending FG 200,000 units, reported cost 1,009,800. Materials enter at the start; OH is 60% of DL. Beginning WIP 200,000 units, 80% conversion complete, materials 200,000, labor 315,000; no beginning FG. Started 1,000,000; completed 900,000. Added materials 1,300,000; labor 1,995,000.

Required

1. Compute material/labor/OH rates, correct ending WIP/FG and cost reconciliation.

> **Source clarification:** The method is unspecified and printed inventory values are inconsistent. Weighted average is the principal solution, with FIFO alternatives provided explicitly.

Q63 · 2013-2014 · Exam Q7 [35 marks] (answer continued)

Model Answer

Using weighted average, beginning OH = 60% x 315,000 = 189,000; added OH = 1,197,000.

Weighted-average calculation	Material	Labor	OH
Total cost	1,500,000	2,310,000	1,386,000
EUs: completed + ending work	1,200,000	1,050,000	1,050,000
Cost/EU	1.25	2.20	1.32

$$\text{Ending WIP} = 300,000[1.25 + 0.5(2.20 + 1.32)] = 903,000\,.$$

$$\text{Ending FG} = 200,000(1.25 + 2.20 + 1.32) = 954,000\,.$$

Total costs 5,196,000 = transferred cost 4,293,000 + ending WIP 903,000. If FIFO is intended instead, current material EU = 1,000,000, labor/OH EU = 890,000; rates = 1.30, 2.241573, 1.344944. FIFO ending WIP = 927,977.53; ending FG (from current units) = 977,303.37.

The two printed inventory amounts are inconsistent with the cost/unit data under either method. Indeed, their per-unit costs would imply a negative material rate if both used one weighted-average rate. They must not be accepted as checked answers.

Weighted-average reconciliation: beginning cost 704,000 + added cost 4,492,000 = 5,196,000; transferred cost 4,293,000 + ending WIP 903,000 = 5,196,000. The paper says no FG at year end but also lists 200,000 ending FG units; interpret the no-FG statement as beginning FG, consistent with the task and parallel papers.

Q64 · 2014-2015 · Exam Q7 [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 37, 36.

Cooperative San Jose's Mixing Department uses weighted-average process costing. Bottles sell at 12 each. Its April report follows.

Reported units/costs	Amount
Beginning WIP units	30,000
Beginning materials/conversion completion	90% / 80%
Units started	200,000
Transferred	190,000

Q64 · 2014-2015 · Exam Q7 [35 marks] (question continued)

Reported units/costs	Amount
Ending WIP units	40,000
Ending materials/conversion completion	75% / 60%
Beginning WIP cost	98,000
Added cost	827,000
Total cost	925,000
Reported transferred cost	805,600
Reported ending WIP cost	119,400

Cost detail	Beginning WIP	Added this month
Materials	67,800	579,000
Conversion	30,200	248,000

Required

1. Find EUs, cost per EU and how many transferred units were started and completed this month.

Model Answer

Weighted average merges beginning WIP costs and current costs. It does not deduct the work already done on beginning WIP from equivalent units.

Physical unit flow	Units
Beginning WIP	30,000
Started	200,000
Total to account for	230,000
Transferred out	190,000
Ending WIP	40,000
Total accounted for	230,000

Q64 · 2014-2015 · Exam Q7 [35 marks] (answer continued)

Equivalent-unit calculation	Materials	Conversion
Transferred out (fully complete)	190,000	190,000
Ending WIP	30,000	24,000
Total equivalent units	220,000	214,000

Cost per equivalent unit	Materials	Conversion
Costs in numerator	646,800.00	278,200.00
Equivalent units	220,000	214,000
Cost per EU (unrounded in assignments)	2.940000	1.300000

Cost assignment	Amount
Transferred out	805,600.00
Ending WIP: materials	88,200.00
Ending WIP: conversion	31,200.00
Ending WIP total	119,400.00
Transferred out + ending WIP	925,000.00
Beginning costs + added costs	925,000.00

Use unrounded rates when assigning costs. The physical units reconcile, and total costs assigned equal total costs to account for.

$$\text{Started and completed} = 190,000 - 30,000 = 160,000\,.$$

Q65 · 2016-2017 · Exam Q4(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 42.

Lindex Cutting adds materials at the start and conversion uniformly. February beginning WIP 50,000, material 100%, conversion 40%; costs material 70,500 and conversion 34,050. Started 225,000; completed 200,000; ending WIP 75,000, material 100%, conversion 20%. Added material 342,000; conversion 352,950.

Required

1. Prepare FIFO quantity/EU schedule and cost reconciliation.

Q65 · 2016-2017 · Exam Q4(a) [20 marks] (answer continued)

Model Answer

FIFO separates prior-period work from work performed this period. Divide only this period's added costs by this period's equivalent units.

Physical unit flow	Units
Beginning WIP	50,000
Started	225,000
Total to account for	275,000
Transferred out	200,000
Ending WIP	75,000
Total accounted for	275,000

Equivalent-unit calculation	Materials	Conversion
Complete beginning WIP	0	30,000
Started and completed	150,000	150,000
Ending WIP	75,000	15,000
Total equivalent units	225,000	195,000

Cost per equivalent unit	Materials	Conversion
Costs in numerator	342,000.00	352,950.00
Equivalent units	225,000	195,000
Cost per EU (unrounded in assignments)	1.520000	1.810000

Cost assignment	Amount
Beginning WIP cost brought forward	104,550.00
Cost to finish beginning WIP	54,300.00
Started and completed this period	499,500.00
Total transferred out	658,350.00
Ending WIP: materials	114,000.00

Q65 · 2016-2017 · Exam Q4(a) [20 marks] (answer continued)

Cost assignment	Amount
Ending WIP: conversion	27,150.00
Ending WIP total	141,150.00
Transferred out + ending WIP	799,500.00
Beginning costs + added costs	799,500.00

Use unrounded rates when assigning costs. The physical units reconcile, and total costs assigned equal total costs to account for.

Q66 · 2016-2017 · Exam Q4(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF pages 42, 43.

A company has ending WIP 300,000 units, 50% conversion complete, reported cost 660,960; ending FG 200,000 units, reported cost 1,009,800. Materials enter at the start; OH is 60% of DL. Beginning WIP 200,000 units, 80% conversion complete, materials 200,000, labor 315,000; no beginning FG. Started 1,000,000; completed 900,000. Added materials 1,300,000; labor 1,995,000.

Required

1. Determine material/labor/OH cost per EU.
2. Determine correct ending WIP and FG cost.

> **Source clarification:** The method is unspecified and printed inventory values are inconsistent. Weighted average is the principal solution, with FIFO alternatives provided explicitly.

Model Answer

Using weighted average, beginning OH = 60% x 315,000 = 189,000; added OH = 1,197,000.

Weighted-average calculation	Material	Labor	OH
Total cost	1,500,000	2,310,000	1,386,000
EUs: completed + ending work	1,200,000	1,050,000	1,050,000
Cost/EU	1.25	2.20	1.32

$$\text{Ending WIP} = 300,000[1.25 + 0.5(2.20 + 1.32)] = 903,000\,.$$

Q66 · 2016-2017 · Exam Q4(b) [15 marks] (answer continued)

$$\text{Ending FG} = 200,000(1.25 + 2.20 + 1.32) = 954,000\,.$$

Total costs 5,196,000 = transferred cost 4,293,000 + ending WIP 903,000. If FIFO is intended instead, current material EU = 1,000,000, labor/OH EU = 890,000; rates = 1.30, 2.241573, 1.344944. FIFO ending WIP = 927,977.53; ending FG (from current units) = 977,303.37.

The two printed inventory amounts are inconsistent with the cost/unit data under either method. Indeed, their per-unit costs would imply a negative material rate if both used one weighted-average rate. They must not be accepted as checked answers.

Q67 · 2017-2018 · Exam Q8(a) [18 marks]

Source: Ipe 2009-25.pdf, PDF page 53.

Builder Products uses weighted-average process costing. Beginning 10,000 units, materials 100% and conversion 80%; started 100,000; transferred 95,000; ending completion materials 60%, conversion 20%. Beginning costs: materials 1,500; labor 1,800; OH 5,400. Added: materials 154,500; labor 22,700; OH 68,100.

Required

1. Compute EUs/unit costs and prepare cost reconciliation.

Model Answer

Weighted average merges beginning WIP costs and current costs. It does not deduct the work already done on beginning WIP from equivalent units.

Physical unit flow	Units
Beginning WIP	10,000
Started	100,000
Total to account for	110,000
Transferred out	95,000
Ending WIP	15,000
Total accounted for	110,000

Equivalent-unit calculation	Materials	Conversion
Transferred out (fully complete)	95,000	95,000

Q67 · 2017-2018 · Exam Q8(a) [18 marks] (answer continued)

Equivalent-unit calculation	Materials	Conversion
Ending WIP	9,000	3,000
Total equivalent units	104,000	98,000

Cost per equivalent unit	Materials	Conversion
Costs in numerator	156,000.00	98,000.00
Equivalent units	104,000	98,000
Cost per EU (unrounded in assignments)	1.500000	1.000000

Cost assignment	Amount
Transferred out	237,500.00
Ending WIP: materials	13,500.00
Ending WIP: conversion	3,000.00
Ending WIP total	16,500.00
Transferred out + ending WIP	254,000.00
Beginning costs + added costs	254,000.00

Use unrounded rates when assigning costs. The physical units reconcile, and total costs assigned equal total costs to account for.

Q68 · 2017-2018 · Exam Q8(b) [17 marks]

Source: Ipe 2009-25.pdf, PDF page 53.

Martin uses weighted-average costing. June beginning units 8,000 (materials 100%, conversion 75%); started 45,000; transferred 48,000; ending completion materials 100%, conversion 40%. Beginning costs: materials 5,150; labor 660; OH 1,320. Added costs: materials 29,300; labor 9,840; OH 19,680.

Required

1. Compute EUs/unit costs and prepare cost reconciliation.

> **Source clarification:** The June report labels its started/added rows May; they are treated as current June activity, consistent with the question heading.

Q68 · 2017-2018 · Exam Q8(b) [17 marks] (answer continued)

Model Answer

Weighted average merges beginning WIP costs and current costs. It does not deduct the work already done on beginning WIP from equivalent units.

Physical unit flow	Units
Beginning WIP	8,000
Started	45,000
Total to account for	53,000
Transferred out	48,000
Ending WIP	5,000
Total accounted for	53,000

Equivalent-unit calculation	Materials	Conversion
Transferred out (fully complete)	48,000	48,000
Ending WIP	5,000	2,000
Total equivalent units	53,000	50,000

Cost per equivalent unit	Materials	Conversion
Costs in numerator	34,450.00	31,500.00
Equivalent units	53,000	50,000
Cost per EU (unrounded in assignments)	0.650000	0.630000

Cost assignment	Amount
Transferred out	61,440.00
Ending WIP: materials	3,250.00
Ending WIP: conversion	1,260.00
Ending WIP total	4,510.00
Transferred out + ending WIP	65,950.00
Beginning costs + added costs	65,950.00

Q68 · 2017-2018 · Exam Q8(b) [17 marks] (answer continued)

Use unrounded rates when assigning costs. The physical units reconcile, and total costs assigned equal total costs to account for.

Q69 · 2018-2019 · Exam Q6 [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 60, 61.

Bohemian Links uses weighted-average costing in Casing/Curing. Each unit is one batch. Beginning WIP 1 batch, ending WIP 1; transferred 60 batches. Completion: beginning transferred-in/mat/conversion = 100%/60%/50%; ending = 100%/20%/10%.

Cost	Transferred-in (Mixing)	Materials	Conversion
Beginning	1,640	26	105
Added	94,740	8,402	61,197

Required

1. Compute equivalent units and unit costs without intermediate rounding.
2. Assign costs to ending WIP and transfers.
3. Prepare cost reconciliation.

Model Answer

Equivalent units and costs	Mixing	Materials	Conversion
Transferred EUs	60	60	60
Ending EUs	1	0.20	0.10
Total EUs	61	60.20	60.10
Total cost	96,380	8,428	61,302
Cost/EU	1,580	140	1,020

Cost assignment	Amount
Transfers: 60 x (1,580 + 140 + 1,020)	164,400
Ending: 1 x 1,580 + 0.20 x 140 + 0.10 x 1,020	1,710
Total assigned	166,110

Q69 · 2018-2019 · Exam Q6 [35 marks] (answer continued)

Cost assignment	Amount
Beginning: 1,640 + 26 + 105	1,771
Added: 94,740 + 8,402 + 61,197	164,339
Total to account for	166,110

Transferred-in costs are separate from added materials/conversion. Physical batches reconcile: beginning 1 + started 60 = transferred 60 + ending 1.

Q70 · 2020-2021 · Exam Q1 [35 marks]

Source: Ipe 2009-25.pdf, PDF page 69.

Luxguard has Base Fab and Finishing departments. April direct materials: 851,000 and 629,000; DL: 330,000 and 270,000; applied OH: 665,000 and 405,000. Base Fab transferred 1,850,000 to Finishing; Finishing transferred 3,200,000 to finished goods.

Base Fab report	Quantity/cost
Beginning units	30,000
Beginning completion: materials / conversion	100% / 60%
Started	420,000
Transferred out	370,000
Ending units	80,000
Ending completion: materials / conversion	50% / 25%
Beginning material/labor/OH cost	92,000 / 21,000 / 37,000
Added material/labor/OH	851,000 / 330,000 / 665,000

Required

1. Determine ending WIP and the cost of units transferred out of Base Fab.

> **Source clarification:** The method is not named. Weighted average is used because it reproduces the printed transfer cost of 1,850,000.

Q70 · 2020-2021 · Exam Q1 [35 marks] (answer continued)

Model Answer

Weighted average merges beginning WIP costs and current costs. It does not deduct the work already done on beginning WIP from equivalent units.

Physical unit flow	Units
Beginning WIP	30,000
Started	420,000
Total to account for	450,000
Transferred out	370,000
Ending WIP	80,000
Total accounted for	450,000

Equivalent-unit calculation	Materials	Conversion
Transferred out (fully complete)	370,000	370,000
Ending WIP	40,000	20,000
Total equivalent units	410,000	390,000

Cost per equivalent unit	Materials	Conversion
Costs in numerator	943,000.00	1,053,000.00
Equivalent units	410,000	390,000
Cost per EU (unrounded in assignments)	2.300000	2.700000

Cost assignment	Amount
Transferred out	1,850,000.00
Ending WIP: materials	92,000.00
Ending WIP: conversion	54,000.00
Ending WIP total	146,000.00
Transferred out + ending WIP	1,996,000.00
Beginning costs + added costs	1,996,000.00

Q70 · 2020-2021 · Exam Q1 [35 marks] (answer continued)

Use unrounded rates when assigning costs. The physical units reconcile, and total costs assigned equal total costs to account for.

Base Fab ending WIP = 146,000 and transferred cost = 1,850,000, agreeing with the stated transfer. Finishing unit/completion data are absent; a separate Finishing EU report is not requested or determinable from these data.

Q71 · 2021-2022 · Exam Q6(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF pages 83, 84.

Selzik cake mixes pass through Blending and Packaging. All materials are added at the start. Blending uses FIFO.

Production/cost data	Amount
Beginning WIP units	10,000
Beginning materials/conversion completion	100% / 30%
Started	170,000
Ending WIP units	20,000
Ending materials/conversion completion	100% / 40%
Beginning WIP materials cost	8,500
Beginning WIP conversion cost	4,900
Materials added	139,400
Conversion added	244,200

Required

1. Compute equivalent units.
2. Compute cost per equivalent unit.
3. Assign costs to ending WIP and transferred units.
4. Prepare a cost reconciliation report.

> **Source clarification:** The source narrative/requirements say March while the table says July. The one-period numerical data are consistent; the calculation uses that single set of balances and additions.

Q71 · 2021-2022 · Exam Q6(b) [15 marks] (answer continued)

Model Answer

FIFO separates prior-period work from work performed this period. Divide only this period's added costs by this period's equivalent units.

Physical unit flow	Units
Beginning WIP	10,000
Started	170,000
Total to account for	180,000
Transferred out	160,000
Ending WIP	20,000
Total accounted for	180,000

Equivalent-unit calculation	Materials	Conversion
Complete beginning WIP	0	7,000
Started and completed	150,000	150,000
Ending WIP	20,000	8,000
Total equivalent units	170,000	165,000

Cost per equivalent unit	Materials	Conversion
Costs in numerator	139,400.00	244,200.00
Equivalent units	170,000	165,000
Cost per EU (unrounded in assignments)	0.820000	1.480000

Cost assignment	Amount
Beginning WIP cost brought forward	13,400.00
Cost to finish beginning WIP	10,360.00
Started and completed this period	345,000.00
Total transferred out	368,760.00
Ending WIP: materials	16,400.00

Q71 · 2021-2022 · Exam Q6(b) [15 marks] (answer continued)

Cost assignment	Amount
Ending WIP: conversion	11,840.00
Ending WIP total	28,240.00
Transferred out + ending WIP	397,000.00
Beginning costs + added costs	397,000.00

Use unrounded rates when assigning costs. The physical units reconcile, and total costs assigned equal total costs to account for.

Q72 · 2022-2023 · Exam Q6(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 93.

Cooperative San Jose's Mixing Department uses weighted-average process costing. Bottles sell at 12 each. Its April report follows.

Reported units/costs	Amount
Beginning WIP units	30,000
Beginning materials/conversion completion	90% / 80%
Units started	200,000
Transferred	190,000
Ending WIP units	40,000
Ending materials/conversion completion	75% / 60%
Beginning WIP cost	98,000
Added cost	827,000
Total cost	925,000
Reported transferred cost	805,600
Reported ending WIP cost	119,400

Cost detail	Beginning WIP	Added this month
Materials	67,800	579,000

Q72 · 2022-2023 · Exam Q6(b) [15 marks] (question continued)

Cost detail	Beginning WIP	Added this month
Conversion	30,200	248,000

Required

1. Compute equivalent units and cost per EU.
2. Find how many transferred units were started and completed this month.
3. The manager says materials rose from about 2.50 per unit to 3.00, but good control held the reported cost below 3.00. Should this claim be rewarded?

Model Answer

Weighted average merges beginning WIP costs and current costs. It does not deduct the work already done on beginning WIP from equivalent units.

Physical unit flow	Units
Beginning WIP	30,000
Started	200,000
Total to account for	230,000
Transferred out	190,000
Ending WIP	40,000
Total accounted for	230,000

Equivalent-unit calculation	Materials	Conversion
Transferred out (fully complete)	190,000	190,000
Ending WIP	30,000	24,000
Total equivalent units	220,000	214,000

Cost per equivalent unit	Materials	Conversion
Costs in numerator	646,800.00	278,200.00
Equivalent units	220,000	214,000
Cost per EU (unrounded in assignments)	2.940000	1.300000

Q72 · 2022-2023 · Exam Q6(b) [15 marks] (answer continued)

Cost assignment	Amount
Transferred out	805,600.00
Ending WIP: materials	88,200.00
Ending WIP: conversion	31,200.00
Ending WIP total	119,400.00
Transferred out + ending WIP	925,000.00
Beginning costs + added costs	925,000.00

Use unrounded rates when assigning costs. The physical units reconcile, and total costs assigned equal total costs to account for.

$$\text{Started and completed} = 190,000 - 30,000 = 160,000\,.$$

Do not reward the claim on this evidence. Weighted-average materials cost is 2.94 because cheaper beginning costs are blended with April costs. April materials work = 190,000 - 30,000 x 90% + 40,000 x 75% = 193,000 EUs; April added materials / April work = 579,000 / 193,000 = 3.00. The reported 2.94 does not demonstrate a current-period price/usage saving.

Q73 · 2022-2023 · Exam Q7(a) [8 marks]

Source: Ipe 2009-25.pdf, PDF page 94.

Why is cost accumulation simpler in process costing than job-order costing?

Model Answer

Process costing accumulates costs for a small number of departments or processes rather than keeping a separate cost sheet for every customer job. Large quantities of similar units share department costs, so records of individual unit identities are unnecessary.

It still requires unit-flow reconciliation, completion percentages, equivalent-unit calculations and cost reconciliation. It is simpler for homogeneous continuous production, but is unsuitable for attributing distinct costs to diverse custom jobs.

Q74 · 2023-2024 · Exam Q6(c) [7 marks]

Source: Ipe 2009-25.pdf, PDF page 105.

Under what conditions is process costing appropriate?

Model Answer

Use process costing when large quantities of similar units are produced through a continuous, repetitive series of operations and costs can be averaged by department. Units within a process receive broadly similar materials and processing.

Examples include oil refining, cement, chemicals, paper and standard food products. Diverse custom orders with materially different costs need job-order or hybrid costing instead.

Q75 · 2024-2025 · Exam Q6(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 115.

Selzik cake mixes pass through Blending and Packaging. All materials are added at the start. Blending uses FIFO.

Production/cost data	Amount
Beginning WIP units	10,000
Beginning materials/conversion completion	100% / 30%
Started	170,000
Ending WIP units	20,000
Ending materials/conversion completion	100% / 40%
Beginning WIP materials cost	8,500
Beginning WIP conversion cost	4,900
Materials added	139,400
Conversion added	244,200

Required

1. Compute equivalent units.
2. Compute cost per equivalent unit.
3. Assign costs to ending WIP and transferred units.
4. Prepare a cost reconciliation report.

Q75 · 2024-2025 · Exam Q6(b) [15 marks] (question continued)

> **Source clarification:** The source narrative/requirements say March while the table says July. The one-period numerical data are consistent; the calculation uses that single set of balances and additions.

Model Answer

FIFO separates prior-period work from work performed this period. Divide only this period's added costs by this period's equivalent units.

Physical unit flow	Units
Beginning WIP	10,000
Started	170,000
Total to account for	180,000
Transferred out	160,000
Ending WIP	20,000
Total accounted for	180,000

Equivalent-unit calculation	Materials	Conversion
Complete beginning WIP	0	7,000
Started and completed	150,000	150,000
Ending WIP	20,000	8,000
Total equivalent units	170,000	165,000

Cost per equivalent unit	Materials	Conversion
Costs in numerator	139,400.00	244,200.00
Equivalent units	170,000	165,000
Cost per EU (unrounded in assignments)	0.820000	1.480000

Cost assignment	Amount
Beginning WIP cost brought forward	13,400.00
Cost to finish beginning WIP	10,360.00
Started and completed this period	345,000.00

Q75 · 2024-2025 · Exam Q6(b) [15 marks] (answer continued)

Cost assignment	Amount
Total transferred out	368,760.00
Ending WIP: materials	16,400.00
Ending WIP: conversion	11,840.00
Ending WIP total	28,240.00
Transferred out + ending WIP	397,000.00
Beginning costs + added costs	397,000.00

Use unrounded rates when assigning costs. The physical units reconcile, and total costs assigned equal total costs to account for.

Activity-based costing and profitability

Q76 · 2009-2010 · Exam Q2(a) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 2.

Transvaal ABC rates: volume 12/DLH; batch 96/batch; order 284/order; customer service 2,620/customer. Cinemas Entertainment orders 2,400 seats in 4 batches, its only annual order. Each uses 0.8 DLH; price 137.95, DM 112, DL cost 14.40/unit.

Required

1. Find product margin ignoring customer service.
2. Find annual customer margin.

Model Answer

Product/customer margin	Amount
Revenue: 2,400 x 137.95	331,080
DM	268,800
DL	34,560
Volume: 1,920 hours x 12	23,040
Four batches	384
One order	284
Product margin	4,012
Customer service	2,620
Customer margin	1,392

Customer service is charged once to the customer. The volume driver is labor-hours, so 2,400 x 0.8 = 1,920 hours.

Q77 · 2009-2010 · Exam Q2(b) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 2, 3.

Silicon Optics resource costs: wages 350,000, other OH 200,000. Wages allocation: volume 30%, orders 35%, customers 25%, Other 10%; other OH allocation: 25%, 15%, 20%, 40%. Annual activity: 10,000 DLHs, 500 orders, 100 customers; Other not assigned. Indus Telecom makes one 100-unit order and no others: price 295, DM 264/unit, 0.50 DLH/unit, wage 25/DLH.

Required

1. Prepare first-stage allocations and activity rates.
2. Compute order OH excluding customer support.

Model Answer

Allocation	Volume	Orders	Customers	Other	Total
Wages	105,000	122,500	87,500	35,000	350,000
Other OH	50,000	30,000	40,000	80,000	200,000
Total	155,000	152,500	127,500	115,000	550,000

Activity	Rate	Order cost
Volume	15.50 / DLH	775
One order	305 / order	305
Customer support	1,275 / customer	1,275

Margin	Amount
Sales	29,500
DM	26,400
Labor: 50 hours x 25	1,250
Order overhead excluding customer	1,080
Product margin	770
Customer support	1,275
Customer margin	-505

Other cost remains outside the customer margin. The one order uses 50 DLHs, not 100 hours.

Q78 · 2011-2012 · Exam Q2(b) [30 marks]

Source: Ipe 2009-25.pdf, PDF pages 10, 11.

Silicon Optics resource costs: wages 350,000, other OH 200,000. Wages allocation: volume 30%, orders 35%, customers 25%, Other 10%; other OH allocation: 25%, 15%, 20%, 40%. Annual activity: 10,000 DLHs, 500 orders, 100 customers; Other not assigned. Indus Telecom makes one 100-unit order and no others: price 295, DM 264/unit, 0.50 DLH/unit, wage 25/DLH.

Required

1. Prepare first-stage allocation, activity rates, order overhead, product margin and customer margin.

> **Source clarification:** This paper omits the resource-consumption percentages. The displayed 30/35/25/10% wages and 25/15/20/40% other-overhead allocation is explicitly assumed from the identical Silicon Optics case in the 2009-2010 and 2012-2013 papers. Without an allocation basis there is no unique numerical solution.

Model Answer

Allocation	Volume	Orders	Customers	Other	Total
Wages	105,000	122,500	87,500	35,000	350,000
Other OH	50,000	30,000	40,000	80,000	200,000
Total	155,000	152,500	127,500	115,000	550,000

Activity	Rate	Order cost
Volume	15.50 / DLH	775
One order	305 / order	305
Customer support	1,275 / customer	1,275

Margin	Amount
Sales	29,500
DM	26,400
Labor: 50 hours x 25	1,250
Order overhead excluding customer	1,080
Product margin	770
Customer support	1,275

Q78 · 2011-2012 · Exam Q2(b) [30 marks] (answer continued)

Margin	Amount
Customer margin	-505

Other cost remains outside the customer margin. The one order uses 50 DLHs, not 100 hours.

Q79 · 2011-2012 · Exam Q8(a) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 16.

Describe limitations of ABC.

Model Answer

Benefits	Limitations
More realistic cost assignment for complex product/customer mixes	Data collection and maintenance can be costly
Identifies non-value-added work and costly activities	Driver selection involves judgment and measurement error
Supports pricing, product mix and process improvements	Allocated costs may remain fixed or unavoidable in the short run
Separates unit, batch, product and customer costs	Employees may resist the additional measurement and changed margins

Use ABC where indirect costs and diversity are significant. For a decision, follow an ABC profitability signal with an analysis of avoidable costs and capacity.

Q80 · 2012-2013 · Exam Q1(a) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 17.

Write the steps for implementing activity-based costing.

Model Answer

1. Identify major activities and define activity cost pools.
2. Assign resource costs to the pools using resource-consumption measures.
3. Select an activity driver for each pool and measure total expected activity.
4. Compute each activity rate: pool cost divided by driver quantity.

Q80 · 2012-2013 · Exam Q1(a) [10 marks] (answer continued)

5. Assign costs to products, orders or customers using the activity each consumes.
6. Combine direct costs with assigned activity costs; interpret product and customer margins.

Keep organization-sustaining costs separate where no useful causal product/customer driver exists. Check that first-stage allocations reconcile to the total resources supplied.

Q81 · 2012-2013 · Exam Q2(b) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 18.

Silicon Optics resource costs: wages 350,000, other OH 200,000. Wages allocation: volume 30%, orders 35%, customers 25%, Other 10%; other OH allocation: 25%, 15%, 20%, 40%. Annual activity: 10,000 DLHs, 500 orders, 100 customers; Other not assigned. Indus Telecom makes one 100-unit order and no others: price 295, DM 264/unit, 0.50 DLH/unit, wage 25/DLH.

Required

1. Prepare first-stage allocation, activity rates, order overhead, product margin and customer margin.

Model Answer

Allocation	Volume	Orders	Customers	Other	Total
Wages	105,000	122,500	87,500	35,000	350,000
Other OH	50,000	30,000	40,000	80,000	200,000
Total	155,000	152,500	127,500	115,000	550,000

Activity	Rate	Order cost
Volume	15.50 / DLH	775
One order	305 / order	305
Customer support	1,275 / customer	1,275

Margin	Amount
Sales	29,500
DM	26,400
Labor: 50 hours x 25	1,250

Q81 · 2012-2013 · Exam Q2(b) [25 marks] (answer continued)

Margin	Amount
Order overhead excluding customer	1,080
Product margin	770
Customer support	1,275
Customer margin	-505

Other cost remains outside the customer margin. The one order uses 50 DLHs, not 100 hours.

Q82 · 2013-2014 · Exam Q1(a) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 24.

Mention benefits and limitations of ABC.

Model Answer

Benefits	Limitations
More realistic cost assignment for complex product/customer mixes	Data collection and maintenance can be costly
Identifies non-value-added work and costly activities	Driver selection involves judgment and measurement error
Supports pricing, product mix and process improvements	Allocated costs may remain fixed or unavoidable in the short run
Separates unit, batch, product and customer costs	Employees may resist the additional measurement and changed margins

Use ABC where indirect costs and diversity are significant. For a decision, follow an ABC profitability signal with an analysis of avoidable costs and capacity.

Q83 · 2013-2014 · Exam Q1(b) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 24.

Product A: 1.8 DLH/unit, 5,000 annual units, DM 72/unit. Product B: 0.9 DLH/unit, 30,000 units, DM 50/unit. Wage 10/DLH. Total annual OH 2,870,000. A uses substantially more special equipment.

Q83 · 2013-2014 · Exam Q1(b) [25 marks] (question continued)

Activity	Cost	Total driver	A driver	B driver
Setups	550,000	225	70	155
Special processing	480,000	12,000	10,000	2,000
General factory	1,260,000	?	?	?
Initial tests	100,000	20	8	12
Transportation	480,000	780	230	550

Drivers: setups; CPU minutes; DLHs; initial product tests; truckloads, respectively.

Required

1. Fill driver blanks, compute traditional rate and unit costs.
2. Classify ABC levels, compute product OH and unit costs.
3. Explain shift from high-volume B to low-volume A.

Model Answer

DLHs are 9,000 for A and 27,000 for B, total 36,000. Traditional rate = 2,870,000/36,000 = 79.722222 per DLH.

Traditional unit cost	A	B
DM	72	50
DL	18	9
OH/unit	143.50	71.75
Total unit cost	233.50	130.75

ABC activity	Rate	A OH	B OH	Level
Setups	2,444.444444	171,111.11	378,888.89	Batch
Special processing	40.000000	400,000.00	80,000.00	Unit
General factory	35.000000	315,000.00	945,000.00	Unit proxy
Initial tests	5,000.000000	40,000.00	60,000.00	Product
Transportation	615.384615	141,538.46	338,461.54	Batch

Q83 · 2013-2014 · Exam Q1(b) [25 marks] (answer continued)

ABC total/unit cost	A	B
Total assigned OH	1,067,649.57	1,802,350.43
OH per unit	213.53	60.08
DM + DL + OH/unit	303.53	119.08

A consumes 83.3% of special processing but only 25% of DLHs. A broad labor rate understates its complexity, subsidizing A with B. ABC moves cost to the low-volume intensive product. General factory uses a unit-level labor proxy here; the underlying pool may contain sustaining costs, so its level designation follows the supplied driver rather than claiming every resource varies per unit.

Q84 · 2014-2015 · Exam Q3(a) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 32.

Briefly distinguish ABC and traditional costing.

Model Answer

Feature	Traditional costing	Activity-based costing
Allocation basis	Usually one or a few volume measures, such as DLHs	Several activity drivers, such as setups, orders and designs
Cost pools	Broad plantwide or departmental overhead pools	Pools built around activities with distinct causes
Low-volume complex products	Can be undercosted when complexity is ignored	Receives order-, batch- and design-related costs
High-volume simple products	Can be overcosted by a broad volume allocation	Receives costs in proportion to activities consumed
Customer/organization costs	Often allocated broadly to products	Can separate product, customer and sustaining costs

ABC improves causal tracing when activities differ substantially. It requires additional data and judgment; an ABC allocation still does not automatically identify which cost is avoidable in a decision.

Q85 · 2014-2015 · Exam Q3(b) [30 marks]

Source: Ipe 2009-25.pdf, PDF pages 32, 33.

Advanced Products has wages/salaries 300,000 and other OH 100,000. Wages allocation: volume 40%, orders 30%, customers 20%, Other 10%; other OH allocation: 30%, 10%, 20%, 40%. Drivers: 20,000 DLHs, 400 orders, 200 customers; Other unassigned. Shenzhen places one 10-unit order and no other orders: 2 DLHs/unit; selling price 300/unit; DM 180/unit; DL 50/unit.

Required

1. Prepare first-stage allocations.
2. Compute activity rates.
3. Find order OH excluding customer support.
4. Find product and customer margins.

Model Answer

Allocate each resource cost to activity pools before computing driver rates.

Resource allocation	Labor support	Orders	Customers	Other	Total
Wages/salaries	120,000	90,000	60,000	30,000	300,000
Other overhead	30,000	10,000	20,000	40,000	100,000
Pool totals	150,000	100,000	80,000	70,000	400,000

Activity pool	Pool cost	Annual activity	Rate
Labor support	150,000	20,000 DLHs	7.50 per DLH
Order processing	100,000	400 orders	250 per order
Customer support	80,000	200 customers	400 per customer
Other	70,000	Not applicable	Not assigned to orders/customers

Order and customer margin	Amount
Sales	3,000.00
Direct materials	1,800.00
Direct labor	500.00
Labor-support cost	150.00

Q85 · 2014-2015 · Exam Q3(b) [30 marks] (answer continued)

Order and customer margin	Amount
One order	250
Product margin before customer support	300.00
Customer support (once)	400
Customer margin	(100.00)

Organization-sustaining costs stay outside product/customer margins. A negative allocated customer margin is a profitability signal; dropping the customer is a separate avoidable-cost decision.

Order OH excluding customer support = 150 volume support + 250 order processing = 400.

Q86 · 2016-2017 · Exam Q5 [8 + 8 + 8 + 11 marks]

Source: Ipe 2009-25.pdf, PDF pages 43, 44.

Advanced Products has wages/salaries 300,000 and other OH 100,000. Wages allocation: volume 40%, orders 30%, customers 20%, Other 10%; other OH allocation: 30%, 10%, 20%, 40%. Drivers: 20,000 DLHs, 400 orders, 200 customers; Other unassigned. Shenzhen places one 10-unit order and no other orders: 2 DLHs/unit; selling price 300/unit; DM 180/unit; DL 50/unit.

Required

1. Prepare first-stage allocations.
2. Compute activity rates.
3. Find order OH excluding customer support.
4. Find product and customer margins.

Model Answer

Allocate each resource cost to activity pools before computing driver rates.

Resource allocation	Labor support	Orders	Customers	Other	Total
Wages/salaries	120,000	90,000	60,000	30,000	300,000
Other overhead	30,000	10,000	20,000	40,000	100,000
Pool totals	150,000	100,000	80,000	70,000	400,000

Q86 · 2016-2017 · Exam Q5 [8 + 8 + 8 + 11 marks] (answer continued)

Activity pool	Pool cost	Annual activity	Rate
Labor support	150,000	20,000 DLHs	7.50 per DLH
Order processing	100,000	400 orders	250 per order
Customer support	80,000	200 customers	400 per customer
Other	70,000	Not applicable	Not assigned to orders/customers

Order and customer margin	Amount
Sales	3,000.00
Direct materials	1,800.00
Direct labor	500.00
Labor-support cost	150.00
One order	250
Product margin before customer support	300.00
Customer support (once)	400
Customer margin	(100.00)

Organization-sustaining costs stay outside product/customer margins. A negative allocated customer margin is a profitability signal; dropping the customer is a separate avoidable-cost decision.

Order OH excluding customer support = 150 volume support + 250 order processing = 400.

Q87 · 2017-2018 · Exam Q4(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 50.

Ferris has manufacturing OH 500,000 and S&A OH 300,000. Activity pools are assembly (units), orders, customers and Other. Other includes unused capacity/organization costs, not assigned to products or customers.

Resource consumption	Assembly	Orders	Customers	Other
Manufacturing	50%	35%	5%	10%
S&A	10%	45%	25%	20%

Q87 · 2017-2018 · Exam Q4(b) [15 marks] (question continued)

Annual activity: 1,000 units, 250 orders, 100 customers. OfficeMart ordered 80 cabinets through four orders.

Required

1. Allocate resources to activity pools.
2. Compute overhead attributable to OfficeMart.

Model Answer

First-stage allocation	Assembly	Orders	Customers	Other	Total
Manufacturing	250,000	175,000	25,000	50,000	500,000
S&A	30,000	135,000	75,000	60,000	300,000
Total	280,000	310,000	100,000	110,000	800,000

Pool	Rate	OfficeMart usage	Assigned cost
Assembly	280/unit	80	22,400
Orders	1,240/order	4	4,960
Customers	1,000/customer	1	1,000
Total OH			28,360

The 110,000 Other pool is excluded. A customer-level charge is applied once, not once per order or cabinet.

Q88 · 2018-2019 · Exam Q3(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 56, 57.

Thermal Rising uses activity rates: volume 26 per DLH; orders 284/order; designs 186/design; customer service 379/customer. Wage 19.50/DLH. Big Sky orders:

Measure	Standard	Custom
Gliders	20	3
Orders	1	3
Designs	0	3

Q88 · 2018-2019 · Exam Q3(a) [20 marks] (question continued)

Measure	Standard	Custom
DLH/glider	26.35	28
Price/glider	1,850	2,400
DM/glider	554	634

Required

1. Compute profitability of each ordered product and the customer overall.

Model Answer

Product/customer margin	Standard	Custom	Total
Revenue	37,000	7,200	44,200
DM	11,080	1,902	12,982
DL	10,276.50	1,638	11,914.50
Volume OH	13,702	2,184	15,886
Order cost	284	852	1,136
Design cost	0	558	558
Product margin	1,657.50	66	1,723.50
Customer service (once)	-	-	379
Customer margin	-	-	1,344.50

Manufacturing volume uses total DLHs (20 x 26.35 and 3 x 28), not the number of units. Service is charged once for the one customer. Allocated profitability is not the same as avoidable profit on discontinuing the customer.

Q89 · January 2020 Term · Exam Q2 [45 marks]

Source: Ipe 2009-25.pdf, PDF pages 64, 65.

ABC operates armored cars. Pools: Travel (miles), Pickup/Delivery (number of pickups/deliveries), Customer Service (customers) and Other (no driver). Resource consumption is below.

Q89 · January 2020 Term · Exam Q2 [45 marks] (question continued)

Resource	Annual cost	Travel	Pickup/delivery	Customer	Other
Driver/guard wages	730,000	50%	35%	10%	5%
Vehicle operation	280,000	70%	5%	0%	25%
Vehicle depreciation	120,000	60%	15%	0%	25%
Customer representatives	170,000	0%	0%	90%	10%
Office expenses	30,000	0%	20%	30%	50%
Administration	320,000	0%	5%	60%	35%

Required

1. Carry out first-stage and second-stage cost allocations.

> **Source clarification:** The paper omits annual activity totals and individual customer usage. The answer gives the complete first stage and the conditional second stage rather than inventing figures.

Model Answer

First-stage allocation	Travel	Pickup/delivery	Customer	Other	Total
Driver/guard wages	365,000.00	255,500.00	73,000.00	36,500.00	730,000
Vehicle operation	196,000.00	14,000.00	0	70,000.00	280,000
Vehicle depreciation	72,000.00	18,000.00	0	30,000.00	120,000
Customer representatives	0	0	153,000.00	17,000.00	170,000
Office expenses	0	6,000.00	9,000.00	15,000.00	30,000
Administration	0	16,000.00	192,000.00	112,000.00	320,000
Pool totals	633,000.00	309,500.00	427,000.00	280,500.00	1,650,000.00

Second-stage rates require total annual miles M, pickups/deliveries P, and customers C, which are not provided in the paper.

$$\text{Travel rate} = 633,000 / M, \quad \text{Pickup rate} = 309,500 / P,$$
$$\text{Customer rate} = 427,000 / C.$$

Assign to a customer using its miles x Travel rate + its pickups/deliveries x Pickup rate + one Customer Service rate. Other cost = 280,500 remains unassigned. A numerical second-stage allocation cannot be uniquely calculated without total activities and a cost object's usage.

Q90 · 2020-2021 · Exam Q4(b) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 72.

Give examples of unit-, batch-, product-, customer-level and organization-sustaining ABC activities.

Model Answer

Activity level	What drives the work	Example
Unit	Each unit made	Machining one component
Batch	Each batch, regardless of batch size	Machine setup or batch inspection
Product	A product line/design	Product engineering and design maintenance
Customer	A customer/account	Account service or a customer-specific visit
Organization sustaining	Company as a whole	General management or factory security

A batch setup is not caused by every unit separately. Sustaining costs need not have a meaningful product/customer driver.

Q91 · 2021-2022 · Exam Q8(c) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 86.

Fogerty makes Hubs and Sprockets, paying 15 per DLH.

Product	Hubs	Sprockets
DLH/unit	0.80	0.40
Annual units	10,000	40,000
DM/unit	32	18

Activity	Cost	Hubs activity	Sprockets activity
Setups	72,000	100	300
Special processing	200,000	5,000 MHs	0
General factory	816,000	Sustaining	Sustaining

Q91 · 2021-2022 · Exam Q8(c) [15 marks] (question continued)

Required

1. Compute ABC overhead rates and unit product costs.

Model Answer

Pool	Rate calculation	Rate
Setups	72,000 / 400	180 per setup
Special processing	200,000 / 5,000	40 per MH

Unit cost calculation	Hubs	Sprockets
DM/unit	32	18
DL/unit	12	6
Assigned setup cost	18,000	54,000
Assigned special cost	200,000	0
Assigned OH/unit	21.80	1.35
ABC product cost/unit	65.80	25.35

General factory cost of 816,000 is organization-sustaining and is not assigned to products in this ABC model. It remains a company-level expense in the profitability reconciliation.

Q92 · 2022-2023 · Exam Q2(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 87, 88.

Thermal Rising uses the following ABC activity rates. Big Sky Outfitters placed the listed standard and custom orders; the direct labor wage is 19.50 per hour.

Activity	Rate
Support direct labor	26 per DLH
Order processing	284 per order
Custom design	186 per design
Customer service	379 per customer

Q92 · 2022-2023 · Exam Q2(a) [20 marks] (question continued)

Customer data	Standard	Custom
Gliders	20	3
Orders	1	3
Designs	0	3
DLHs per glider	26.35	28
Selling price per glider	1,850	2,400
Materials per glider	564	634

Required

1. Use ABC to compute the customer margin of Big Sky Outfitters.

Model Answer

Multiply each activity driver by its activity rate. Product margins exclude the one annual customer-service charge; the customer margin deducts it once, not once per product.

Cost/margin line	Standard	Custom	Total
Sales	37,000.00	7,200.00	44,200.00
Direct materials	11,280.00	1,902.00	13,182.00
Direct labor	10,276.50	1,638.00	11,914.50
Labor-support overhead	13,702.00	2,184.00	15,886.00
Order costs	284.00	852.00	1,136.00
Design costs	0.00	558.00	558.00
Product margin	1,457.50	66.00	1,523.50

Customer margin	Amount
Total product margin	1,523.50
Customer service	379
Customer margin	1,144.50

Q92 · 2022-2023 · Exam Q2(a) [20 marks] (answer continued)

The labor hours are 20 x 26.35 = 527 for the standard gliders and 3 x 28 = 84 for custom gliders. This customer consumes 611 support DLHs, four orders, three custom designs and one customer-service activity.

Q93 · 2023-2024 · Exam Q3(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 100, 101.

Advanced Products has wages/salaries of 300,000 and other overhead of 100,000. The ABC annual drivers are 20,000 DLHs, 400 orders and 200 customers. Other is organization-sustaining and not assigned.

Resource allocation	Labor support	Orders	Customers	Other
Wages/salaries	40%	30%	20%	10%
Other overhead	30%	10%	20%	40%

Shenzhen Enterprises placed one order for 10 units and no other orders this year. Per unit: 2 DLHs, selling price 300, materials 180, direct labor 50.

Required

1. Prepare a customer-margin report for Shenzhen.

Model Answer

Allocate each resource cost to activity pools before computing driver rates.

Resource allocation	Labor support	Orders	Customers	Other	Total
Wages/salaries	120,000	90,000	60,000	30,000	300,000
Other overhead	30,000	10,000	20,000	40,000	100,000
Pool totals	150,000	100,000	80,000	70,000	400,000

Activity pool	Pool cost	Annual activity	Rate
Labor support	150,000	20,000 DLHs	7.50 per DLH
Order processing	100,000	400 orders	250 per order
Customer support	80,000	200 customers	400 per customer
Other	70,000	Not applicable	Not assigned to orders/customers

Q93 · 2023-2024 · Exam Q3(a) [20 marks] (answer continued)

Order and customer margin	Amount
Sales	3,000.00
Direct materials	1,800.00
Direct labor	500.00
Labor-support cost	150.00
One order	250
Product margin before customer support	300.00
Customer support (once)	400
Customer margin	(100.00)

Organization-sustaining costs stay outside product/customer margins. A negative allocated customer margin is a profitability signal; dropping the customer is a separate avoidable-cost decision.

Q94 · 2024-2025 · Exam Q4(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 112.

Thermal Rising uses the following ABC activity rates. Big Sky Outfitters placed the listed standard and custom orders; the direct labor wage is 19.50 per hour.

Activity	Rate
Support direct labor	26 per DLH
Order processing	284 per order
Custom design	186 per design
Customer service	379 per customer

Customer data	Standard	Custom
Gliders	20	3
Orders	1	3
Designs	0	3
DLHs per glider	26.35	28

Q94 · 2024-2025 · Exam Q4(a) [20 marks] (question continued)

Customer data	Standard	Custom
Selling price per glider	1,850	2,400
Materials per glider	564	634

Required

1. Use ABC to compute the customer margin of Big Sky Outfitters.

Model Answer

Multiply each activity driver by its activity rate. Product margins exclude the one annual customer-service charge; the customer margin deducts it once, not once per product.

Cost/margin line	Standard	Custom	Total
Sales	37,000.00	7,200.00	44,200.00
Direct materials	11,280.00	1,902.00	13,182.00
Direct labor	10,276.50	1,638.00	11,914.50
Labor-support overhead	13,702.00	2,184.00	15,886.00
Order costs	284.00	852.00	1,136.00
Design costs	0.00	558.00	558.00
Product margin	1,457.50	66.00	1,523.50

Customer margin	Amount
Total product margin	1,523.50
Customer service	379
Customer margin	1,144.50

The labor hours are 20 x 26.35 = 527 for the standard gliders and 3 x 28 = 84 for custom gliders. This customer consumes 611 support DLHs, four orders, three custom designs and one customer-service activity.

Variable and absorption costing

Q95 · 2009-2010 · Exam Q1(a) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 1.

Dexter looms sell at 50. Unit DM 11, DL 6, variable MOH 3; fixed MOH 120,000/year. Variable S&A 5/sold unit, fixed S&A 70,000/year. Production/sales: Year 1 10,000/8,000; Year 2 6,000/8,000. Beginning Year 1 zero, ending Year 1 2,000 units, ending Year 2 zero.

Required

1. Find unit product cost and annual income statements using absorption and variable costing.

Model Answer

Variable product cost is 20.00 per unit. Fixed MOH is assigned to units produced, not units sold. The absorption inventory flow below uses FIFO when year-specific fixed overhead rates differ.

Year 1 unit cost	Variable	Absorption
Variable manufacturing	20.00	20.00
Fixed MOH / production	0	12.00
Unit product cost	20.00	32.00

Year 1 income statement	Variable	Absorption
Sales	400,000.00	400,000.00
Beginning inventory cost	0.00	0.00
Cost of goods manufactured	200,000.00	320,000.00
Ending inventory	40,000.00	64,000.00
COGS	160,000.00	256,000.00
Manufacturing contribution / gross margin	240,000.00	144,000.00
Variable S&A	40,000.00	40,000.00
Contribution margin (variable column)	200,000.00	
Fixed MOH period expense	120,000.00	0
Fixed S&A	70,000.00	70,000.00

Q95 · 2009-2010 · Exam Q1(a) [25 marks] (answer continued)

Year 1 income statement	Variable	Absorption
NOI	10,000.00	34,000.00

Year 2 unit cost	Variable	Absorption
Variable manufacturing	20.00	20.00
Fixed MOH / production	0	20.00
Unit product cost	20.00	40.00

Year 2 income statement	Variable	Absorption
Sales	400,000.00	400,000.00
Beginning inventory cost	40,000.00	64,000.00
Cost of goods manufactured	120,000.00	240,000.00
Ending inventory	0.00	0.00
COGS	160,000.00	304,000.00
Manufacturing contribution / gross margin	240,000.00	96,000.00
Variable S&A	40,000.00	40,000.00
Contribution margin (variable column)	200,000.00	
Fixed MOH period expense	120,000.00	0
Fixed S&A	70,000.00	70,000.00
NOI	10,000.00	(14,000.00)

Income reconciliation	Beginning inventory fixed MOH	Ending inventory fixed MOH	Absorption minus variable NOI
Year 1	0.00	24,000.00	24,000.00
Year 2	24,000.00	0.00	(24,000.00)

Each period's difference equals ending inventory fixed MOH less beginning inventory fixed MOH. When older inventory is sold, release its original fixed-overhead rate rather than replacing it with the current production rate.

Q96 · 2009-2010 · Exam Q3(b) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 3.

Compare income effects of absorption and variable costing.

Model Answer

Absorption costing treats fixed manufacturing overhead as a product cost. The portion attached to unsold production remains in inventory until those units are sold. Variable costing expenses all current-period fixed manufacturing overhead immediately.

Inventory movement	Usual income result	Reason
Production > sales; inventory increases	Absorption NOI > variable NOI	Some current fixed MOH is deferred in inventory
Sales > production; inventory decreases	Absorption NOI < variable NOI	Fixed MOH from an earlier period is released to expense
Inventory fixed-MOH balance unchanged	Same NOI	No net deferral or release

$$\text{Income difference} = \text{Fixed MOH in ending inventory} - \text{Fixed MOH in beginning inv}$$

Fixed selling and administrative expenses are period costs under both methods. Use inventory fixed-overhead amounts, rather than only unit counts, when per-unit overhead rates change.

Q97 · 2011-2012 · Exam Q7(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 15.

Matador first-year price BDT 50; produce 20,000, sell 16,000; no beginning FG. Unit DM 18, DL 7, variable MOH 2, variable S&A 5. Fixed MOH 160,000; fixed S&A 110,000.

Required

1. Compute unit product costs and income statements under absorption and variable costing.

Model Answer

No beginning inventory is assumed as stated. Variable product cost includes direct materials, direct labor and variable manufacturing overhead; absorption product cost also includes fixed manufacturing overhead assigned to production.

Unit product cost (currency per unit)	Variable	Absorption
Variable manufacturing cost	27.00	27.00

Q97 · 2011-2012 · Exam Q7(a) [20 marks] (answer continued)

Unit product cost (currency per unit)	Variable	Absorption
Fixed MOH per produced unit	0	8.00
Unit product cost	27.00	35.00

Income statement	Absorption	Variable
Sales	800,000.00	800,000.00
Cost of goods sold	560,000.00	432,000.00
Gross margin / manufacturing contribution	240,000.00	368,000.00
Variable selling and administrative	80,000.00	80,000.00
Fixed manufacturing overhead (period expense)	0	160,000.00
Fixed selling and administrative	110,000.00	110,000.00
Net operating income	50,000.00	18,000.00

In the absorption column, fixed manufacturing overhead is already included in product cost/COGS; do not expense it a second time. The variable-costing contribution margin after all variable costs is 288,000.00.

Inventory and reconciliation	Amount
Ending units	4,000
Variable-cost inventory	108,000.00
Absorption-cost inventory	140,000.00
Fixed MOH deferred / absorption NOI less variable NOI	32,000.00

The two income figures differ because fixed manufacturing overhead attached to unsold goods remains in absorption inventory but is expensed immediately under variable costing.

Q98 · 2012-2013 · Exam Q1(b) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 17.

Denton produces 25,000 units annually, selling 20,000 in Year 1 and 30,000 in Year 2 at 50. No beginning Year 1 inventory. Unit absorption cost 34: DM 8, DL 10, variable MOH 2, fixed MOH 14 from annual 350,000. S&A 3 per sold unit + 250,000 fixed. Reported absorption NOI 10,000 / 140,000; ending Year 1 inventory 5,000 at 34 = 170,000, Year 2 zero.

Required

1. Prepare contribution statements under variable costing.
2. Reconcile incomes each year.

Model Answer

Variable product cost is 20.00 per unit. Fixed MOH is assigned to units produced, not units sold. The absorption inventory flow below uses FIFO when year-specific fixed overhead rates differ.

Year 1 unit cost	Variable	Absorption
Variable manufacturing	20.00	20.00
Fixed MOH / production	0	14.00
Unit product cost	20.00	34.00

Year 1 income statement	Variable	Absorption
Sales	1,000,000.00	1,000,000.00
Beginning inventory cost	0.00	0.00
Cost of goods manufactured	500,000.00	850,000.00
Ending inventory	100,000.00	170,000.00
COGS	400,000.00	680,000.00
Manufacturing contribution / gross margin	600,000.00	320,000.00
Variable S&A	60,000.00	60,000.00
Contribution margin (variable column)	540,000.00	
Fixed MOH period expense	350,000.00	0
Fixed S&A	250,000.00	250,000.00
NOI	(60,000.00)	10,000.00

Q98 · 2012-2013 · Exam Q1(b) [25 marks] (answer continued)

Year 2 unit cost	Variable	Absorption
Variable manufacturing	20.00	20.00
Fixed MOH / production	0	14.00
Unit product cost	20.00	34.00

Year 2 income statement	Variable	Absorption
Sales	1,500,000.00	1,500,000.00
Beginning inventory cost	100,000.00	170,000.00
Cost of goods manufactured	500,000.00	850,000.00
Ending inventory	0.00	0.00
COGS	600,000.00	1,020,000.00
Manufacturing contribution / gross margin	900,000.00	480,000.00
Variable S&A	90,000.00	90,000.00
Contribution margin (variable column)	810,000.00	
Fixed MOH period expense	350,000.00	0
Fixed S&A	250,000.00	250,000.00
NOI	210,000.00	140,000.00

Income reconciliation	Beginning inventory fixed MOH	Ending inventory fixed MOH	Absorption minus variable NOI
Year 1	0.00	70,000.00	70,000.00
Year 2	70,000.00	0.00	(70,000.00)

Each period's difference equals ending inventory fixed MOH less beginning inventory fixed MOH. When older inventory is sold, release its original fixed-overhead rate rather than replacing it with the current production rate.

Q99 · 2012-2013 · Exam Q3(a) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 19.

Briefly discuss comparative income effects of absorption and variable costing.

Q99 · 2012-2013 · Exam Q3(a) [10 marks] (answer continued)

Model Answer

Absorption costing treats fixed manufacturing overhead as a product cost. The portion attached to unsold production remains in inventory until those units are sold. Variable costing expenses all current-period fixed manufacturing overhead immediately.

Inventory movement	Usual income result	Reason
Production > sales; inventory increases	Absorption NOI > variable NOI	Some current fixed MOH is deferred in inventory
Sales > production; inventory decreases	Absorption NOI < variable NOI	Fixed MOH from an earlier period is released to expense
Inventory fixed-MOH balance unchanged	Same NOI	No net deferral or release

$$\text{Income difference} = \text{Fixed MOH in ending inventory} - \text{Fixed MOH in beginning inv}$$

Fixed selling and administrative expenses are period costs under both methods. Use inventory fixed-overhead amounts, rather than only unit counts, when per-unit overhead rates change.

Q100 · 2013-2014 · Exam Q4(c) [24 marks]

Source: Ipe 2009-25.pdf, PDF pages 26, 27.

Company B uses FIFO. Price 25; variable manufacturing 9/unit; variable S&A 6/unit. Opening October FG 5,000 units at absorption cost 16. Quarterly fixed MOH 1,680,000, allocated equally 560,000/month and applied at 7/unit using budgeted 80,000 monthly production.

Reported absorption statement	October	November	December
Production units	85,000	80,000	60,000
Sales	1,750,000	1,875,000	2,000,000
Beginning FG cost	80,000	320,000	400,000
Variable production cost	765,000	720,000	540,000
Applied fixed OH	595,000	560,000	420,000
COGM	1,360,000	1,280,000	960,000
Ending FG cost	320,000	400,000	80,000
Unadjusted COGS	1,120,000	1,200,000	1,280,000

Q100 · 2013-2014 · Exam Q4(c) [24 marks] (question continued)

Reported absorption statement	October	November	December
Under/(over) OH	-35,000	0	140,000
Adjusted COGS	1,085,000	1,200,000	1,420,000
Gross margin	665,000	675,000	580,000
S&A	620,000	650,000	680,000
NOI	45,000	25,000	-100,000

Required

1. Prepare monthly variable income statements.
2. Compute monthly BE under both methods.
3. Explain erratic absorption profit versus rising sales.
4. Reconcile both incomes each month.

Model Answer

Variable statement	October	November	December
Sales units	70,000	75,000	80,000
Sales	1,750,000	1,875,000	2,000,000
Variable COGS	630,000	675,000	720,000
Variable S&A	420,000	450,000	480,000
CM	700,000	750,000	800,000
Fixed MOH	560,000	560,000	560,000
Fixed S&A	200,000	200,000	200,000
Variable NOI	-60,000	-10,000	40,000

Fixed OH inventory reconciliation	October	November	December
Beginning FG units	5,000	20,000	25,000
Ending FG units	20,000	25,000	5,000
Beginning inventory fixed OH	35,000	140,000	175,000
Ending inventory fixed OH	140,000	175,000	35,000

Q100 · 2013-2014 · Exam Q4(c) [24 marks] (answer continued)

Fixed OH inventory reconciliation	October	November	December
Fixed OH deferred/(released)	105,000	35,000	-140,000
Variable NOI + movement = absorption NOI	45,000	25,000	-100,000

Variable BE = (560,000 + 200,000)/(25 - 9 - 6) = 76,000 units every month. If each month's actual production is held constant, adjusted absorption NOI = 3 x sales units - 760,000 + 7 x production units. Absorption BE is therefore 55,000; 66,666.667 (at least 66,667); 113,333.333 (at least 113,334). These are accounting BE quantities conditional on those production plans.

Absorption income rises when fixed overhead is deferred into inventory and falls when it is released. December releases 140,000 of fixed OH, overwhelming the sales improvement. Variable income follows sales contribution. If production always equals sales, both BE quantities become 76,000; absorption BE is not a unique quantity without a production/inventory assumption.

Q101 · 2014-2015 · Exam Q2(a) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 32.

Explain causes of NOI differences under variable and absorption costing.

Model Answer

Absorption costing treats fixed manufacturing overhead as a product cost. The portion attached to unsold production remains in inventory until those units are sold. Variable costing expenses all current-period fixed manufacturing overhead immediately.

Inventory movement	Usual income result	Reason
Production > sales; inventory increases	Absorption NOI > variable NOI	Some current fixed MOH is deferred in inventory
Sales > production; inventory decreases	Absorption NOI < variable NOI	Fixed MOH from an earlier period is released to expense
Inventory fixed-MOH balance unchanged	Same NOI	No net deferral or release

$$\text{Income difference} = \text{Fixed MOH in ending inventory} - \text{Fixed MOH in beginning inv}$$

Fixed selling and administrative expenses are period costs under both methods. Use inventory fixed-overhead amounts, rather than only unit counts, when per-unit overhead rates change.

Q102 · 2014-2015 · Exam Q2(b) [30 marks]

Source: Ipe 2009-25.pdf, PDF page 32.

Fine Producers first-month sales 400,000 (40,000 units); production 50,000; no opening inventory. Unit DM 2, DL 1.60, variable MOH 0.40, variable S&A 1.50; fixed monthly MOH 150,000 and fixed S&A 40,000. Reported variable COGS 160,000, CM after variable S&A 180,000 and loss 10,000. Month 2 sells 60,000 units and has no closing inventory; price and costs continue unchanged.

Required

1. Identify the method and show whether absorption costing can report profit.
2. Prepare Month 2 income statements under both methods.
3. Reconcile Month 2 incomes.

Model Answer

The given statement uses variable costing. Month 1 absorption product cost = 4 + 150,000/50,000 = 7; income = 400,000 - 280,000 - 60,000 - 40,000 = 20,000 profit. The 30,000 difference is 10,000 ending units x fixed OH 3.

Variable product cost is 4.00 per unit. Fixed MOH is assigned to units produced, not units sold. The absorption inventory flow below uses FIFO when year-specific fixed overhead rates differ.

Year 1 unit cost	Variable	Absorption
Variable manufacturing	4.00	4.00
Fixed MOH / production	0	3.00
Unit product cost	4.00	7.00

Year 1 income statement	Variable	Absorption
Sales	400,000.00	400,000.00
Beginning inventory cost	0.00	0.00
Cost of goods manufactured	200,000.00	350,000.00
Ending inventory	40,000.00	70,000.00
COGS	160,000.00	280,000.00
Manufacturing contribution / gross margin	240,000.00	120,000.00
Variable S&A	60,000.00	60,000.00

Q102 · 2014-2015 · Exam Q2(b) [30 marks] (answer continued)

Year 1 income statement	Variable	Absorption
Contribution margin (variable column)	180,000.00	
Fixed MOH period expense	150,000.00	0
Fixed S&A	40,000.00	40,000.00
NOI	(10,000.00)	20,000.00

Year 2 unit cost	Variable	Absorption
Variable manufacturing	4.00	4.00
Fixed MOH / production	0	3.00
Unit product cost	4.00	7.00

Year 2 income statement	Variable	Absorption
Sales	600,000.00	600,000.00
Beginning inventory cost	40,000.00	70,000.00
Cost of goods manufactured	200,000.00	350,000.00
Ending inventory	0.00	0.00
COGS	240,000.00	420,000.00
Manufacturing contribution / gross margin	360,000.00	180,000.00
Variable S&A	90,000.00	90,000.00
Contribution margin (variable column)	270,000.00	
Fixed MOH period expense	150,000.00	0
Fixed S&A	40,000.00	40,000.00
NOI	80,000.00	50,000.00

Income reconciliation	Beginning inventory fixed MOH	Ending inventory fixed MOH	Absorption minus variable NOI
Year 1	0.00	30,000.00	30,000.00
Year 2	30,000.00	0.00	(30,000.00)

Q102 · 2014-2015 · Exam Q2(b) [30 marks] (answer continued)

Each period's difference equals ending inventory fixed MOH less beginning inventory fixed MOH. When older inventory is sold, release its original fixed-overhead rate rather than replacing it with the current production rate.

In the tables Year 1/Year 2 denote Month 1/Month 2 for this problem. Month 2 production = 60,000 sales - 10,000 beginning = 50,000. Variable NOI 80,000 exceeds absorption NOI 50,000 by the 30,000 beginning inventory overhead released.

Q103 · 2016-2017 · Exam Q3(b) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 41, 42.

Lynch produces 25,000 units and sells 20,000 at 50 each, with no beginning inventory. Per-unit costs: DM 6, DL 9, variable MOH 3, variable S&A 4. Fixed annual MOH 300,000; fixed S&A 190,000.

Required

1. Compute unit product cost under both methods.
2. Prepare both income statements.
3. Reconcile the operating incomes.

Model Answer

No beginning inventory is assumed as stated. Variable product cost includes direct materials, direct labor and variable manufacturing overhead; absorption product cost also includes fixed manufacturing overhead assigned to production.

Unit product cost (currency per unit)	Variable	Absorption
Variable manufacturing cost	18.00	18.00
Fixed MOH per produced unit	0	12.00
Unit product cost	18.00	30.00

Income statement	Absorption	Variable
Sales	1,000,000.00	1,000,000.00
Cost of goods sold	600,000.00	360,000.00
Gross margin / manufacturing contribution	400,000.00	640,000.00

Q103 · 2016-2017 · Exam Q3(b) [20 marks] (answer continued)

Income statement	Absorption	Variable
Variable selling and administrative	80,000.00	80,000.00
Fixed manufacturing overhead (period expense)	0	300,000.00
Fixed selling and administrative	190,000.00	190,000.00
Net operating income	130,000.00	70,000.00

In the absorption column, fixed manufacturing overhead is already included in product cost/COGS; do not expense it a second time. The variable-costing contribution margin after all variable costs is 560,000.00.

Inventory and reconciliation	Amount
Ending units	5,000
Variable-cost inventory	90,000.00
Absorption-cost inventory	150,000.00
Fixed MOH deferred / absorption NOI less variable NOI	60,000.00

The two income figures differ because fixed manufacturing overhead attached to unsold goods remains in absorption inventory but is expensed immediately under variable costing.

Q104 · 2017-2018 · Exam Q3(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 48.

Dexter hand looms sell for 50. Unit DM 11, DL 6, variable MOH 3; annual fixed MOH 120,000. S&A 4/unit sold + 70,000 fixed. Year 1 production/sales = 10,000/8,000; Year 2 = 6,000/8,000; beginning Year 1 FG zero. Ending units Year 1 2,000; Year 2 zero.

Required

1. Compute product costs and annual income statements using absorption and variable costing.

Model Answer

Variable product cost is 20.00 per unit. Fixed MOH is assigned to units produced, not units sold. The absorption inventory flow below uses FIFO when year-specific fixed overhead rates differ.

Q104 · 2017-2018 · Exam Q3(a) [20 marks] (answer continued)

Year 1 unit cost	Variable	Absorption
Variable manufacturing	20.00	20.00
Fixed MOH / production	0	12.00
Unit product cost	20.00	32.00

Year 1 income statement	Variable	Absorption
Sales	400,000.00	400,000.00
Beginning inventory cost	0.00	0.00
Cost of goods manufactured	200,000.00	320,000.00
Ending inventory	40,000.00	64,000.00
COGS	160,000.00	256,000.00
Manufacturing contribution / gross margin	240,000.00	144,000.00
Variable S&A	32,000.00	32,000.00
Contribution margin (variable column)	208,000.00	
Fixed MOH period expense	120,000.00	0
Fixed S&A	70,000.00	70,000.00
NOI	18,000.00	42,000.00

Year 2 unit cost	Variable	Absorption
Variable manufacturing	20.00	20.00
Fixed MOH / production	0	20.00
Unit product cost	20.00	40.00

Year 2 income statement	Variable	Absorption
Sales	400,000.00	400,000.00
Beginning inventory cost	40,000.00	64,000.00
Cost of goods manufactured	120,000.00	240,000.00
Ending inventory	0.00	0.00

Q104 · 2017-2018 · Exam Q3(a) [20 marks] (answer continued)

Year 2 income statement	Variable	Absorption
COGS	160,000.00	304,000.00
Manufacturing contribution / gross margin	240,000.00	96,000.00
Variable S&A	32,000.00	32,000.00
Contribution margin (variable column)	208,000.00	
Fixed MOH period expense	120,000.00	0
Fixed S&A	70,000.00	70,000.00
NOI	18,000.00	(6,000.00)

Income reconciliation	Beginning inventory fixed MOH	Ending inventory fixed MOH	Absorption minus variable NOI
Year 1	0.00	24,000.00	24,000.00
Year 2	24,000.00	0.00	(24,000.00)

Each period's difference equals ending inventory fixed MOH less beginning inventory fixed MOH. When older inventory is sold, release its original fixed-overhead rate rather than replacing it with the current production rate.

Q105 · 2018-2019 · Exam Q2(b) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 55, 56.

Tami's Creations makes swimsuits. First-quarter output 30,000, sales 28,000 at 40, no beginning inventory. Unit costs: DM 3.50, DL 12, variable MOH 1, variable S&A 6. Fixed MOH 300,000; fixed S&A 200,000. The contribution statement reports sales 1,120,000, variable COGS 462,000, variable S&A 168,000, CM 490,000 and operating loss 10,000. A CPA suggests absorption costing may report profit.

Required

1. Compute absorption unit product cost.
2. Reconcile variable and absorption operating income/loss.

Q105 · 2018-2019 · Exam Q2(b) [20 marks] (answer continued)

Model Answer

No beginning inventory is assumed as stated. Variable product cost includes direct materials, direct labor and variable manufacturing overhead; absorption product cost also includes fixed manufacturing overhead assigned to production.

Unit product cost (currency per unit)	Variable	Absorption
Variable manufacturing cost	16.50	16.50
Fixed MOH per produced unit	0	10.00
Unit product cost	16.50	26.50

Income statement	Absorption	Variable
Sales	1,120,000.00	1,120,000.00
Cost of goods sold	742,000.00	462,000.00
Gross margin / manufacturing contribution	378,000.00	658,000.00
Variable selling and administrative	168,000.00	168,000.00
Fixed manufacturing overhead (period expense)	0	300,000.00
Fixed selling and administrative	200,000.00	200,000.00
Net operating income	10,000.00	(10,000.00)

In the absorption column, fixed manufacturing overhead is already included in product cost/COGS; do not expense it a second time. The variable-costing contribution margin after all variable costs is 490,000.00.

Inventory and reconciliation	Amount
Ending units	2,000
Variable-cost inventory	33,000.00
Absorption-cost inventory	53,000.00
Fixed MOH deferred / absorption NOI less variable NOI	20,000.00

The two income figures differ because fixed manufacturing overhead attached to unsold goods remains in absorption inventory but is expensed immediately under variable costing.

Q105 · 2018-2019 · Exam Q2(b) [20 marks] (answer continued)

Absorption profit is 10,000 because 20,000 fixed overhead remains in 2,000 unsold swimsuits. This is a timing difference, not an additional sale or cash inflow; it does not itself solve the cash strain.

Q106 · January 2020 Term · Exam Q1 [45 marks]

Source: Ipe 2009-25.pdf, PDF page 64.

XYZ makes wooden hand looms. Price 60; per-produced-unit DM 12, DL 6, variable MOH 2. Annual fixed MOH 120,000; S&A 5 per sold unit plus 70,000 annually.

Inventory data	Year 1	Year 2
Beginning units	0	2,000
Production	10,000	6,000
Sales	8,000	8,000
Ending units	2,000	0

Required

1. Compute unit product costs and prepare annual income statements under absorption and variable costing.

Model Answer

Variable product cost is 20.00 per unit. Fixed MOH is assigned to units produced, not units sold. The absorption inventory flow below uses FIFO when year-specific fixed overhead rates differ.

Year 1 unit cost	Variable	Absorption
Variable manufacturing	20.00	20.00
Fixed MOH / production	0	12.00
Unit product cost	20.00	32.00

Year 1 income statement	Variable	Absorption
Sales	480,000.00	480,000.00
Beginning inventory cost	0.00	0.00
Cost of goods manufactured	200,000.00	320,000.00

Q106 · January 2020 Term · Exam Q1 [45 marks] (answer continued)

Year 1 income statement	Variable	Absorption
Ending inventory	40,000.00	64,000.00
COGS	160,000.00	256,000.00
Manufacturing contribution / gross margin	320,000.00	224,000.00
Variable S&A	40,000.00	40,000.00
Contribution margin (variable column)	280,000.00	
Fixed MOH period expense	120,000.00	0
Fixed S&A	70,000.00	70,000.00
NOI	90,000.00	114,000.00

Year 2 unit cost	Variable	Absorption
Variable manufacturing	20.00	20.00
Fixed MOH / production	0	20.00
Unit product cost	20.00	40.00

Year 2 income statement	Variable	Absorption
Sales	480,000.00	480,000.00
Beginning inventory cost	40,000.00	64,000.00
Cost of goods manufactured	120,000.00	240,000.00
Ending inventory	0.00	0.00
COGS	160,000.00	304,000.00
Manufacturing contribution / gross margin	320,000.00	176,000.00
Variable S&A	40,000.00	40,000.00
Contribution margin (variable column)	280,000.00	
Fixed MOH period expense	120,000.00	0
Fixed S&A	70,000.00	70,000.00
NOI	90,000.00	66,000.00

Q106 · January 2020 Term · Exam Q1 [45 marks] (answer continued)

Income reconciliation	Beginning inventory fixed MOH	Ending inventory fixed MOH	Absorption minus variable NOI
Year 1	0.00	24,000.00	24,000.00
Year 2	24,000.00	0.00	(24,000.00)

Each period's difference equals ending inventory fixed MOH less beginning inventory fixed MOH. When older inventory is sold, release its original fixed-overhead rate rather than replacing it with the current production rate.

Q107 · 2020-2021 · Exam Q3(a) [25 marks]

Source: Ipe 2009-25.pdf, PDF pages 70, 71.

Tami's Creations makes swimsuits. First-quarter output 30,000, sales 28,000 at 40, no beginning inventory. Unit costs: DM 3.50, DL 12, variable MOH 1, variable S&A 6. Fixed MOH 300,000; fixed S&A 200,000. The contribution statement reports sales 1,120,000, variable COGS 462,000, variable S&A 168,000, CM 490,000 and operating loss 10,000. A CPA suggests absorption costing may report profit.

Required

1. Prepare the absorption income statement.

Model Answer

No beginning inventory is assumed as stated. Variable product cost includes direct materials, direct labor and variable manufacturing overhead; absorption product cost also includes fixed manufacturing overhead assigned to production.

Unit product cost (currency per unit)	Variable	Absorption
Variable manufacturing cost	16.50	16.50
Fixed MOH per produced unit	0	10.00
Unit product cost	16.50	26.50

Income statement	Absorption	Variable
Sales	1,120,000.00	1,120,000.00
Cost of goods sold	742,000.00	462,000.00

Q107 · 2020-2021 · Exam Q3(a) [25 marks] (answer continued)

Income statement	Absorption	Variable
Gross margin / manufacturing contribution	378,000.00	658,000.00
Variable selling and administrative	168,000.00	168,000.00
Fixed manufacturing overhead (period expense)	0	300,000.00
Fixed selling and administrative	200,000.00	200,000.00
Net operating income	10,000.00	(10,000.00)

In the absorption column, fixed manufacturing overhead is already included in product cost/COGS; do not expense it a second time. The variable-costing contribution margin after all variable costs is 490,000.00.

Inventory and reconciliation	Amount
Ending units	2,000
Variable-cost inventory	33,000.00
Absorption-cost inventory	53,000.00
Fixed MOH deferred / absorption NOI less variable NOI	20,000.00

The two income figures differ because fixed manufacturing overhead attached to unsold goods remains in absorption inventory but is expensed immediately under variable costing.

Absorption profit is 10,000 because 20,000 fixed overhead remains in 2,000 unsold swimsuits. This is a timing difference, not an additional sale or cash inflow; it does not itself solve the cash strain.

Q108 · 2021-2022 · Exam Q8(a) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 85.

State the basic difference between variable and absorption costing.

Model Answer

Feature	Variable costing	Absorption costing
Unit product cost	DM + DL + variable MOH	DM + DL + variable MOH + fixed MOH

Q108 · 2021-2022 · Exam Q8(a) [5 marks] (answer continued)

Model Answer

Feature	Variable costing	Absorption costing
Fixed manufacturing overhead	Entire amount is a period expense	Included in production inventory, then COGS as sold
Income-statement format	Contribution format: variable vs fixed	Functional format: COGS vs S&A
S&A costs	Period costs, separated by behavior	Period costs, excluded from product inventory
Common use	Internal contribution/CVP analysis	External product/inventory costing in the textbook model

The source of income differences is the timing of fixed manufacturing overhead, not a different total cash cost.

Q109 · 2021-2022 · Exam Q8(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 85.

Lynch produces 25,000 units and sells 20,000 at 50 each, with no beginning inventory. Per-unit costs: DM 6, DL 9, variable MOH 3, variable S&A 4. Fixed annual MOH 300,000; fixed S&A 190,000.

Required

1. Compute unit product cost under both methods.
2. Prepare both income statements.
3. Reconcile the operating incomes.

Model Answer

No beginning inventory is assumed as stated. Variable product cost includes direct materials, direct labor and variable manufacturing overhead; absorption product cost also includes fixed manufacturing overhead assigned to production.

Unit product cost (currency per unit)	Variable	Absorption
Variable manufacturing cost	18.00	18.00
Fixed MOH per produced unit	0	12.00

Q109 · 2021-2022 · Exam Q8(b) [15 marks] (answer continued)

Unit product cost (currency per unit)	Variable	Absorption
Unit product cost	18.00	30.00

Income statement	Absorption	Variable
Sales	1,000,000.00	1,000,000.00
Cost of goods sold	600,000.00	360,000.00
Gross margin / manufacturing contribution	400,000.00	640,000.00
Variable selling and administrative	80,000.00	80,000.00
Fixed manufacturing overhead (period expense)	0	300,000.00
Fixed selling and administrative	190,000.00	190,000.00
Net operating income	130,000.00	70,000.00

In the absorption column, fixed manufacturing overhead is already included in product cost/COGS; do not expense it a second time. The variable-costing contribution margin after all variable costs is 560,000.00.

Inventory and reconciliation	Amount
Ending units	5,000
Variable-cost inventory	90,000.00
Absorption-cost inventory	150,000.00
Fixed MOH deferred / absorption NOI less variable NOI	60,000.00

The two income figures differ because fixed manufacturing overhead attached to unsold goods remains in absorption inventory but is expensed immediately under variable costing.

Q110 · 2022-2023 · Exam Q8(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF pages 96, 97.

Heaton reports absorption income of 70,000 in Year 1 and 120,000 in Year 2. Its unit absorption cost is 18: DM 4, DL 7, variable MOH 1 and fixed MOH 6 (270,000 / 45,000 units). S&A is 2 per sold unit plus 130,000 fixed per year. Fixed MOH comprises 40% salaries and 60% depreciation.

Q110 · 2022-2023 · Exam Q8(b) [15 marks] (question continued)

Production/sales	Year 1	Year 2
Units produced	45,000	45,000
Units sold	40,000	50,000
Sales at 25	1,000,000	1,250,000
Absorption COGS	720,000	900,000
Gross margin	280,000	350,000
S&A	210,000	230,000
Absorption NOI	70,000	120,000

Required

1. Prepare a variable-costing contribution income statement for each year.
2. Reconcile absorption and variable NOI for each year.

Model Answer

Variable product cost is 12.00 per unit. Fixed MOH is assigned to units produced, not units sold. The absorption inventory flow below uses FIFO when year-specific fixed overhead rates differ.

Year 1 unit cost	Variable	Absorption
Variable manufacturing	12.00	12.00
Fixed MOH / production	0	6.00
Unit product cost	12.00	18.00

Year 1 income statement	Variable	Absorption
Sales	1,000,000.00	1,000,000.00
Beginning inventory cost	0.00	0.00
Cost of goods manufactured	540,000.00	810,000.00
Ending inventory	60,000.00	90,000.00
COGS	480,000.00	720,000.00
Manufacturing contribution / gross margin	520,000.00	280,000.00
Variable S&A	80,000.00	80,000.00

Q110 · 2022-2023 · Exam Q8(b) [15 marks] (answer continued)

Year 1 income statement	Variable	Absorption
Contribution margin (variable column)	440,000.00	
Fixed MOH period expense	270,000.00	0
Fixed S&A	130,000.00	130,000.00
NOI	40,000.00	70,000.00

Year 2 unit cost	Variable	Absorption
Variable manufacturing	12.00	12.00
Fixed MOH / production	0	6.00
Unit product cost	12.00	18.00

Year 2 income statement	Variable	Absorption
Sales	1,250,000.00	1,250,000.00
Beginning inventory cost	60,000.00	90,000.00
Cost of goods manufactured	540,000.00	810,000.00
Ending inventory	0.00	0.00
COGS	600,000.00	900,000.00
Manufacturing contribution / gross margin	650,000.00	350,000.00
Variable S&A	100,000.00	100,000.00
Contribution margin (variable column)	550,000.00	
Fixed MOH period expense	270,000.00	0
Fixed S&A	130,000.00	130,000.00
NOI	150,000.00	120,000.00

Income reconciliation	Beginning inventory fixed MOH	Ending inventory fixed MOH	Absorption minus variable NOI
Year 1	0.00	30,000.00	30,000.00
Year 2	30,000.00	0.00	(30,000.00)

Q110 · 2022-2023 · Exam Q8(b) [15 marks] (answer continued)

Each period's difference equals ending inventory fixed MOH less beginning inventory fixed MOH. When older inventory is sold, release its original fixed-overhead rate rather than replacing it with the current production rate.

Q111 · 2023-2024 · Exam Q8(a) [18 marks]

Source: Ipe 2009-25.pdf, PDF page 107.

Walsh's first two years have the following costs. Year 1: produced 50,000, sold 40,000. Year 2: produced 40,000, sold 50,000. Selling price = 60; there was no inventory when operations began.

Cost	Amount
DM per unit	25
DL per unit	15
Variable MOH per unit	5
Variable S&A per sold unit	2
Annual fixed MOH	250,000
Annual fixed S&A	80,000

Required

1. Compute unit product cost and income statements under variable costing for both years.
2. Repeat under absorption costing.
3. Explain both years' income differences.

Model Answer

Variable product cost is 45.00 per unit. Fixed MOH is assigned to units produced, not units sold. The absorption inventory flow below uses FIFO when year-specific fixed overhead rates differ.

Year 1 unit cost	Variable	Absorption
Variable manufacturing	45.00	45.00
Fixed MOH / production	0	5.00
Unit product cost	45.00	50.00

Q111 · 2023-2024 · Exam Q8(a) [18 marks] (answer continued)

Year 1 income statement	Variable	Absorption
Sales	2,400,000.00	2,400,000.00
Beginning inventory cost	0.00	0.00
Cost of goods manufactured	2,250,000.00	2,500,000.00
Ending inventory	450,000.00	500,000.00
COGS	1,800,000.00	2,000,000.00
Manufacturing contribution / gross margin	600,000.00	400,000.00
Variable S&A	80,000.00	80,000.00
Contribution margin (variable column)	520,000.00	
Fixed MOH period expense	250,000.00	0
Fixed S&A	80,000.00	80,000.00
NOI	190,000.00	240,000.00

Year 2 unit cost	Variable	Absorption
Variable manufacturing	45.00	45.00
Fixed MOH / production	0	6.25
Unit product cost	45.00	51.25

Year 2 income statement	Variable	Absorption
Sales	3,000,000.00	3,000,000.00
Beginning inventory cost	450,000.00	500,000.00
Cost of goods manufactured	1,800,000.00	2,050,000.00
Ending inventory	0.00	0.00
COGS	2,250,000.00	2,550,000.00
Manufacturing contribution / gross margin	750,000.00	450,000.00
Variable S&A	100,000.00	100,000.00
Contribution margin (variable column)	650,000.00	
Fixed MOH period expense	250,000.00	0

Q111 · 2023-2024 · Exam Q8(a) [18 marks] (answer continued)

Year 2 income statement	Variable	Absorption
Fixed S&A	80,000.00	80,000.00
NOI	320,000.00	270,000.00

Income reconciliation	Beginning inventory fixed MOH	Ending inventory fixed MOH	Absorption minus variable NOI
Year 1	0.00	50,000.00	50,000.00
Year 2	50,000.00	0.00	(50,000.00)

Each period's difference equals ending inventory fixed MOH less beginning inventory fixed MOH. When older inventory is sold, release its original fixed-overhead rate rather than replacing it with the current production rate.

Q112 · 2023-2024 · Exam Q8(b) [7 marks]

Source: Ipe 2009-25.pdf, PDF page 107.

Differentiate variable and absorption costing.

Model Answer

Feature	Variable costing	Absorption costing
Unit product cost	DM + DL + variable MOH	DM + DL + variable MOH + fixed MOH
Fixed manufacturing overhead	Entire amount is a period expense	Included in production inventory, then COGS as sold
Income-statement format	Contribution format: variable vs fixed	Functional format: COGS vs S&A
S&A costs	Period costs, separated by behavior	Period costs, excluded from product inventory
Common use	Internal contribution/CVP analysis	External product/inventory costing in the textbook model

The source of income differences is the timing of fixed manufacturing overhead, not a different total cash cost.

Q113 · 2024-2025 · Exam Q5(c) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 113.

High Country's new camp-cot plant has no beginning inventory. May production is 10,000 units and sales are 8,000 units at 75 each.

Cost	Amount
Direct materials per unit	20
Direct labor per unit	8
Variable MOH per unit	2
Fixed MOH per month	100,000
Variable selling/admin per sold unit	6
Fixed selling/admin per month	200,000

Required

1. For absorption costing, compute unit product cost and an income statement.
2. For variable costing, compute unit product cost and a contribution income statement.
3. Explain the inventory-value and operating-income differences.

Model Answer

No beginning inventory is assumed as stated. Variable product cost includes direct materials, direct labor and variable manufacturing overhead; absorption product cost also includes fixed manufacturing overhead assigned to production.

Unit product cost (currency per unit)	Variable	Absorption
Variable manufacturing cost	30.00	30.00
Fixed MOH per produced unit	0	10.00
Unit product cost	30.00	40.00

Income statement	Absorption	Variable
Sales	600,000.00	600,000.00
Cost of goods sold	320,000.00	240,000.00
Gross margin / manufacturing contribution	280,000.00	360,000.00

Q113 · 2024-2025 · Exam Q5(c) [15 marks] (answer continued)

Income statement	Absorption	Variable
Variable selling and administrative	48,000.00	48,000.00
Fixed manufacturing overhead (period expense)	0	100,000.00
Fixed selling and administrative	200,000.00	200,000.00
Net operating income	32,000.00	12,000.00

In the absorption column, fixed manufacturing overhead is already included in product cost/COGS; do not expense it a second time. The variable-costing contribution margin after all variable costs is 312,000.00.

Inventory and reconciliation	Amount
Ending units	2,000
Variable-cost inventory	60,000.00
Absorption-cost inventory	80,000.00
Fixed MOH deferred / absorption NOI less variable NOI	20,000.00

The two income figures differ because fixed manufacturing overhead attached to unsold goods remains in absorption inventory but is expensed immediately under variable costing.

Q114 · 2024-2025 · Exam Q8(c) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 117.

If units produced equal unit sales, which method gives the higher net operating income: variable or absorption costing? Why?

Model Answer

Under the usual assumption of an unchanged fixed manufacturing overhead cost per unit, the two methods report the same operating income when production equals sales and inventory does not change. No net fixed manufacturing overhead is deferred in, or released from, inventory.

$$\text{Absorption NOI} - \text{Variable NOI} = \text{Ending inventory fixed MOH} - \text{Beginning invent}$$

Q114 · 2024-2025 · Exam Q8(c) [10 marks] (answer continued)

Equal unit quantities alone do not guarantee equality if beginning and ending inventories contain different fixed-overhead rates. The standard exam conclusion assumes the same per-unit rate and no other inventory-cost adjustment.

Cost-volume-profit and operating leverage

Q115 · 2009-2010 · Exam Q4(b) [23 marks]

Source: Ipe 2009-25.pdf, PDF pages 4, 5.

Telephone answering machines: annual sales 1,200,000 for 20,000 units at 60; variable expenses 900,000 at 45; CM 300,000 at 15; fixed expenses 240,000; NOI 60,000.

Required

1. Compute CM and variable-expense ratios.
2. Compute BE units/sales by equation.
3. Find profit increase from 400,000 extra sales.
4. Find units for 90,000 profit.
5. Find margin of safety in dollars and %.
6. Compute DOL and profit % change from 8% higher sales.
7. Verify with a contribution statement.

Model Answer

$$\text{CM ratio} = 25\%, \quad \text{Variable ratio} = 75\%.$$

$$60Q - 45Q - 240,000 = 0 \Longrightarrow Q = 16,000, \quad S = 960,000.$$

Extra profit = 400,000 x 25% = 100,000. Target units = (240,000 + 90,000)/15 = 22,000. The apparent 860 in the unit-sales scan is 60, as confirmed by total sales / 20,000.

Margin of safety = 1,200,000 - 960,000 = 240,000, or 20% of sales. DOL = 300,000/60,000 = 5; 8% higher sales implies 40% higher NOI.

Statement at 8% extra volume	Amount
Sales: 21,600 x 60	1,296,000
Variable costs: 21,600 x 45	972,000
CM	324,000
Fixed expense	240,000
NOI	84,000

Profit rises 24,000, exactly 40% of 60,000.

Q116 · 2011-2012 · Exam Q2(a) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 10.

Explain operating leverage.

Model Answer

$$\text{DOL} = \frac{\text{Contribution margin}}{\text{Net operating income}}.$$

At a given profitable activity level, a higher DOL means a percentage change in sales creates a larger percentage change in operating profit. It offers greater upside when sales grow and greater downside when sales fall. A lower DOL means profit is less sensitive to sales changes.

$$\%\Delta\text{NOI} \approx \text{DOL} \times \%\Delta\text{Sales}.$$

The comparison applies at the specified sales levels and cost structure. DOL changes as sales move away from break-even; it is not a permanent company constant. At zero or negative operating income, the usual percentage interpretation is unreliable.

Q117 · 2011-2012 · Exam Q8(b) [8 marks]

Source: Ipe 2009-25.pdf, PDF page 16.

Does a 10% price decrease have the same NOI effect as a 10% increase in variable expenses? Explain.

Model Answer

No, generally. At unchanged volume Q, a 10% price decrease reduces profit by 0.10PQ; a 10% increase in variable expense reduces profit by 0.10vQ. These are equal only when P = v. A profitable positive-CM product has P > v, so the price cut causes the larger decline. Example: price 20, variable cost 12 and 1,000 units: price-cut loss 2,000 versus cost-rise loss 1,200. Changes in demand would require a separate volume analysis.

Q118 · 2011-2012 · Exam Q8(c) [7 marks]

Source: Ipe 2009-25.pdf, PDF page 16.

State the significance of the break-even point.

Q118 · 2011-2012 · Exam Q8(c) [7 marks] (answer continued)

Model Answer

Break-even is the sales activity where total contribution exactly covers total fixed costs and operating income is zero. It identifies minimum activity to avoid a loss under the model and supports target-profit, pricing, cost-structure and safety-margin analysis.

$$Q_{BE} = F \,/\, (P - v), \quad S_{BE} = F \,/\, \text{CM ratio}\,.$$

It is not automatically a cash break-even point because fixed costs may include noncash depreciation. Constant price/variable cost, fixed cost, relevant range and stable sales mix are essential assumptions.

Q119 · 2011-2012 · Exam Q8(d) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 16.

Sheer Elegance stocking price 2/pair; variable expense 0.80; fixed annual rent 12,000, depreciation 3,000, selling 30,000, admin 15,000, total 60,000. Turning a part-time salesperson full-time adds 8,000 annual fixed expense and is expected to add 20,000 sales.

Required

1. Find BE pairs and sales.
2. Find pairs for 9,000 profit.
3. Should the staffing change be adopted?

Model Answer

$$\text{CM} = 1.20, \quad \text{CM ratio} = 60\%, \quad Q_{BE} = 50,000, \quad S_{BE} = 100,000\,.$$

Target pairs = (60,000 + 9,000)/1.20 = 57,500. Staffing adds 20,000 x 60% - 8,000 = 4,000 profit, so adopt if the sales estimate and operating assumptions are credible.

Q120 · 2012-2013 · Exam Q8(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 23.

Telephone answering machines: annual sales 1,200,000 for 20,000 units at 60; variable expenses 900,000 at 45; CM 300,000 at 15; fixed expenses 240,000; NOI 60,000.

Required

1. Compute CM and variable-expense ratios.

Q120 · 2012-2013 · Exam Q8(b) [15 marks] (question continued)

2. Find BE units/sales using the equation method.
3. Find profit increase from 400,000 extra sales with unchanged cost behavior.
4. Find units for minimum profit 90,000.

Model Answer

$$\text{CM ratio} = 25\%, \quad \text{Variable ratio} = 75\%.$$

$$60Q - 45Q - 240,000 = 0 \Longrightarrow Q = 16,000, \quad S = 960,000.$$

Extra profit = 400,000 x 25% = 100,000. Target units = (240,000 + 90,000)/15 = 22,000. The apparent 860 in the unit-sales scan is 60, as confirmed by total sales / 20,000.

Q121 · 2013-2014 · Exam Q3(a) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 25.

Define operating leverage and explain its significance.

Model Answer

$$\text{DOL} = \frac{\text{Contribution margin}}{\text{Net operating income}}.$$

At a given profitable activity level, a higher DOL means a percentage change in sales creates a larger percentage change in operating profit. It offers greater upside when sales grow and greater downside when sales fall. A lower DOL means profit is less sensitive to sales changes.

$$\%\Delta\text{NOI} \approx \text{DOL} \times \%\Delta\text{Sales}.$$

The comparison applies at the specified sales levels and cost structure. DOL changes as sales move away from break-even; it is not a permanent company constant. At zero or negative operating income, the usual percentage interpretation is unreliable.

Q122 · 2013-2014 · Exam Q3(b) [30 marks]

Source: Ipe 2009-25.pdf, PDF pages 25, 26.

Cricket balls sell at 25; variable cost 15; annual fixed cost 210,000; 30,000 sold last year.

Required

1. Find CM, NOI, CM ratio, BE units/sales and DOL.

Q122 · 2013-2014 · Exam Q3(b) [30 marks] (question continued)

2. If labor raises variable cost by 3/unit with price unchanged, find CM ratio, BE and profit at the same volume.
3. Find new volume to retain last year's income.
4. Find price retaining the original CM ratio after the labor rise.
5. Independently, automation cuts original variable cost 40% and doubles fixed cost: find CM ratio, BE and volume for original profit.
6. Prepare automated statement at 30,000 units, compute DOL and recommend.

Model Answer

Measure	Original	Higher labor	Automated
Price	25	25	25
Variable cost/unit	15	18	9
CM/unit	10	7	16
CM ratio	40%	28%	64%
Fixed cost	210,000	210,000	420,000
BE units	21,000	30,000	26,250
BE sales	525,000	750,000	656,250
Sales at 30,000 units	750,000	750,000	750,000
Variable expense	450,000	540,000	270,000
CM	300,000	210,000	480,000
NOI	90,000	0	60,000
DOL	3.3333	Undefined at BE	8

Higher-labor target = (210,000 + 90,000)/7 = 42,857.143; at least 42,858 balls. To retain 40% CM ratio, (P - 18)/P = 0.40, so P = 30. Automated target = (420,000 + 90,000)/16 = 31,875 balls.

At 30,000 balls automation reduces NOI by 30,000 and raises DOL to 8: profit becomes more sales-sensitive. Cost indifference = (420,000 - 210,000)/(15 - 9) = 35,000 balls. Recommend only if sustained sales/capacity/quality benefits justify the higher commitment, not on a variable-cost reduction alone.

Q123 · 2014-2015 · Exam Q1(a) [22 marks]

Source: Ipe 2009-25.pdf, PDF page 31.

Superior Door price 100; variable DM 30, DL 12, MOH 8, selling commission 10% of sales. Fixed manufacturing 1,250 and fixed S&A 2,500.

Required

1. Find CM/door.
2. Find profit at 200 doors.
3. Find break-even doors and sales dollars.
4. Find doors and sales dollars for 7,500 target profit.

Model Answer

$$v = 30 + 12 + 8 + 10 = 60, \quad \text{CM} = 40, \quad F = 3,750\,.$$

At 200 doors profit = 200 x 40 - 3,750 = 4,250. Continuous break-even = 3,750/40 = 93.75 doors, sales 9,375; at least 94 whole doors (9,400 sales) avoid a loss. Target = (3,750 + 7,500)/40 = 281.25 doors, theoretical sales 28,125; sell at least 282 whole doors, sales 28,200, for the minimum target profit.

Q124 · 2014-2015 · Exam Q1(b) [13 marks]

Source: Ipe 2009-25.pdf, PDF page 31.

Hermann price 90, variable expense 63, CM 27 (30%); monthly fixed expense 30,000; sales 2,000 units. Consider independently: (i) extra advertising 5,000 raises sales by 9,000; (ii) better components add 2/unit variable cost and raise unit sales 10%.

Required

1. Should each proposal be adopted?

Model Answer

Advertising: extra CM = 9,000 x 30% = 2,700, less 5,000 expense gives a 2,300 decrease; reject. Components: new sales 2,200 units, CM 25/unit; new CM 55,000 vs old 54,000 gives 1,000 extra profit; accept, assuming fixed costs and price remain unchanged.

Q125 · 2016-2017 · Exam Q3(a) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 41.

Engberg monthly statement: sales 80,000; variable expenses 32,000; CM 48,000; fixed expenses 38,000; NOI 10,000.

Required

1. Find DOL.
2. Estimate profit change from 5% higher sales.
3. Verify with a contribution statement.

Model Answer

$$\text{DOL} = 48,000 / 10,000 = 4.8; \quad \%\Delta\text{NOI} = 4.8(5\%) = 24\%.$$

Statement after sales increase	Amount
Sales	84,000
Variable expenses (40%)	33,600
CM	50,400
Fixed expenses	38,000
NOI	12,400

Profit increases 2,400, or 24%; the sales change is 4,000, not the same as the profit change.

Q126 · 2017-2018 · Exam Q7 [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 54, 53.

PEM's original monthly contribution statement is below. Treat each proposed change independently, referring back to the original data.

Original statement	Amount
Sales: 19,500 units at 30	585,000
Variable expenses	409,500
Contribution margin	175,500
Fixed expenses	180,000
Operating loss	-4,500

Q126 · 2017-2018 · Exam Q7 [35 marks] (question continued)

Required

1. Compute CM ratio and break-even units and sales.
2. Evaluate 16,000 extra advertising producing 80,000 extra sales.
3. Prepare a new statement if price falls 10%, advertising rises 60,000, and unit sales double.
4. If new packaging adds 0.75 variable cost per unit, find units to earn 9,750 profit.
5. Automation reduces variable cost by 3/unit and raises fixed cost by 72,000: find CM ratio and BE units/sales, compare contribution statements at 26,000 units (total, per unit and %), and recommend.

Model Answer

Base variable expense per unit is 409,500 / 19,500 = 21; unit CM = 30 - 21 = 9.

Original CVP measure	Calculation	Result
CM ratio	175,500 / 585,000	30%
Break-even units	180,000 / 9	20,000 units
Break-even sales	180,000 / 0.30	600,000

Extra advertising: added contribution = 80,000 x 30% = 24,000; added cost = 16,000. Profit rises by 8,000, changing the 4,500 loss to a 3,500 profit.

Price-cut scenario	Amount
New unit selling price	27
Units sold	39,000
Sales	1,053,000
Variable expenses: 39,000 x 21	819,000
Contribution margin	234,000
Fixed expenses: 180,000 + 60,000	240,000
Operating loss	-6,000

$$\text{Packaging target units} = \frac{180,000 + 9,750}{30 - 21 - 0.75} = 23,000\,.$$

Q126 · 2017-2018 · Exam Q7 [35 marks] (answer continued)

Automation scenario	Calculation	Result
New variable cost per unit	21 - 3	18
New unit CM	30 - 18	12
New CM ratio	12 / 30	40%
New fixed expenses	180,000 + 72,000	252,000
Break-even units	252,000 / 12	21,000
Break-even sales	252,000 / 0.40	630,000
NOI at current 19,500 units	19,500 x 12 - 252,000	(18,000)

At present volume, automation lowers profit by 13,500, so do not adopt it on current-volume evidence. Its extra 72,000 fixed cost is recovered through 3 savings per unit at 24,000 units. Above 24,000 units it improves profit relative to the original process; it also increases operating risk.

At 26,000 units	Original total	Original/unit	Automated total	Automated/unit
Sales	780,000	30	780,000	30
Variable expense	546,000	21	468,000	18
CM	234,000	9	312,000	12
Fixed expense	180,000	6.92	252,000	9.69
NOI	54,000	2.08	60,000	2.31

Percentage of sales	Original	Automated
Sales	100%	100%
Variable expenses	70%	60%
CM	30%	40%
Fixed expenses	23.08%	32.31%
NOI	6.92%	7.69%

At the specified 26,000-unit forecast, recommend automation on profit grounds: it improves monthly NOI by 6,000. The earlier current-volume comparison explains the risk; the recommendation changes because forecast volume exceeds the 24,000-unit indifference point.

Q127 · 2018-2019 · Exam Q7 [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 61, 62.

Memofax monthly sales 13,500 units at 20 = 270,000; variable expenses 189,000; fixed expenses 90,000.

Required

1. Compute CM ratio and break-even units/sales.
2. Find NOI effect of 8,000 extra advertising and 70,000 extra sales.
3. Independently, cut price 10%, add 35,000 advertising and double unit sales: prepare a contribution statement.
4. Independently, add 0.60/unit packaging: find units for 4,500 profit.
5. Automation halves variable expenses/unit and adds 118,000 fixed costs: find CM ratio/BE, compare statements at 20,000 units and recommend.

Model Answer

$$v = 14, \quad \text{CM / unit} = 6, \quad \text{CM ratio} = 30\%.$$

Break-even = 90,000 / 6 = 15,000 units, or 300,000 sales. Current loss = 9,000. Advertising proposal adds 70,000 x 30% - 8,000 = 13,000 profit; new NOI = 4,000.

Contribution statement	Price-cut proposal	20,000 units: original	20,000 units: automated
Units	27,000	20,000	20,000
Price	18	20	20
Sales	486,000	400,000	400,000
Variable expense	378,000	280,000	140,000
CM	108,000	120,000	260,000
Fixed expense	125,000	90,000	208,000
NOI	-17,000	30,000	52,000

Packaging target: (90,000 + 4,500) / (20 - 14 - 0.60) = 17,500 units. Automation CM ratio = 13/20 = 65%; break-even = 208,000/13 = 16,000 units or 320,000 sales. At 20,000 units automation raises profit by 22,000, so recommend at that sustained volume, subject to investment/operational risks. Indifference volume is 118,000 / 7 = 16,857.14 units; below it the larger fixed cost can hurt.

Q128 · January 2020 Term · Exam Q6(a) [22 marks]

Source: Ipe 2009-25.pdf, PDF pages 67, 68.

Outback camp stoves sell for 50; variable expense 32; fixed expense 108,000 monthly. Current sales 8,000 units. Proposed price cut 10% would raise unit sales 25%.

Required

1. Find break-even units and dollars.
2. Explain the effect if variable expense rises as a proportion of price.
3. Prepare present/proposed contribution statements, total and per unit.
4. Find units at the new price to earn at least 35,000.

Model Answer

$$\text{BE units} = 108,000 \,/\, (50 - 32) = 6,000; \quad \text{BE sales} = 300,000\,.$$

A higher variable-cost ratio lowers the CM ratio, so break-even sales rise when fixed expense is unchanged.

Line	Present total	Present/unit	Proposed total	Proposed/unit
Units	8,000	-	10,000	-
Sales	400,000	50	450,000	45
Variable expense	256,000	32	320,000	32
Contribution	144,000	18	130,000	13
Fixed expense	108,000	13.50	108,000	10.80
NOI	36,000	4.50	22,000	2.20

$$Q = (108,000 + 35,000) \,/\, (45 - 32) = 11,000\,.$$

The proposed price cut reduces profit by 14,000 at the projected volume; 11,000 units are needed for the target.

Q129 · January 2020 Term · Exam Q6(b) [23 marks]

Source: Ipe 2009-25.pdf, PDF page 68.

Menlo quarterly sales 450,000 at 30/unit; variable expenses 180,000 at 12/unit; CM 270,000; fixed costs 216,000; NOI 54,000.

Q129 · January 2020 Term · Exam Q6(b) [23 marks] (question continued)

Required

1. Find break-even units and dollars.
2. State CM at break-even without calculation.
3. Find units for 90,000 target profit and verify with a contribution statement.
4. Find margin of safety in dollars and percentage.
5. Find CM ratio and profit effect of 50,000 extra sales with unchanged fixed cost.

Model Answer

$$\text{CM / unit} = 18, \quad \text{CM ratio} = 60\%, \quad Q_{BE} = 12,000, \quad S_{BE} = 360,000.$$

At break-even total CM equals fixed costs: 216,000. Target units = (216,000 + 90,000) / 18 = 17,000.

Target-profit statement	Amount
Sales: 17,000 x 30	510,000
Variable expenses: 17,000 x 12	204,000
CM	306,000
Fixed expenses	216,000
NOI	90,000

$$\text{MOS} = 450,000 - 360,000 = 90,000 = 20\% \text{ of sales}.$$

Extra profit = 50,000 x 60% = 30,000.

Q130 · 2021-2022 · Exam Q7(a) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 84.

PEM's original monthly contribution statement is below. Treat each proposed change independently, referring back to the original data.

Original statement	Amount
Sales: 19,500 units at 30	585,000
Variable expenses	409,500
Contribution margin	175,500

Q130 · 2021-2022 · Exam Q7(a) [25 marks] (question continued)

Original statement	Amount
Fixed expenses	180,000
Operating loss	-4,500

Required

1. Compute CM ratio and break-even units and sales.
2. Evaluate 16,000 extra advertising producing 80,000 extra sales.
3. Prepare a new statement if price falls 10%, advertising rises 60,000, and unit sales double.
4. If new packaging adds 0.75 variable cost per unit, find units to earn 9,750 profit.
5. If automation reduces variable cost by 3 per unit and raises fixed costs by 72,000 monthly, compute the new CM ratio and break-even point; recommend whether to automate.

Model Answer

Base variable expense per unit is 409,500 / 19,500 = 21; unit CM = 30 - 21 = 9.

Original CVP measure	Calculation	Result
CM ratio	175,500 / 585,000	30%
Break-even units	180,000 / 9	20,000 units
Break-even sales	180,000 / 0.30	600,000

Extra advertising: added contribution = 80,000 x 30% = 24,000; added cost = 16,000. Profit rises by 8,000, changing the 4,500 loss to a 3,500 profit.

Price-cut scenario	Amount
New unit selling price	27
Units sold	39,000
Sales	1,053,000
Variable expenses: 39,000 x 21	819,000
Contribution margin	234,000
Fixed expenses: 180,000 + 60,000	240,000
Operating loss	-6,000

Q130 · 2021-2022 · Exam Q7(a) [25 marks] (answer continued)

$$\text{Packaging target units} = \frac{180,000 + 9,750}{30 - 21 - 0.75} = 23,000\,.$$

Automation scenario	Calculation	Result
New variable cost per unit	21 - 3	18
New unit CM	30 - 18	12
New CM ratio	12 / 30	40%
New fixed expenses	180,000 + 72,000	252,000
Break-even units	252,000 / 12	21,000
Break-even sales	252,000 / 0.40	630,000
NOI at current 19,500 units	19,500 x 12 - 252,000	(18,000)

At present volume, automation lowers profit by 13,500, so do not adopt it on current-volume evidence. Its extra 72,000 fixed cost is recovered through 3 savings per unit at 24,000 units. Above 24,000 units it improves profit relative to the original process; it also increases operating risk.

Q131 · 2022-2023 · Exam Q8(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 95, 96.

Feather Friends sells birdhouses for 20 each, with variable expenses of 8 each and annual fixed expenses of 180,000. Answer the proposals independently.

Operating results for the leverage subpart	Amount
Sales	400,000
Variable expenses	160,000
CM	240,000
Fixed expenses	180,000
NOI	60,000

Required

1. Compute the CM ratio.
2. Use it to compute break-even sales dollars.

Q131 · 2022-2023 · Exam Q8(a) [20 marks] (question continued)

3. Find profit increase if sales rise by 75,000 and fixed costs do not change.
4. Using the provided statement, compute DOL and the percentage profit increase for 20% sales growth.
5. At original sales of 18,000 units, compare the old statement with one based on a 10% price cut, 30,000 more advertising, and one-third more unit sales. Recommend.
6. At original sales of 18,000 units, suppose price stays 20, sales commissions rise by 1 per unit and sales volume rises 25%. Find maximum extra advertising with profit unchanged, using incremental analysis.

Model Answer

Base measure	Calculation	Answer
Unit CM	20 - 8	12
CM ratio	12 / 20	60%
Break-even dollars	180,000 / 0.60	300,000
Profit from 75,000 extra sales	75,000 x 0.60	45,000
DOL in supplied statement	240,000 / 60,000	4.00
Profit increase for 20% sales growth	4 x 20%	80%

18,000-unit pricing proposal	Original	Proposed
Unit price	20	18
Unit sales	18,000	24,000
Sales	360,000	432,000
Variable expenses	144,000	192,000
Contribution margin	216,000	240,000
Fixed expenses	180,000	210,000
NOI	36,000	30,000

Reject the price/advertising proposal on these estimates; operating income falls by 6,000.

$$\text{New contribution} = 18,000(1.25)(20 - 8 - 1) = 247,500\,.$$

$$\text{Maximum extra advertising} = 247,500 - 18,000(20 - 8) = 31,500\,.$$

Q131 · 2022-2023 · Exam Q8(a) [20 marks] (answer continued)

The extra 1 commission applies to all 22,500 units in that scenario, not only the additional units. Spending at most 31,500 on advertising preserves the original 36,000 profit.

Q132 · 2023-2024 · Exam Q7(a) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 105.

Explain the importance of cost-volume-profit analysis.

Model Answer

CVP links selling prices, variable cost, fixed cost, sales volume and profit. It helps management estimate break-even activity, target-profit sales, safety margins and the profit effect of pricing, advertising or cost changes.

$$\text{NOI} = Q(P - v) - F\,.$$

Its basic form assumes a relevant range with constant unit price/variable cost, constant total fixed cost, and a stable sales mix for multiple products. Test those assumptions before extending a simple CVP result to a large capacity or mix change.

Q133 · 2023-2024 · Exam Q7(b) [25 marks]

Source: Ipe 2009-25.pdf, PDF pages 105, 106.

Feather Friends sells birdhouses for 20 each, with variable expenses of 8 each and annual fixed expenses of 180,000. Answer the proposals independently.

Operating results for the leverage subpart	Amount
Sales	400,000
Variable expenses	160,000
CM	240,000
Fixed expenses	180,000
NOI	60,000

Required

1. Compute the CM ratio.
2. Use it to compute break-even sales dollars.

Q133 · 2023-2024 · Exam Q7(b) [25 marks] (question continued)

3. Find profit increase if sales rise by 75,000 and fixed costs do not change.
4. Using the provided statement, compute DOL and the percentage profit increase for 20% sales growth.
5. At original sales of 18,000 units, compare the old statement with one based on a 10% price cut, 30,000 more advertising, and one-third more unit sales. Recommend.
6. At original sales of 18,000 units, suppose price stays 20, sales commissions rise by 1 per unit and sales volume rises 25%. Find maximum extra advertising with profit unchanged, using incremental analysis.

Model Answer

Base measure	Calculation	Answer
Unit CM	20 - 8	12
CM ratio	12 / 20	60%
Break-even dollars	180,000 / 0.60	300,000
Profit from 75,000 extra sales	75,000 x 0.60	45,000
DOL in supplied statement	240,000 / 60,000	4.00
Profit increase for 20% sales growth	4 x 20%	80%

18,000-unit pricing proposal	Original	Proposed
Unit price	20	18
Unit sales	18,000	24,000
Sales	360,000	432,000
Variable expenses	144,000	192,000
Contribution margin	216,000	240,000
Fixed expenses	180,000	210,000
NOI	36,000	30,000

Reject the price/advertising proposal on these estimates; operating income falls by 6,000.

$$\text{New contribution} = 18,000(1.25)(20 - 8 - 1) = 247,500\,.$$

$$\text{Maximum extra advertising} = 247,500 - 18,000(20 - 8) = 31,500\,.$$

Q133 · 2023-2024 · Exam Q7(b) [25 marks] (answer continued)

The extra 1 commission applies to all 22,500 units in that scenario, not only the additional units. Spending at most 31,500 on advertising preserves the original 36,000 profit.

Q134 · 2023-2024 · Exam Q7(c) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 106.

Compare a high operating-leverage ratio with a low operating-leverage ratio.

Model Answer

$$\text{DOL} = \frac{\text{Contribution margin}}{\text{Net operating income}}.$$

At a given profitable activity level, a higher DOL means a percentage change in sales creates a larger percentage change in operating profit. It offers greater upside when sales grow and greater downside when sales fall. A lower DOL means profit is less sensitive to sales changes.

$$\%\Delta\text{NOI} \approx \text{DOL} \times \%\Delta\text{Sales}.$$

The comparison applies at the specified sales levels and cost structure. DOL changes as sales move away from break-even; it is not a permanent company constant. At zero or negative operating income, the usual percentage interpretation is unreliable.

Q135 · 2024-2025 · Exam Q5(a) [8 marks]

Source: Ipe 2009-25.pdf, PDF page 113.

What is meant by a product's contribution margin ratio? How is it useful in planning operations?

Model Answer

Contribution margin is sales less all variable expenses. It first covers fixed expenses; any excess becomes operating profit.

$$\text{CM ratio} = \frac{\text{Sales} - \text{Variable expenses}}{\text{Sales}} = \frac{P - v}{P}.$$

It measures the contribution from each extra dollar of sales, assuming the variable-cost ratio is constant. For example, a 40% ratio means an additional 10,000 in sales creates 4,000 in contribution before any change in fixed costs. Use it for break-even sales, target-profit sales and evaluating advertising or pricing changes.

Q135 · 2024-2025 · Exam Q5(a) [8 marks] (answer continued)

$$\text{Break} - \text{even sales} = \frac{F}{\text{CM ratio}}, \quad \text{Target sales} = \frac{F + \text{Target profit}}{\text{CM ratio}}.$$

Q136 · 2024-2025 · Exam Q7(a) [25 marks]

Source: Ipe 2009-25.pdf, PDF pages 115, 116.

PEM's original monthly contribution statement is below. Treat each proposed change independently, referring back to the original data.

Original statement	Amount
Sales: 19,500 units at 30	585,000
Variable expenses	409,500
Contribution margin	175,500
Fixed expenses	180,000
Operating loss	-4,500

Required

1. Compute CM ratio and break-even units and sales.
2. Evaluate 16,000 extra advertising producing 80,000 extra sales.
3. Prepare a new statement if price falls 10%, advertising rises 60,000, and unit sales double.
4. If new packaging adds 0.75 variable cost per unit, find units to earn 9,750 profit.
5. If automation reduces variable cost by 3 per unit and raises fixed costs by 72,000 monthly, compute the new CM ratio and break-even point; recommend whether to automate.

Model Answer

Base variable expense per unit is 409,500 / 19,500 = 21; unit CM = 30 - 21 = 9.

Original CVP measure	Calculation	Result
CM ratio	175,500 / 585,000	30%
Break-even units	180,000 / 9	20,000 units
Break-even sales	180,000 / 0.30	600,000

Q136 · 2024-2025 · Exam Q7(a) [25 marks] (answer continued)

Extra advertising: added contribution = 80,000 x 30% = 24,000; added cost = 16,000. Profit rises by 8,000, changing the 4,500 loss to a 3,500 profit.

Price-cut scenario	Amount
New unit selling price	27
Units sold	39,000
Sales	1,053,000
Variable expenses: 39,000 x 21	819,000
Contribution margin	234,000
Fixed expenses: 180,000 + 60,000	240,000
Operating loss	-6,000

$$\text{Packaging target units} = \frac{180,000 + 9,750}{30 - 21 - 0.75} = 23,000\,.$$

Automation scenario	Calculation	Result
New variable cost per unit	21 - 3	18
New unit CM	30 - 18	12
New CM ratio	12 / 30	40%
New fixed expenses	180,000 + 72,000	252,000
Break-even units	252,000 / 12	21,000
Break-even sales	252,000 / 0.40	630,000
NOI at current 19,500 units	19,500 x 12 - 252,000	(18,000)

At present volume, automation lowers profit by 13,500, so do not adopt it on current-volume evidence. Its extra 72,000 fixed cost is recovered through 3 savings per unit at 24,000 units. Above 24,000 units it improves profit relative to the original process; it also increases operating risk.

Relevant costs and managerial decisions

Q137 · 2013-2014 · Exam Q8(a) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 30.

Discuss which costs matter when adding or dropping a product line.

Model Answer

Compare lost/gained contribution margin with avoidable fixed costs and capacity opportunity costs. Include effects on complementary products, customers and service. Exclude sunk costs and common allocations that remain after the decision.

$$\Delta \text{Profit}_{\text{drop}} = \text{Avoidable fixed savings} - \text{Lost CM} + \text{Alternative} - \text{use benefit}.$$

A line with a loss after allocated common costs can still benefit the company. Verify whether traceable costs actually disappear; being traceable does not automatically make depreciation or a committed salary avoidable.

Q138 · 2013-2014 · Exam Q8(b) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 30.

Define relevant, differential, sunk and opportunity cost.

Model Answer

A relevant cost is a future cost that differs between alternatives, such as extra material for a special order.

Term	Meaning	Example
Differential cost/revenue	The future difference in cost/revenue between alternatives	An order creates 12,000 revenue and 8,000 extra costs: 4,000 benefit
Opportunity cost	Benefit sacrificed by choosing one alternative	Using a room internally sacrifices rent obtainable from leasing it
Sunk cost	A past cost that cannot be changed by the current decision	The original purchase price of equipment already owned

Only future amounts that differ between alternatives affect the decision. An opportunity cost need not appear in the accounting ledger to be relevant; a sunk cost appears in the ledger but cannot be changed.

Q139 · 2014-2015 · Exam Q4(a) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 34.

Climate Control produces 15,000 thermostats annually. Supplier quote 20/unit. Unit cost: DM 6, DL 8, variable MOH 1, traceable fixed MOH 5 (40% supervisor salary, 60% special-equipment depreciation with no resale), allocated common fixed MOH 10. No alternative use for the facilities.

Required

1. Should the company buy rather than make?

Model Answer

Relevant annual cost	Make	Buy
DM	90,000	0
DL	120,000	0
Variable OH	15,000	0
Avoidable supervisor cost	30,000	0
Supplier purchase	0	300,000
Total	255,000	300,000

Continue making: relevant cost 17/unit is 3 less than buying, saving 45,000 annually. Depreciation 3/unit and common fixed allocation 10/unit remain regardless of choice. This assumes supervisor salary is avoidable; no capacity opportunity cost exists.

Q140 · 2017-2018 · Exam Q3(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 49.

Regal Cycle is considering dropping Racing bikes. Special equipment has no resale value and does not wear out.

Quarterly statement	Dirt	Mountain	Racing	Total
Sales	90,000	150,000	60,000	300,000
Variable expenses	27,000	60,000	33,000	120,000
CM	63,000	90,000	27,000	180,000
Traceable advertising	10,000	14,000	6,000	30,000

Q140 · 2017-2018 · Exam Q3(b) [15 marks] (question continued)

Quarterly statement	Dirt	Mountain	Racing	Total
Special-equipment depreciation	6,000	9,000	8,000	23,000
Line-manager salaries	12,000	13,000	10,000	35,000
Allocated common fixed	18,000	30,000	12,000	60,000
NOI	17,000	24,000	-9,000	32,000

Required

1. Should Racing bikes be discontinued? Explain with calculations.

Model Answer

Effect of dropping Racing	Profit change
Lost CM	-27,000
Avoided advertising	6,000
Avoided manager salary	10,000
Net decrease	-11,000

Retain the line: company NOI would fall from 32,000 to 21,000. The 8,000 depreciation and 12,000 common allocation are unavoidable under the stated assumptions. If the manager salary cannot actually be avoided or redeployed productively, dropping is even less attractive.

Q141 · 2017-2018 · Exam Q4(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 49, 50.

Come-Clean makes Grit 337 at 1.60 per pound and can sell it for 2.00 per pound. A jar of Sparkle polish sells for 4.00 and uses one-quarter pound of Grit. Extra ingredients cost 0.65 and direct labor 1.48 per jar. Variable MOH = 25% of direct labor; variable selling cost = 7.5% of polish sales. Monthly supervisor salary = 3,000; special mixing-equipment depreciation = 1,400 with negligible resale value and no wear from use; polish advertising = 4,000.

Required

1. Compute incremental contribution per jar from further processing.
2. Find the minimum jars/month needed to justify continued processing.

Q141 · 2017-2018 · Exam Q4(a) [20 marks] (answer continued)

Model Answer

The choice is to sell the quarter-pound of Grit or process it into polish. The original Grit manufacturing cost is common to both choices. Use the lost Grit sales revenue as the opportunity cost.

Incremental contribution per jar	Amount
Polish revenue	4
Lost Grit revenue: 0.25 x 2.00	-0.50
Extra ingredients	-0.65
Extra direct labor	-1.48
Variable MOH: 25% x 1.48	-0.37
Variable selling: 7.5% x 4.00	-0.30
Incremental contribution margin	0.70

Depreciation of the existing special equipment is sunk, not incremental. Monthly supervisor/advertising costs are not part of per-jar contribution; if both are avoidable, the positive 0.70 contribution must cover 7,000 monthly fixed costs before further processing increases total profit. That requires 10,000 jars, with all other assumptions unchanged.

$$\text{Minimum jars} = \frac{3,000 + 4,000}{0.70} = 10,000\,.$$

At exactly 10,000 jars the incremental alternatives are indifferent; above it processing adds profit. Depreciation remains excluded because the equipment has negligible resale and does not wear out.

Q142 · 2018-2019 · Exam Q1(a) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 55.

Hallas sells glue at 35/gallon; variable cost 21. Normal sales 40,000 gallons/month. Strikes reduce sales to 11,000 monthly for two months. Fixed monthly manufacturing/selling costs are 230,000 / 310,000. Closing saves 60,000 manufacturing cost and 10% selling cost each month, but restarting costs 14,000. Lean production means no inventory.

Q142 · 2018-2019 · Exam Q1(a) [25 marks] (question continued)

Required

1. Should Hallas close for the two-month strike? Support the recommendation with calculations.

Model Answer

Two-month closing effect	Amount
Lost CM: 11,000 x (35 - 21) x 2	-308,000
Factory fixed savings: 60,000 x 2	120,000
Selling savings: 310,000 x 10% x 2	62,000
Restart cost	-14,000
Net effect on profit	-140,000

Stay open. Closing makes the two-month result 140,000 worse. Unavoidable costs continue under either alternative; the low-volume accounting loss alone does not justify shutting down.

Q143 · 2018-2019 · Exam Q4(b) [10 marks]

Source: Ipe 2009-25.pdf, PDF pages 58, 59.

Banner faces more demand than available capacity. Wage 8/hour; only 3,000 DLHs weekly.

Per unit	A	B	C
Price	60	90	80
DM	27	14	40
DL	12	32	16
Variable MOH	3	8	4
CM	18	36	20

Required

1. Compute CM per labor-hour.
2. Which product orders should be worked on next week?

Q143 · 2018-2019 · Exam Q4(b) [10 marks] (answer continued)

Model Answer

Product	Hours/unit	CM/hour	Rank
A	1.50	12	1
B	4	9	3
C	2	10	2

Work on A first, C second, B third. If A alone has demand for at least 2,000 units, all 3,000 hours should produce A, yielding 36,000 CM. Otherwise satisfy A demand, then C, then B. Individual demand limits are not supplied.

Q144 · January 2020 Term · Exam Q3(a) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 65.

UVX normally sells 60,000 units at 21; capacity 75,000. Listed unit costs: DM 5.10; DL 3.30; variable MOH 1; a second row also labeled variable MOH 4; variable S&A 1.50; fixed S&A 2.40. Special order: 15,000 at 14, with no regular-sales displacement or fixed-cost change.

Required

1. Determine the annual profit increase/decrease if accepted.

> **Source clarification:** Both manufacturing-overhead rows are printed as variable. The usual intended fixed-overhead interpretation and the literal alternative are stated explicitly.

Model Answer

The duplicate MOH label creates two possible readings. Under the customary intended reading, the 4 row is fixed manufacturing overhead; relevant unit cost = 5.10 + 3.30 + 1 + 1.50 = 10.90.

$$\Delta\text{Profit} = 15,000(14 - 10.90) = 46,500 \text{ increase}.$$

If the printed 4 is literally an additional variable overhead cost, relevant cost is 14.90 and profit decreases by 13,500; reject. The normal 21 price is not an opportunity cost because spare capacity covers the order and regular sales do not change.

Q145 · 2020-2021 · Exam Q7(b) [17 marks]

Source: Ipe 2009-25.pdf, PDF pages 74, 75.

Glade normally sells 8,000 units at 15; capacity 10,000. Unit costs: DM 2.50, DL 3, variable MOH 0.50, fixed MOH 4.25, variable S&A 1.50, fixed S&A 2. A 2,000-unit overseas order offers 12; regular sales and total fixed costs do not change.

Required

1. Find the monthly profit effect of accepting.
2. For 500 inferior units left from last year sold through regular channels at reduced prices, identify the relevant unit cost for minimum price.

Model Answer

$$\text{Incremental cost / unit} = 2.50 + 3 + 0.50 + 1.50 = 7.50\,.$$

$$\Delta\text{Profit} = 2,000(12 - 7.50) = 9,000\,.$$

Accept if qualitative/market considerations are satisfactory. The order exactly uses spare capacity and displaces no regular sales. For the old inferior units, manufacturing cost is sunk: the relevant cost is the 1.50 variable selling/admin expense per unit, assuming no additional rework/disposal cost. Their past fixed or variable manufacturing costs do not determine a minimum incremental selling price.

Q146 · 2021-2022 · Exam Q2(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 79, 80.

Scottie makes wool yarn and sweaters; one spindle of yarn makes one sweater. Sweaters normally sell for 40, but warm weather forces a price of 30. Sweater cost: buttons/thread/lining 2; yarn 16; DL 5.80; MOH 8.70; total 32.50. A spindle of yarn sells for 20 and costs raw wool 7, DL 3.60 and MOH 5.40, total 16. MOH is allocated at 150% of DL and is entirely fixed; it will not change if sweaters are discontinued. Materials and labor are variable.

Required

1. Should the company sell yarn or process it into sweaters?
2. What is the lowest acceptable sweater price?

Q146 · 2021-2022 · Exam Q2(a) [20 marks] (answer continued)

Model Answer

Incremental analysis: one sweater	Amount
Sweater revenue	30
Yarn sales forgone	-20
Additional materials	-2
Additional labor	-5.80
Incremental profit from processing	2.20

Process the yarn: sweaters add 2.20 profit per unit despite the reported absorption loss. The yarn manufacturing cost is incurred under either alternative; fixed MOH is unchanged.

$$\text{Minimum price} = 20 + 2 + 5.80 = 27.80\,.$$

At 27.80 the alternatives have equal profit. Above it, further processing is preferable under the given assumptions.

Q147 · 2021-2022 · Exam Q3(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 81.

Benoit has strong demand for all three products. The same material costs 3 per pound; only 5,000 pounds are available monthly.

Per unit	A	B	C
Selling price	80	56	70
Direct materials	24	15	9
Other variable expenses	24	27	40
Contribution margin	32	14	21
CM ratio	40%	25%	30%

Required

1. Which orders should be accepted first, second and third?

Q147 · 2021-2022 · Exam Q3(a) [20 marks] (answer continued)

Model Answer

Product	Material pounds/unit	CM/unit	CM/pound	Priority
A	8	32	4	2
B	5	14	2.80	3
C	3	21	7	1

Accept C first, A second and B third. Rank profit contribution per scarce pound, not contribution per product or CM ratio. If C demand alone exceeds capacity, use all material for C; specific order quantities require demand limits.

Q148 · 2022-2023 · Exam Q3 [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 88, 89.

Pegasus considers dropping flight 482. A round trip has 175 seats, 40% occupancy and a ticket price of 200. Variable passenger expense = 15 per passenger. Flight-crew salaries are fixed annual salaries; assistants are paid per trip. One-third of insurance is destination-specific and avoidable. Baggage/loading costs are allocations that do not change. No replacement flight is authorized; no aircraft or flight crew can be eliminated. Depreciation is obsolescence rather than wear.

Typical round-trip income/cost	Amount
Ticket revenue	14,000
Variable passenger expenses	1,050
Contribution margin	12,950
Flight-crew salaries	1,800
Flight promotion	750
Aircraft depreciation	1,550
Fuel	5,800
Liability insurance	4,200
Flight assistants	1,500
Baggage/loading allocation	1,700

Q148 · 2022-2023 · Exam Q3 [35 marks] (question continued)

Typical round-trip income/cost	Amount
Overnight crew/assistant cost	300
Total flight expenses	17,600
Operating loss	-4,650

Required

1. Analyze the effect of dropping flight 482 on airline profit.

Model Answer

Dropping the flight saves only costs that disappear. It also loses the entire passenger contribution margin.

Avoidable cost per round trip	Saving
Promotion	750
Fuel	5,800
One-third of insurance	1,400
Flight-assistant wages	1,500
Overnight cost	300
Total savings	9,750
Lost contribution margin	-12,950
Change in airline profit	-3,200

Keep the flight under the stated conditions. Dropping it decreases company profit by 3,200 per round trip. The reported loss includes 7,850 of unavoidable allocated/committed costs, which will continue without the flight. Multiply the 3,200 decline by the number of canceled trips to obtain a period total; that number is not provided.

Q149 · 2022-2023 · Exam Q4 [35 marks]

Source: Ipe 2009-25.pdf, PDF page 90.

Walton Toy has 130,000 DLHs of capacity. Direct labor costs 8 per hour; variable MOH is 2 per DLH. Fixed costs total 520,000; all nonmanufacturing costs are fixed. Any product can be made using the existing employees/equipment.

Product	Demand units	Price	DM per unit	DL per unit
Debbie	50,000	13.50	4.30	3.20
Trish	42,000	5.50	1.10	2
Sarah	35,000	21	6.44	5.60
Mike	40,000	10	2	4
Sewing kit	325,000	8	3.20	1.60

Required

1. Compute contribution margin per direct labor-hour for each product.
2. Compute total labor-hours needed for the forecast demand.
3. Allocate the 130,000 available labor-hours to the products.

Model Answer

Convert direct labor cost per unit into hours by dividing by the wage rate. Deduct variable overhead before ranking products by contribution per scarce hour.

Product	Hours/unit	Variable OH/unit	CM/unit	CM/DLH	Demand hours
Debbie	0.40	0.80	5.20	13.00	20,000
Trish	0.25	0.50	1.90	7.60	10,500
Sarah	0.70	1.40	7.56	10.80	24,500
Mike	0.50	1.00	3.00	6.00	20,000
Sewing kit	0.20	0.40	2.80	14.00	65,000

Full demand requires 140,000 hours, exceeding capacity by 10,000. Allocate first to the highest contribution per constrained hour, stopping at each product's demand.

Ranked product	CM/hour	Allocated hours	Units produced	Contribution
Sewing kit	14.00	65,000	325,000	910,000.00

Q149 · 2022-2023 · Exam Q4 [35 marks] (answer continued)

Ranked product	CM/hour	Allocated hours	Units produced	Contribution
Debbie	13.00	20,000	50,000	260,000.00
Sarah	10.80	24,500	35,000	264,600.00
Trish	7.60	10,500	42,000	79,800.00
Mike	6.00	10,000	20,000	60,000.00

Total contribution = 1,574,400.00; less 520,000 fixed costs gives operating income 1,054,400.00. Mike receives the remaining 10,000 hours, producing 20,000 of its 40,000 demanded units. Fixed cost is unchanged across feasible mixes and does not determine the ranking.

Q150 · 2023-2024 · Exam Q3(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF pages 101, 102.

Come-Clean makes Grit 337 at 1.60 per pound and can sell it for 2.00 per pound. A jar of Sparkle polish sells for 4.00 and uses one-quarter pound of Grit. Extra ingredients cost 0.65 and direct labor 1.48 per jar. Variable MOH = 25% of direct labor; variable selling cost = 7.5% of polish sales. Monthly supervisor salary = 3,000; special mixing-equipment depreciation = 1,400 with negligible resale value and no wear from use; polish advertising = 4,000.

Required

1. Compute incremental contribution margin per jar from further processing Grit into silver polish.

Model Answer

The choice is to sell the quarter-pound of Grit or process it into polish. The original Grit manufacturing cost is common to both choices. Use the lost Grit sales revenue as the opportunity cost.

Incremental contribution per jar	Amount
Polish revenue	4
Lost Grit revenue: 0.25 x 2.00	-0.50
Extra ingredients	-0.65
Extra direct labor	-1.48

Q150 · 2023-2024 · Exam Q3(b) [15 marks] (answer continued)

Incremental contribution per jar	Amount
Variable MOH: 25% x 1.48	-0.37
Variable selling: 7.5% x 4.00	-0.30
Incremental contribution margin	0.70

Depreciation of the existing special equipment is sunk, not incremental. Monthly supervisor/advertising costs are not part of per-jar contribution; if both are avoidable, the positive 0.70 contribution must cover 7,000 monthly fixed costs before further processing increases total profit. That requires 10,000 jars, with all other assumptions unchanged.

Q151 · 2023-2024 · Exam Q4(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 102.

Portsmouth's bottleneck is upholstery time; demand exceeds capacity.

Product	Recliner	Sofa	Love seat
Selling price	1,400	1,800	1,500
Variable cost	800	1,200	1,000
Upholstery hours per unit	8	10	5

Required

1. If overtime would produce sofas, find the maximum additional payment per upholstery hour.
2. A nearby company offers high-quality love-seat upholstery at 45 per hour and works as fast as Portsmouth. Should the offer be accepted?

Model Answer

Product	Unit CM	Hours	CM per constrained hour
Recliner	600	8	75
Sofa	600	10	60
Love seat	500	5	100

Q151 · 2023-2024 · Exam Q4(a) [20 marks] (answer continued)

For the stated sofa overtime use, the maximum additional cost is 600 / 10 = 60 per hour; at exactly 60, the extra sofa contribution is fully consumed. This is an additional payment above costs already included in the unit-variable-cost data.

Accept the outside love-seat capacity at 45 per hour on the given conditions. Each love seat supplies 500 contribution before the outside charge and uses five hours; the extra charge is 225, leaving 275 additional contribution per love seat, or 55 per outsourced hour. Demand exceeds capacity, so these extra units can be sold. The statement assumes the listed baseline variable costs still apply.

Q152 · 2024-2025 · Exam Q3(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 111.

XYZ has Hardware and Linens departments. Of Linens' fixed expenses, 340,000 is sunk or allocated and will continue if Linens is dropped. Dropping Linens also reduces Hardware sales by 10%.

Contribution income statement	Total	Hardware	Linens
Sales	4,000,000	3,000,000	1,000,000
Variable expenses	1,300,000	900,000	400,000
Contribution margin	2,700,000	2,100,000	600,000
Fixed expenses	2,200,000	1,400,000	800,000
Operating income (loss)	500,000	700,000	-200,000

Required

1. Determine the effect of dropping Linens on the company's total net operating income.

Model Answer

Ignore the allocated loss as a decision rule; compare the contribution lost with the fixed costs actually avoided. Hardware's contribution ratio is 2,100,000 / 3,000,000 = 70%.

Incremental effect	Amount
Avoidable Linens fixed costs: 800,000 - 340,000	460,000
Lost Linens contribution margin	-600,000

Q152 · 2024-2025 · Exam Q3(b) [15 marks] (answer continued)

Incremental effect	Amount
Lost Hardware contribution: 10% x 3,000,000 x 70%	-210,000
Change in total operating income	-350,000
New company operating income: 500,000 - 350,000	150,000

Keep Linens under the stated conditions. Dropping it lowers total company profit by 350,000 even though its reported departmental statement shows a loss.

Q153 · 2024-2025 · Exam Q4(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 112.

Andretti sells 60,000 Daks annually at 32 each. Capacity is 90,000 units without additional fixed manufacturing overhead. A proposed 80,000 increase in fixed selling costs would increase sales by 25%.

Unit cost	Amount
Direct materials	10
Direct labor	4.50
Variable MOH	2.30
Fixed MOH	5
Variable selling	1.20
Fixed selling	3.50
Total	26.50

Fixed MOH is 300,000 total; fixed selling costs are 210,000 total.

Required

1. Would the additional fixed selling expenditure be justified?

Model Answer

Existing fixed costs continue and are not incremental. Variable cost of an extra unit is 10 + 4.50 + 2.30 + 1.20 = 18.00; extra contribution is 32 - 18 = 14.

$$\text{Extra units} = 60,000 \times 25\% = 15,000\,.$$

Q153 · 2024-2025 · Exam Q4(b) [15 marks] (answer continued)

$$\Delta\text{NOI} = 15,000 \times 14 - 80,000 = 130,000\,.$$

Accept the proposal on the stated data. Total sales become 75,000 units, within the 90,000-unit capacity, and annual operating income increases by 130,000.

Master budgets and cash planning

Q154 · 2009-2010 · Exam Q8(a) [8 marks]

Source: Ipe 2009-25.pdf, PDF page 8.

Distinguish continuous and self-imposed budgets.

Model Answer

Concept	Meaning	Example
Continuous/rolling budget	Adds a new period as the oldest ends to maintain a constant planning horizon	Update a 12-month budget each month
Self-imposed/participative budget	Responsible managers contribute to preparing targets	Department managers estimate their own resources for review

They address different dimensions: updating horizon versus who prepares it. A company can use a participative rolling budget at the same time.

Q155 · 2009-2010 · Exam Q8(b) [27 marks]

Source: Ipe 2009-25.pdf, PDF pages 8, 9.

Prepare a third-quarter retail cash plan. June 30 cash 8,000, RM/merchandise inventory 18,000, inventory AP 11,700. Sales are 20% cash, 80% credit; credit collections 10% current month, 70% next, 20% second following. May/June sales 30,000 / 36,000. Ending inventory = 75% of next month COGS; purchases paid 50% current and 50% next. Monthly operating expenses include 2,000 depreciation. Buy land 4,500 in July; pay dividends 1,000 in September. Minimum cash 8,000; loans/principal repayments in 1,000 multiples, borrow beginning month and repay at end. Annual interest 12%, using whole months.

Budgeted income	July	August	September	October
Sales	40,000	70,000	50,000	45,000
COGS	24,000	42,000	30,000	27,000
Selling expense	7,200	11,700	8,500	7,300
Admin expense	5,600	7,200	6,100	5,900
Total operating expense	12,800	18,900	14,600	13,200

Q155 · 2009-2010 · Exam Q8(b) [27 marks] (question continued)

Budgeted income	July	August	September	October
Net income	3,200	9,100	5,400	4,800

Required

1. Prepare monthly/quarter collections.
2. Prepare inventory purchases and payments.
3. Prepare monthly/quarter cash budget.

Model Answer

Collections	July	August	September	Quarter
Cash sales: 20%	8,000	14,000	10,000	32,000
Current credit: 8% total sales	3,200	5,600	4,000	12,800
Prior month credit: 56% total sales	20,160	22,400	39,200	81,760
Second prior month: 16%	4,800	5,760	6,400	16,960
Total	36,160	47,760	59,600	143,520

Inventory budget	July	August	September	Quarter
COGS	24,000	42,000	30,000	96,000
Desired ending inventory	31,500	22,500	20,250	20,250
Beginning inventory	18,000	31,500	22,500	18,000
Purchases	37,500	33,000	27,750	98,250
Cash paid	30,450	35,250	30,375	96,075

Cash budget	July	August	September	Quarter
Beginning cash	8,000	8,410	8,020	8,000
Collections	36,160	47,760	59,600	143,520
Available cash	44,160	56,170	67,620	151,520
Inventory payments	30,450	35,250	30,375	96,075
Cash operating expenses	10,800	16,900	12,600	40,300

Q155 · 2009-2010 · Exam Q8(b) [27 marks] (answer continued)

Cash budget	July	August	September	Quarter
Land	4,500	0	0	4,500
Dividends	0	0	1,000	1,000
Total disbursements before financing	45,750	52,150	43,975	141,875
Cash before financing	-1,590	4,020	23,645	9,645
Borrowing	10,000	4,000	0	14,000
Principal repayment	0	0	14,000	14,000
Interest	0	0	380	380
Ending cash	8,410	8,020	9,265	9,265

September interest = 10,000 x 1% x 3 months + 4,000 x 1% x 2 = 380. Repaying all 14,000 principal leaves 9,265, above the minimum. Quarter cash uses the opening July and closing September balances; intermediate balances cancel. Inventory AP ends 13,875, so payments reconcile: 98,250 + 11,700 - 13,875 = 96,075.

Q156 · 2012-2013 · Exam Q2(a) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 18.

What are advantages of a self-imposed budget?

Model Answer

A self-imposed or participative budget is prepared with input from managers responsible for carrying it out. It uses their operating knowledge, encourages ownership and effort, improves communication, and produces more realistic commitments. Management still coordinates targets and reviews assumptions to prevent slack and local goals that conflict with strategy.

Q157 · 2012-2013 · Exam Q4(b) [25 marks]

Source: Ipe 2009-25.pdf, PDF pages 19, 20.

Crydon swim-fin price 50/pair. Sales forecast July 6,000; August 7,000; September 5,000; October 4,000; November/December 3,000 each. Collections 40% current, 50% next month, 10% bad; beginning collectible AR 130,000. FG target 10% next sales; beginning FG 600. Material 2 lb/pair at 2.50; RM target 20% next production needs; beginning RM 2,440 lb. Pay 60% current, 40% next; opening AP 11,400.

Required

1. Prepare third-quarter monthly/total sales and collections.
2. Prepare production July-October.
3. Prepare third-quarter materials purchases and cash payments.

Model Answer

Sales/collections	July	August	September	Quarter
Sales pairs	6,000	7,000	5,000	18,000
Sales dollars	300,000	350,000	250,000	900,000
Current collections	120,000	140,000	100,000	360,000
Prior receivables	130,000	150,000	175,000	455,000
Total collections	250,000	290,000	275,000	815,000

First derive production from sales and the finished-goods policy. Raw-material purchases depend on production needs, not directly on sales.

Production budget	July	August	September	Quarter
Budgeted sales	6,000	7,000	5,000	18,000
Desired ending FG	700	500	400	400
Beginning FG	600	700	500	600
Required production	6,100	6,800	4,900	17,800

The month after the quarter is needed to determine the final raw-material inventory. Its production is 3,900 units and its materials need is 7,800.

Q157 · 2012-2013 · Exam Q4(b) [25 marks] (answer continued)

Direct-materials purchases (quantity)	July	August	September	Quarter
Materials for production	12,200	13,600	9,800	35,600
Desired ending materials	2,720	1,960	1,560	1,560
Beginning materials	2,440	2,720	1,960	2,440
Materials to purchase	12,480	12,840	9,400	34,720

For the quarter column, use the quarter-opening and quarter-closing inventory balances. Do not add the internal month-to-month inventory balances; those transfers cancel.

October production = 4,000 + 300 - 400 = 3,900 pairs.

Material cost and payments	July	August	September	Quarter
Purchases, lb	12,480	12,840	9,400	34,720
Purchases at 2.50	31,200	32,100	23,500	86,800
Cash payments	30,120	31,740	26,940	88,800

Closing AP = 40% x 23,500 = 9,400; 86,800 + 11,400 - 9,400 = 88,800 cash paid.

Q158 · 2013-2014 · Exam Q4(a) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 26.

Mention advantages of budgeting.

Model Answer

1. Makes managers plan ahead and test assumptions.
2. Coordinates sales, production, purchasing, staffing and financing.
3. Identifies shortages and capacity conflicts before they occur.
4. Communicates goals and assigns responsibility.
5. Provides a basis for performance comparison and corrective action.

Benefits depend on realistic assumptions, participation and comparison at the appropriate actual activity level.

Q159 · 2013-2014 · Exam Q4(b) [$2 + 4$ marks]

Source: Ipe 2009-25.pdf, PDF page 26.

Define a master budget and show the interrelations among its component budgets.

Model Answer

A master budget is an integrated set of operating and financial budgets for a planning period. The sales budget usually initiates it. It coordinates planned operations, resource purchases, financing and the expected financial statements.



For a retailer, a merchandise-purchases budget replaces the production budgets. Ending balances in one budget become beginning balances in the next period; the schedules must agree.

Q160 · 2013-2014 · Exam Q8(d) [8 marks]

Source: Ipe 2009-25.pdf, PDF page 30.

Mention benefits and limitations of participative budgets.

Q160 · 2013-2014 · Exam Q8(d) [8 marks] (answer continued)

Model Answer

Benefits	Limitations
Uses local operating knowledge	Employees may build budget slack
Improves ownership, communication and motivation	Negotiation and consolidation take time
Creates more credible, attainable targets	Departments may optimize locally rather than company-wide

Senior managers should review assumptions, coordinate departments, align budgets with strategy and avoid treating every estimate as an inflexible performance promise.

Q161 · 2014-2015 · Exam Q4(b) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 34.

Hareston quarterly production 7,000; 8,000; 6,000; 5,000. Each unit requires 2 lb material at 1.40; beginning RM 1,400 lb; beginning AP 2,940. Ending RM target 10% next-quarter need, except Q4 ending target 1,500 lb. Pay 80% current purchases, 20% following quarter. DL 0.60 h/unit at 14; labor workforce adjusts to required hours.

Required

1. Prepare quarterly/annual materials and cash-payment budgets.
2. Prepare labor budget.

Model Answer

Materials/payments	Q1	Q2	Q3	Q4	Year
Production material needs	14,000	16,000	12,000	10,000	52,000
Desired ending RM	1,600	1,200	1,000	1,500	1,500
Beginning RM	1,400	1,600	1,200	1,000	1,400
Purchases, lb	14,200	15,600	11,800	10,500	52,100
Purchase cost	19,880	21,840	16,520	14,700	72,940
Cash payments	18,844	21,448	17,584	15,064	72,940

Q161 · 2014-2015 · Exam Q4(b) [25 marks] (answer continued)

Labor budget	Q1	Q2	Q3	Q4	Year
Hours: units x 0.60	4,200	4,800	3,600	3,000	15,600
Wage cost at 14	58,800	67,200	50,400	42,000	218,400

Q1 payments = beginning payable 2,940 + 80% of 19,880. Subsequent quarters pay 20% of prior purchases + 80% current. Ending AP is 2,940, equal to opening AP, so annual payments equal purchases.

Q162 · 2016-2017 · Exam Q6(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 44.

Fill this quarterly retail cash budget (amounts in thousands). Minimum beginning cash each quarter is 5; annual interest paid is 1.

Line	Q1	Q2	Q3	Q4	Year
Beginning cash	6	?	?	?	?
Collections	?	?	96	?	323
Available cash	71	?	?	?	?
Inventory purchases	35	45	?	35	?
Operating expenses	?	30	30	?	113
Equipment	8	8	10	?	36
Dividends	2	2	2	2	?
Total payments	?	85	?	?	?
Excess/deficiency	-2	?	11	?	?
Borrowing	?	15	0	0	?
Repayment incl. interest	0	0	?	-17	?
Total financing	?	?	?	?	?
Ending cash	?	?	?	?	?

Source clarification: The standard minimum-cash policy is interpreted as borrowing only what is needed and repaying surplus debt; otherwise the blanks are not all uniquely determined.

Q162 · 2016-2017 · Exam Q6(a) [20 marks] (answer continued)

Model Answer

Completed cash budget (000)	Q1	Q2	Q3	Q4	Year
Beginning cash	6	5	5	5	6
Collections	65	70	96	92	323
Available cash	71	75	101	97	329
Inventory purchases	35	45	48	35	163
Operating expenses	28	30	30	25	113
Equipment purchases	8	8	10	10	36
Dividends	2	2	2	2	8
Total payments	73	85	90	72	320
Excess/deficiency	-2	-10	11	25	9
Borrowing	7	15	0	0	22
Repayments incl. interest	0	0	-6	-17	-23
Total financing	7	15	-6	-17	-1
Ending cash	5	5	5	8	8

Borrow just enough to retain the 5 minimum and repay available surplus: Q1 7; Q2 15; Q3 repayment 6. Q4 repayment 17 clears the remaining 16 principal plus annual 1 interest. Annual beginning/ending cash are 6 and 8, not the sum of quarterly cash balances. Check 6 + 323 - 320 + 22 - 23 = 8.

Q163 · 2016-2017 · Exam Q6(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 44.

What is a master budget? Illustrate with a suitable diagram.

Model Answer

A master budget is an integrated set of operating and financial budgets for a planning period. The sales budget usually initiates it. It coordinates planned operations, resource purchases, financing and the expected financial statements.

Q163 · 2016-2017 · Exam Q6(b) [15 marks] (answer continued)



For a retailer, a merchandise-purchases budget replaces the production budgets. Ending balances in one budget become beginning balances in the next period; the schedules must agree.

Q164 · 2017-2018 · Exam Q2(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 48.

Define a master budget and explain its importance with a suitable diagram.

Model Answer

A master budget is an integrated set of operating and financial budgets for a planning period. The sales budget usually initiates it. It coordinates planned operations, resource purchases, financing and the expected financial statements.

Q164 · 2017-2018 · Exam Q2(b) [15 marks] (answer continued)



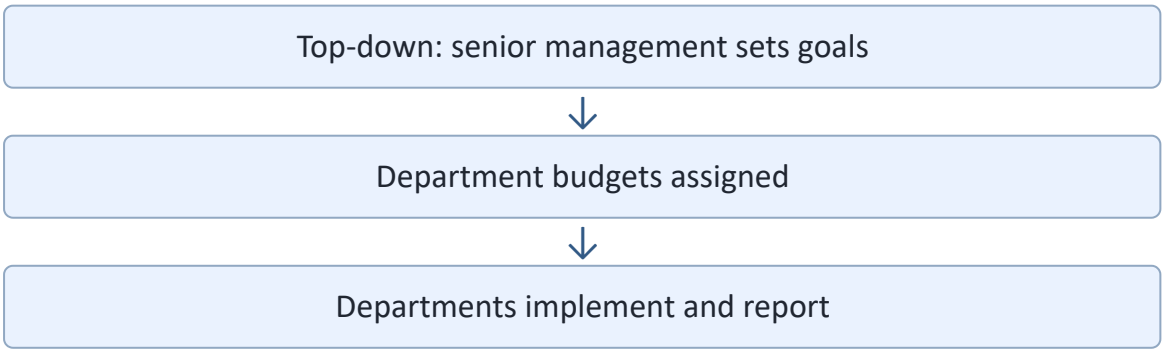
For a retailer, a merchandise-purchases budget replaces the production budgets. Ending balances in one budget become beginning balances in the next period; the schedules must agree.

Q165 · 2018-2019 · Exam Q1(b) [10 marks]

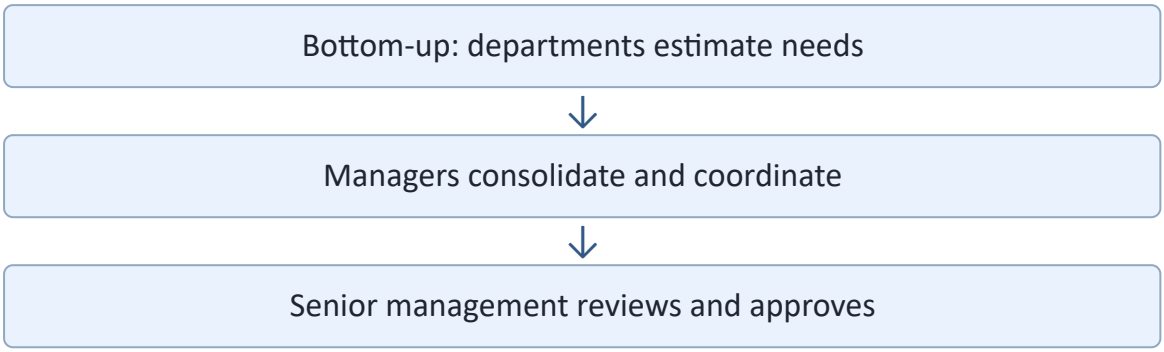
Source: Ipe 2009-25.pdf, PDF page 55.

Distinguish top-down from bottom-up budget preparation with diagrams.

Model Answer



Q165 · 2018-2019 · Exam Q1(b) [10 marks] (answer continued)



Aspect	Top-down	Bottom-up
Strength	Fast, strategically consistent	Local knowledge, ownership and motivation
Risk	Unrealistic targets; weak employee commitment	Budget slack, negotiations and slower preparation
Good control	Consult operating managers; test feasibility	Challenge assumptions and reconcile strategic targets

A participative hybrid combines strategic direction with realistic operating input and final management review.

Q166 · 2018-2019 · Exam Q4(a) [25 marks]

Source: Ipe 2009-25.pdf, PDF pages 57, 58.

Mynor sells at 8. Quarterly sales Year 2: 40,000; 60,000; 100,000; 50,000; Year 3 Q1/Q2: 70,000 / 80,000. Collect 75% currently, 25% next quarter; beginning AR 65,000 collected Q1. FG target 30% next-quarter sales; beginning FG 12,000. Each unit needs 5 lb material at 0.80/lb. RM target 10% next-quarter production needs; beginning RM 23,000 lb. Pay 60% currently, 40% next quarter; beginning AP 81,500.

Required

1. Prepare sales and collections budgets.
2. Prepare production budget.
3. Prepare materials and payment budgets, with annual totals.

Model Answer

Sales and collections	Q1	Q2	Q3	Q4	Year
Units	40,000	60,000	100,000	50,000	250,000

Q166 · 2018-2019 · Exam Q4(a) [25 marks] (answer continued)

Model Answer

Sales and collections	Q1	Q2	Q3	Q4	Year
Sales	320,000	480,000	800,000	400,000	2,000,000
75% current sales	240,000	360,000	600,000	300,000	1,500,000
Prior AR / 25% prior sales	65,000	80,000	120,000	200,000	465,000
Collections	305,000	440,000	720,000	500,000	1,965,000

First derive production from sales and the finished-goods policy. Raw-material purchases depend on production needs, not directly on sales.

Production budget	Q1	Q2	Q3	Q4	Year
Budgeted sales	40,000	60,000	100,000	50,000	250,000
Desired ending FG	18,000	30,000	15,000	21,000	21,000
Beginning FG	12,000	18,000	30,000	15,000	12,000
Required production	46,000	72,000	85,000	56,000	259,000

The quarter after the year is needed to determine the final raw-material inventory. Its production is 73,000 units and its materials need is 365,000.

Direct-materials purchases (quantity)	Q1	Q2	Q3	Q4	Year
Materials for production	230,000	360,000	425,000	280,000	1,295,000
Desired ending materials	36,000	42,500	28,000	36,500	36,500
Beginning materials	23,000	36,000	42,500	28,000	23,000
Materials to purchase	243,000	366,500	410,500	288,500	1,308,500

For the year column, use the year-opening and year-closing inventory balances. Do not add the internal quarter-to-quarter inventory balances; those transfers cancel.

Materials purchases/payments	Q1	Q2	Q3	Q4	Year
Pounds purchased	243,000	366,500	410,500	288,500	1,308,500
Cost at 0.80	194,400	293,200	328,400	230,800	1,046,800
Cash paid	198,140	253,680	314,320	269,840	1,035,980

Q166 · 2018-2019 · Exam Q4(a) [25 marks] (answer continued)

Year 3 Q1 production = 70,000 + 24,000 - 21,000 = 73,000, giving Q4 ending RM of 36,500 lb. Ending AP = 92,320; cash payments check: 1,046,800 + 81,500 - 92,320 = 1,035,980.

Q167 · January 2020 Term · Exam Q3(b) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 65.

How does a master budget help an industrial manager make decisions? Relate its components to a balanced scorecard and justify.

Model Answer

A master budget is an integrated set of operating and financial budgets for a planning period. The sales budget usually initiates it. It coordinates planned operations, resource purchases, financing and the expected financial statements.



For a retailer, a merchandise-purchases budget replaces the production budgets. Ending balances in one budget become beginning balances in the next period; the schedules must agree.

Q167 · January 2020 Term · Exam Q3(b) [20 marks] (answer continued)

A master budget coordinates sales, capacity, staffing, purchases and financing; identifies cash shortages in advance; sets accountable targets; and allows actual results to be compared at the appropriate activity level.

Scorecard perspective	Budget connection	Complementary measure
Financial	Cash, income and balance-sheet budgets	ROI, operating cash flow, margin
Customer	Sales, service and marketing budgets	Retention, satisfaction, on-time delivery
Internal processes	Production/material/labor/overhead budgets	Yield, throughput, defects
Learning and growth	Training/technology spending	Skills, employee development, system reliability

The budget supplies quantified resource plans, while the scorecard links financial and nonfinancial results to strategy. Spending within budget is not sufficient if customer service, quality or learning deteriorates.

Q168 · 2020-2021 · Exam Q2 [35 marks]

Source: Ipe 2009-25.pdf, PDF page 70.

Each bottle of Mink Caress needs 3 grams of musk oil costing 150 roubles per kilogram. Year 2 quarterly production: 60,000; 90,000; 150,000; 100,000 bottles. Year 3 Q1 production: 70,000. Ending oil inventory is 20% of next quarter production needs; beginning oil is 36,000 grams.

Required

1. Prepare the quarterly and annual direct-materials budget, including purchase cost in roubles.

Model Answer

$$150 \text{ roubles / kg} = 0.15 \text{ rouble / gram}.$$

Direct-materials budget	Q1	Q2	Q3	Q4	Year
Production bottles	60,000	90,000	150,000	100,000	400,000
Grams needed	180,000	270,000	450,000	300,000	1,200,000
Desired ending grams	54,000	90,000	60,000	42,000	42,000

Q168 · 2020-2021 · Exam Q2 [35 marks] (answer continued)

Direct-materials budget	Q1	Q2	Q3	Q4	Year
Beginning grams	36,000	54,000	90,000	60,000	36,000
Grams purchased	198,000	306,000	420,000	282,000	1,206,000
Purchase cost, roubles	29,700.00	45,900.00	63,000.00	42,300.00	180,900.00

Annual purchases are 1,200,000 production grams + 42,000 ending - 36,000 opening = 1,206,000 grams; annual cost = 180,900 roubles. Do not multiply grams directly by the kilogram price.

Q169 · 2021-2022 · Exam Q1(a) [18 marks]

Source: Ipe 2009-25.pdf, PDF pages 78, 79.

Milo sells beach umbrellas for 12 each, all on credit. Forecast sales units: July 30,000; August 70,000; September 50,000; October 20,000; November and December 10,000 each. June sales were 300,000. Collections: 30% in the sale month, 65% next month; 5% uncollectible. Finished-goods inventory is 15% of next month sales; June met the target. Each umbrella requires 4 feet of Gilden at 0.80 per foot. Ending materials are 50% of next month production needs. Purchases are paid 50% currently and 50% next month; beginning July accounts payable is 76,000.

Required

1. Prepare the July-September sales budget and schedule of expected cash collections, with quarter totals.

Model Answer

Sales budget	July	August	September	Quarter
Units	30,000	70,000	50,000	150,000
Sales at 12	360,000	840,000	600,000	1,800,000

Cash collections	July	August	September	Quarter
Current-month sales: 30%	108,000	252,000	180,000	540,000
Prior-month sales: 65%	195,000	234,000	546,000	975,000
Total collections	303,000	486,000	726,000	1,515,000

Q169 · 2021-2022 · Exam Q1(a) [18 marks] (answer continued)

July includes 65% of June sales. The 5% bad-debt portion is never collected; it is not a cash payment.

Q170 · 2021-2022 · Exam Q1(b) [17 marks]

Source: Ipe 2009-25.pdf, PDF pages 78, 79.

Milo sells beach umbrellas for 12 each, all on credit. Forecast sales units: July 30,000; August 70,000; September 50,000; October 20,000; November and December 10,000 each. June sales were 300,000. Collections: 30% in the sale month, 65% next month; 5% uncollectible. Finished-goods inventory is 15% of next month sales; June met the target. Each umbrella requires 4 feet of Gilden at 0.80 per foot. Ending materials are 50% of next month production needs. Purchases are paid 50% currently and 50% next month; beginning July accounts payable is 76,000.

Required

1. Prepare the direct-materials budget and expected cash-disbursement schedule for July-September.

> **Source clarification:** The printed beginning/ending materials line mistakenly repeats collection percentages. Materials balances are derived from the stated inventory policy, assuming June met that policy: beginning July 72,000 feet; end September 37,000 feet.

Model Answer

First derive production from sales and the finished-goods policy. Raw-material purchases depend on production needs, not directly on sales.

Production budget	July	August	September	Quarter
Budgeted sales	30,000	70,000	50,000	150,000
Desired ending FG	10,500	7,500	3,000	3,000
Beginning FG	4,500	10,500	7,500	4,500
Required production	36,000	67,000	45,500	148,500

The month after the quarter is needed to determine the final raw-material inventory. Its production is 18,500 units and its materials need is 74,000.

Q170 · 2021-2022 · Exam Q1(b) [17 marks] (answer continued)

Direct-materials purchases (quantity)	July	August	September	Quarter
Materials for production	144,000	268,000	182,000	594,000
Desired ending materials	134,000	91,000	37,000	37,000
Beginning materials	72,000	134,000	91,000	72,000
Materials to purchase	206,000	225,000	128,000	559,000

For the quarter column, use the quarter-opening and quarter-closing inventory balances. Do not add the internal month-to-month inventory balances; those transfers cancel.

Purchases and payments	July	August	September	Quarter
Feet purchased	206,000	225,000	128,000	559,000
Cost at 0.80	164,800	180,000	102,400	447,200
50% current purchases	82,400	90,000	51,200	223,600
Prior-month payable	76,000	82,400	90,000	248,400
Cash payments	158,400	172,400	141,200	472,000

Ending payable = 51,200. Check: 447,200 purchases + 76,000 beginning payable - 51,200 ending payable = 472,000 cash paid.

Q171 · 2022-2023 · Exam Q1(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 87.

Zan forecasts quarterly production of 5,000, 8,000, 7,000 and 6,000 units. Each unit needs 8 grams of material at 1.20 per gram. Beginning Q1 materials = 6,000 grams; beginning accounts payable = 2,880. Desired ending materials = 25% of the following quarter's production need; Q4 desired ending materials = 8,000 grams. Payments are 60% in the purchase quarter and 40% next quarter. Each unit needs 0.20 labor-hour at 11.50 per hour.

Required

1. Prepare the direct-materials budget.

Model Answer

$$\text{Purchases} = \text{Production needs} + \text{Desired ending RM} - \text{Beginning RM}.$$

Q171 · 2022-2023 · Exam Q1(a) [20 marks] (answer continued)

Direct-materials budget	Q1	Q2	Q3	Q4	Year
Production units	5,000	8,000	7,000	6,000	26,000
Grams needed	40,000	64,000	56,000	48,000	208,000
Desired ending grams	16,000	14,000	12,000	8,000	8,000
Beginning grams	6,000	16,000	14,000	12,000	6,000
Grams to purchase	50,000	62,000	54,000	44,000	210,000
Price per gram	1.20	1.20	1.20	1.20	1.20
Purchase cost	60,000.00	74,400.00	64,800.00	52,800.00	252,000.00

The annual inventory calculation uses 6,000 opening grams and 8,000 closing grams. The intermediate inventory balances cancel; annual purchases are 208,000 production grams + 8,000 - 6,000 = 210,000 grams, costing 252,000.

Q172 · 2023-2024 · Exam Q1(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 98.

Weller's budgeted unit sales are 15,000, 16,000, 14,000 and 13,000 for the four quarters. Variable S&A is 2.50 per unit. Quarterly advertising = 8,000; executive salaries = 35,000; depreciation = 20,000. Insurance payments are 5,000 in Q1 and Q3; property tax of 8,000 is paid in Q2.

Required

1. Prepare the annual selling and administrative expense budget.

Model Answer

Use budgeted units sold for variable selling/admin expense. The scheduled insurance and property-tax amounts are included in the stated quarters; no separate coverage/accrual schedule is provided.

S&A budget	Q1	Q2	Q3	Q4	Year
Units sold	15,000	16,000	14,000	13,000	58,000
Variable S&A	37,500.00	40,000.00	35,000.00	32,500.00	145,000.00
Advertising	8,000	8,000	8,000	8,000	32,000

Q172 · 2023-2024 · Exam Q1(a) [20 marks] (answer continued)

S&A budget	Q1	Q2	Q3	Q4	Year
Executive salaries	35,000	35,000	35,000	35,000	140,000
Depreciation	20,000	20,000	20,000	20,000	80,000
Insurance	5,000	0	5,000	0	10,000
Property tax	0	8,000	0	0	8,000
Total S&A	105,500.00	111,000.00	103,000.00	95,500.00	415,000.00
Less noncash depreciation	20,000	20,000	20,000	20,000	80,000
Cash S&A payments	85,500.00	91,000.00	83,000.00	75,500.00	335,000.00

Depreciation is an expense but not a cash payment. If a coverage schedule for prepaid insurance or accrued tax were supplied, the expense budget and cash-payment schedule would need separate timing.

Q173 · 2024-2025 · Exam Q1(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 108.

Pearl Products Limited requires 3 cc of solvent H300 per unit of Supermix. Desired month-end finished goods = 3,000 units plus 20% of the next month's sales; June 30 finished goods = 10,000 units. Desired month-end raw materials = one-half of the following month's production requirements; June 30 raw materials = 54,000 cc. There is no work in process.

Month	Budgeted sales (units)
July	35,000
August	40,000
September	50,000
October	30,000
November	20,000
December	10,000

Required

1. Prepare a direct-materials budget showing cc of solvent to purchase in July, August and September and the quarter total.

Q173 · 2024-2025 · Exam Q1(a) [20 marks] (answer continued)

Model Answer

First derive production from sales and the finished-goods policy. Raw-material purchases depend on production needs, not directly on sales.

Production budget	July	August	September	Quarter
Budgeted sales	35,000	40,000	50,000	125,000
Desired ending FG	11,000	13,000	9,000	9,000
Beginning FG	10,000	11,000	13,000	10,000
Required production	36,000	42,000	46,000	124,000

The month after the quarter is needed to determine the final raw-material inventory. Its production is 28,000 units and its materials need is 84,000.

Direct-materials purchases (quantity)	July	August	September	Quarter
Materials for production	108,000	126,000	138,000	372,000
Desired ending materials	63,000	69,000	42,000	42,000
Beginning materials	54,000	63,000	69,000	54,000
Materials to purchase	117,000	132,000	111,000	360,000

For the quarter column, use the quarter-opening and quarter-closing inventory balances. Do not add the internal month-to-month inventory balances; those transfers cancel.

Flexible budgets and performance reports

Q174 · 2020-2021 · Exam Q6 [35 marks]

Source: Ipe 2009-25.pdf, PDF pages 73, 74.

Milano Pizza budgets 1,200 pizzas at 13.50 and 180 deliveries. Actual pizzas 1,240; deliveries 174.

Cost	Fixed/month	Per pizza	Per delivery	Actual
Ingredients	0	3.80	0	4,985
Kitchen staff	5,220	0	0	5,281
Utilities	630	0.05	0	984
Delivery person	0	0	3.50	609
Delivery vehicle	540	0	1.50	655
Depreciation	275	0	0	275
Rent	1,830	0	0	1,830
Miscellaneous	820	0.15	0	954

Actual revenue is 17,420.

Required

1. Prepare the performance report with activity and revenue/spending variances.
2. Explain the activity variances.

Model Answer

The planning budget uses planned activity. The flexible budget applies the same cost formulas to actual activity; it does not replace formula rates with actual costs.

Driver	Planned	Actual
Deliveries	180	174
Pizzas	1,200	1,240

Budget line	Planning	Flexible
Revenue	16,200.00	16,740.00

Q174 · 2020-2021 · Exam Q6 [35 marks] (answer continued)

Budget line	Planning	Flexible
Ingredients	4,560.00	4,712.00
Kitchen staff	5,220.00	5,220.00
Utilities	690.00	692.00
Delivery person	630.00	609.00
Delivery vehicle	810.00	801.00
Depreciation	275.00	275.00
Rent	1,830.00	1,830.00
Miscellaneous	1,000.00	1,006.00
Total expenses	15,015.00	15,145.00
Net operating income	1,185.00	1,595.00

Activity variances compare flexible with planning budgets. Revenue/spending variances compare actual with flexible budgets. The F/U designation measures whether the change improves or reduces profit.

Line	Planning	Flexible	Actual	Activity variance	Revenue/spending
Revenue	16,200.00	16,740.00	17,420.00	540.00 F	680.00 F
Ingredients	4,560.00	4,712.00	4,985.00	152.00 U	273.00 U
Kitchen staff	5,220.00	5,220.00	5,281.00	0.00 -	61.00 U
Utilities	690.00	692.00	984.00	2.00 U	292.00 U
Delivery person	630.00	609.00	609.00	21.00 F	0.00 -
Delivery vehicle	810.00	801.00	655.00	9.00 F	146.00 F
Depreciation	275.00	275.00	275.00	0.00 -	0.00 -
Rent	1,830.00	1,830.00	1,830.00	0.00 -	0.00 -
Miscellaneous	1,000.00	1,006.00	954.00	6.00 U	52.00 F
Total expenses	15,015.00	15,145.00	15,573.00	130.00 U	428.00 U
Net operating income	1,185.00	1,595.00	1,847.00	410.00 F	252.00 F

Q174 · 2020-2021 · Exam Q6 [35 marks] (answer continued)

Forty more pizzas increase expected revenue by 540 and pizza-related costs by 160. Six fewer deliveries decrease expected delivery costs by 30. Net activity effect on profit = 540 - 160 + 30 = 410 F. These are volume effects, not evidence that actual costs were well or badly controlled.

Q175 · 2021-2022 · Exam Q4 [10 + 10 + 15 marks]

Source: Ipe 2009-25.pdf, PDF pages 81, 82.

Harrald's Fish House plans 1,800 meals; actual activity is 1,700 meals.

Line	Monthly formula	Actual
Revenue	16.50 x meals	27,920
Ingredients	6.25 x meals	11,110
Wages	10,400	10,130
Utilities	800 + 0.20 x meals	1,080
Rent	2,200	2,200
Miscellaneous	600 + 0.80 x meals	2,240

Required

1. Prepare the planning budget.
2. Prepare the flexible budget.
3. Prepare the flexible-budget performance report.

Model Answer

The planning budget uses planned activity. The flexible budget applies the same cost formulas to actual activity; it does not replace formula rates with actual costs.

Driver	Planned	Actual
Meals	1,800	1,700

Budget line	Planning	Flexible
Revenue	29,700.00	28,050.00
Ingredients	11,250.00	10,625.00

Q175 · 2021-2022 · Exam Q4 [10 + 10 + 15 marks] (answer continued)

Budget line	Planning	Flexible
Wages	10,400.00	10,400.00
Utilities	1,160.00	1,140.00
Rent	2,200.00	2,200.00
Miscellaneous	2,040.00	1,960.00
Total expenses	27,050.00	26,325.00
Net operating income	2,650.00	1,725.00

Activity variances compare flexible with planning budgets. Revenue/spending variances compare actual with flexible budgets. The F/U designation measures whether the change improves or reduces profit.

Line	Planning	Flexible	Actual	Activity variance	Revenue/spending
Revenue	29,700.00	28,050.00	27,920.00	1,650.00 U	130.00 U
Ingredients	11,250.00	10,625.00	11,110.00	625.00 F	485.00 U
Wages	10,400.00	10,400.00	10,130.00	0.00 -	270.00 F
Utilities	1,160.00	1,140.00	1,080.00	20.00 F	60.00 F
Rent	2,200.00	2,200.00	2,200.00	0.00 -	0.00 -
Miscellaneous	2,040.00	1,960.00	2,240.00	80.00 F	280.00 U
Total expenses	27,050.00	26,325.00	26,760.00	725.00 F	435.00 U
Net operating income	2,650.00	1,725.00	1,160.00	925.00 U	565.00 U

Q176 · 2022-2023 · Exam Q2(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 88.

Via Gelato planned 6,000 liters but actually sold 6,200 in June.

Line	Fixed per month	Per liter	Actual June
Revenue	0	12	71,540
Raw materials	0	4.65	29,230
Wages	5,600	1.40	13,860

Q176 · 2022-2023 · Exam Q2(b) [15 marks] (question continued)

Line	Fixed per month	Per liter	Actual June
Utilities	1,630	0.20	3,270
Rent	2,600	0	2,600
Insurance	1,350	0	1,350
Miscellaneous	650	0.35	2,590

Required

1. Prepare a report of revenue and spending variances for June.

Model Answer

The planning budget uses planned activity. The flexible budget applies the same cost formulas to actual activity; it does not replace formula rates with actual costs.

Driver	Planned	Actual
Liters sold	6,000	6,200

Budget line	Planning	Flexible
Revenue	72,000.00	74,400.00
Raw materials	27,900.00	28,830.00
Wages	14,000.00	14,280.00
Utilities	2,830.00	2,870.00
Rent	2,600.00	2,600.00
Insurance	1,350.00	1,350.00
Miscellaneous	2,750.00	2,820.00
Total expenses	51,430.00	52,750.00
Net operating income	20,570.00	21,650.00

Activity variances compare flexible with planning budgets. Revenue/spending variances compare actual with flexible budgets. The F/U designation measures whether the change improves or reduces profit.

Q176 · 2022-2023 · Exam Q2(b) [15 marks] (answer continued)

Line	Planning	Flexible	Actual	Activity variance	Revenue/spending
Revenue	72,000.00	74,400.00	71,540.00	2,400.00 F	2,860.00 U
Raw materials	27,900.00	28,830.00	29,230.00	930.00 U	400.00 U
Wages	14,000.00	14,280.00	13,860.00	280.00 U	420.00 F
Utilities	2,830.00	2,870.00	3,270.00	40.00 U	400.00 U
Rent	2,600.00	2,600.00	2,600.00	0.00 -	0.00 -
Insurance	1,350.00	1,350.00	1,350.00	0.00 -	0.00 -
Miscellaneous	2,750.00	2,820.00	2,590.00	70.00 U	230.00 F
Total expenses	51,430.00	52,750.00	52,900.00	1,320.00 U	150.00 U
Net operating income	20,570.00	21,650.00	18,640.00	1,080.00 F	3,010.00 U

Q177 · 2023-2024 · Exam Q2(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 100.

Via Gelato planned 6,000 liters but actually sold 6,200 in June.

Line	Fixed per month	Per liter	Actual June
Revenue	0	12	71,540
Raw materials	0	4.65	29,230
Wages	5,600	1.40	13,860
Utilities	1,630	0.20	3,270
Rent	2,600	0	2,600
Insurance	1,350	0	1,350
Miscellaneous	650	0.35	2,590

Required

1. Prepare a report of revenue and spending variances for June.

Model Answer

The planning budget uses planned activity. The flexible budget applies the same cost formulas to actual activity; it does not replace formula rates with actual costs.

Q177 · 2023-2024 · Exam Q2(b) [15 marks] (answer continued)

Driver	Planned	Actual
Liters sold	6,000	6,200

Budget line	Planning	Flexible
Revenue	72,000.00	74,400.00
Raw materials	27,900.00	28,830.00
Wages	14,000.00	14,280.00
Utilities	2,830.00	2,870.00
Rent	2,600.00	2,600.00
Insurance	1,350.00	1,350.00
Miscellaneous	2,750.00	2,820.00
Total expenses	51,430.00	52,750.00
Net operating income	20,570.00	21,650.00

Activity variances compare flexible with planning budgets. Revenue/spending variances compare actual with flexible budgets. The F/U designation measures whether the change improves or reduces profit.

Line	Planning	Flexible	Actual	Activity variance	Revenue/spending
Revenue	72,000.00	74,400.00	71,540.00	2,400.00 F	2,860.00 U
Raw materials	27,900.00	28,830.00	29,230.00	930.00 U	400.00 U
Wages	14,000.00	14,280.00	13,860.00	280.00 U	420.00 F
Utilities	2,830.00	2,870.00	3,270.00	40.00 U	400.00 U
Rent	2,600.00	2,600.00	2,600.00	0.00 -	0.00 -
Insurance	1,350.00	1,350.00	1,350.00	0.00 -	0.00 -
Miscellaneous	2,750.00	2,820.00	2,590.00	70.00 U	230.00 F
Total expenses	51,430.00	52,750.00	52,900.00	1,320.00 U	150.00 U
Net operating income	20,570.00	21,650.00	18,640.00	1,080.00 F	3,010.00 U

Q178 · 2024-2025 · Exam Q2(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 109, 110.

Gourmand Cooking School uses courses and students as its cost drivers. Revenue averages 800 per student. September was planned for three courses and 45 students; actual activity was three courses and 42 students.

Cost formula	Fixed per month	Per course	Per student
Instructor wages	0	3,080	0
Classroom supplies	0	0	260
Utilities	870	130	0
Campus rent	4,200	0	0
Insurance	1,890	0	0
Administrative expenses	3,270	15	4

Actual September results	Amount
Revenue	32,400
Instructor wages	9,080
Classroom supplies	8,540
Utilities	1,530
Campus rent	4,200
Insurance	1,890
Administrative expenses	3,790

Required

1. Prepare the planning budget at three courses and 45 students.
2. Prepare the flexible budget at three courses and 42 students.

Model Answer

The planning budget uses planned activity. The flexible budget applies the same cost formulas to actual activity; it does not replace formula rates with actual costs.

Q178 · 2024-2025 · Exam Q2(a) [20 marks] (answer continued)

Driver	Planned	Actual
Courses	3	3
Students	45	42

Budget line	Planning	Flexible
Revenue	36,000.00	33,600.00
Instructor wages	9,240.00	9,240.00
Classroom supplies	11,700.00	10,920.00
Utilities	1,260.00	1,260.00
Campus rent	4,200.00	4,200.00
Insurance	1,890.00	1,890.00
Administrative expenses	3,495.00	3,483.00
Total expenses	31,785.00	30,993.00
Net operating income	4,215.00	2,607.00

Standard costs and variances

Q179 · 2009-2010 · Exam Q4(a) [12 marks]

Source: Ipe 2009-25.pdf, PDF pages 3, 4.

Sonne Whim standard materials 7.2 oz at 2.50; labor 0.4 h at 10. Purchased/used 20,000 oz at 2.40 to make 2,500 bottles. Labor 900 hours costing 10,800.

Required

1. Compute material price/quantity and labor rate/efficiency variances.

Model Answer

Compute the standard quantity and hours allowed for actual output, then compare actual inputs and prices/rates. For cost variances: positive = unfavorable (U); negative = favorable (F).

Standard allowed for actual output	Calculation	Allowed
Materials	2,500 x 7.20	18,000.00
Labor-hours	2,500 x 0.40	1,000.00

Variance	Calculation	Result
Materials price	20,000 x (2.40 - 2.50)	2,000.00 F
Materials quantity	2.50 x (20,000 - 18,000)	5,000.00 U
Labor rate	900.00 x (12.00 - 10.00)	1,800.00 U
Labor efficiency	10.00 x (900.00 - 1,000.00)	1,000.00 F

A favorable rate or price variance can offset an unfavorable usage variance. Evaluate the causes, input quality, waste and rework; the small net variance does not prove that every component is well controlled.

Q180 · 2011-2012 · Exam Q5(b) [27 marks]

Source: Ipe 2009-25.pdf, PDF page 13.

Shirt Makers standards per dozen long-sleeve shirts: 24 yards at BDT 140 = 3,360; labor 3 h at BDT 120 = 360. April has no opening WIP. Lots 30/31 complete; Lot 32 material 100%, labor 80%. Purchased 95,000 yards for BDT 13,965,000; labor cost BDT 1,353,000.

Q180 · 2011-2012 · Exam Q5(b) [27 marks] (question continued)

Lot	Dozens	Material used yards	Labor-hours
30	1,000	24,100	2,980
31	1,700	40,440	5,130
32	1,200	28,825	2,890

Required

1. Compute material price variance.
2. Compute total/lot material quantity variances in yards and BDT.
3. Compute labor rate variance.
4. Compute total/lot efficiency variances in hours and BDT.
5. When are physical-unit versus monetary variances preferable?

Model Answer

Actual material price = 13,965,000/95,000 = 147; MPV = 95,000 x (147 - 140) = 665,000 U.
Actual labor-hours = 11,000; wage = 123; LRV = 11,000 x (123 - 120) = 33,000 U.

Lot	Standard yards	Used - standard yards	Quantity variance
30	24,000	100 U	14,000 U
31	40,800	360 F	50,400 F
32	28,800	25 U	3,500 U
Total	93,600	235 F	32,900 F

Lot	Standard hours for completed work	Actual - standard	Efficiency variance
30	3,000	20 F	2,400 F
31	5,100	30 U	3,600 U
32	2,880	10 U	1,200 U
Total	10,980	20 U	2,400 U

Q180 · 2011-2012 · Exam Q5(b) [27 marks] (answer continued)

Lot 32 labor standard is 1,200 dozen x 3 h x 80% = 2,880; its materials standard is not reduced because materials are complete. Physical variances help supervisors locate waste or bottlenecks without price changes; monetary variances compare financial significance across different inputs and support profitability analysis. Use both to avoid confusing cheap inputs with efficient work.

Q181 · 2012-2013 · Exam Q3(b) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 19.

Topaz material standards are missing. Labor standard is 2.5 h/unit at 9 (22.50/unit). Output 1,400; purchased/used material 6,000 lb costing 16,500; actual labor cost 28,500. Material quantity variance 1,200 U; total material variance 300 F; labor efficiency variance 4,500 F.

Required

1. Find material standard price, total standard quantity allowed and standard quantity/unit.
2. Find actual hourly wage and labor rate variance.

Model Answer

$$\text{MPV} = -300 - 1,200 = -1,500\,.$$
$$16,500 - 6,000SP = -1,500 \Longrightarrow SP = 3\,.$$
$$3(6,000 - SQ) = 1,200 \Longrightarrow SQ = 5,600\ \text{lb}\,.$$

Standard quantity/unit = 5,600 / 1,400 = 4 lb; standard material cost = 4 x 3 = 12/unit.

$$SH = 1,400(2.5) = 3,500, \quad 9(AH - 3,500) = -4,500 \Longrightarrow AH = 3,000\,.$$
$$AR = 28,500 \,/\, 3,000 = 9.50, \quad \text{LRV} = 3,000(9.50 - 9) = 1,500\ U\,.$$

Check labor total variance = 28,500 - 31,500 = 3,000 F = 1,500 U - 4,500 F.

Q182 · 2014-2015 · Exam Q8(b) [12 marks]

Source: Ipe 2009-25.pdf, PDF page 38.

P&G standard bag materials 7.2 lb at 5; labor 0.4 h at 20. Output 2,500 bags. Purchased/used 20,000 lb at 4.80, no opening/closing materials. Actual labor 900 hours at 24.

Required

1. Compute material price/quantity and labor rate/efficiency variances.

Q182 · 2014-2015 · Exam Q8(b) [12 marks] (answer continued)

Model Answer

Compute the standard quantity and hours allowed for actual output, then compare actual inputs and prices/rates. For cost variances: positive = unfavorable (U); negative = favorable (F).

Standard allowed for actual output	Calculation	Allowed
Materials	2,500 x 7.20	18,000.00
Labor-hours	2,500 x 0.40	1,000.00

Variance	Calculation	Result
Materials price	20,000 x (4.80 - 5.00)	4,000.00 F
Materials quantity	5.00 x (20,000 - 18,000)	10,000.00 U
Labor rate	900.00 x (24.00 - 20.00)	3,600.00 U
Labor efficiency	20.00 x (900.00 - 1,000.00)	2,000.00 F

A favorable rate or price variance can offset an unfavorable usage variance. Evaluate the causes, input quality, waste and rework; the small net variance does not prove that every component is well controlled.

Q183 · 2016-2017 · Exam Q7 [10 + 15 + 10 marks]

Source: Ipe 2009-25.pdf, PDF page 45.

Valley View performed 1,800 blood tests and 2,400 smears. Purchased 12,000 plates for 28,200 net of a 6% discount; unused 1,500, no opening plates. Actual labor 1,150 hours costing 13,800; VOH 7,820. Industry: 2 plates/test at 2.50 each; blood labor 0.30 h/test; smear 0.15 h/test; wage 14/h; VOH 6/h. Industry staffing is 50% senior technicians; Valley View uses 25% seniors and 75% assistants.

Required

1. Compute material price/quantity variances.
2. Compute labor rate/efficiency and evaluate staffing policy.
3. Compute VOH spending/efficiency and relate efficiency to labor efficiency.

Q183 · 2016-2017 · Exam Q7 [10 + 15 + 10 marks] (answer continued)

Model Answer

Used plates = 12,000 - 1,500 = 10,500; standard plates = 4,200 tests x 2 = 8,400. SH = 1,800 x 0.30 + 2,400 x 0.15 = 900. Actual wage = 12; actual plate price = 2.35.

Compute the standard quantity and hours allowed for actual output, then compare actual inputs and prices/rates. For cost variances: positive = unfavorable (U); negative = favorable (F).

Standard allowed for actual output	Calculation	Allowed
Materials	4,200 x 2.00	8,400.00
Labor-hours	4,200 x 0.21	900.00

Variance	Calculation	Result
Materials price	12,000 x (2.35 - 2.50)	1,800.00 F
Materials quantity	2.50 x (10,500 - 8,400)	5,250.00 U
Labor rate	1,150.00 x (12.00 - 14.00)	2,300.00 F
Labor efficiency	14.00 x (1,150.00 - 900.00)	3,500.00 U

Variable overhead variance	Calculation	Result
Spending	7,820.00 - 1,150.00 x 6.00	920.00 U
Efficiency	6.00 x (1,150.00 - 900.00)	1,500.00 U

A favorable rate or price variance can offset an unfavorable usage variance. Evaluate the causes, input quality, waste and rework; the small net variance does not prove that every component is well controlled.

More assistants plausibly explain the 2,300 F wage-rate variance, but 250 extra hours create 3,500 U efficiency: labor is 1,200 U overall. Material waste is also high. Do not continue merely because assistants cost less; investigate competence, supervision, retests and quality. Both efficiency variances use the same 250 extra hours: 3,500 U at wage 14 and 1,500 U at OH rate 6. Industry averages must be validated for the hospital's test complexity before judging staff.

Q184 · 2017-2018 · Exam Q2(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 48.

Marvel standards for 1,900 seat-cover sets: 2,850 labor-hours; materials 42,560 (22.40/set), labor 17,100 (9/set), VOH 6,840 (3.60/set). Standard material is 5.6 yards/set. Actual output 2,000; labor-hours 2,800; purchased and used 12,000 yards costing 45,600; DL 18,200; VOH 7,000.

Required

1. Compute materials price/quantity and variable-overhead rate/efficiency variances.

Model Answer

Standard material price = 22.40 / 5.6 = 4 per yard; actual price = 45,600/12,000 = 3.80. Standard hours = 2,000 x 1.5 = 3,000; standard VOH rate = 6,840/2,850 = 2.40.

Variance	Calculation	Result
Material price	12,000 x (3.80 - 4)	2,400 F
Material quantity	4 x (12,000 - 11,200)	3,200 U
VOH rate/spending	7,000 - 2,800 x 2.40	280 U
VOH efficiency	2.40 x (2,800 - 3,000)	480 F

Q185 · 2018-2019 · Exam Q3(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 57.

Xavier standards per unit: materials 6 oz at 0.50; DL 1.8 hours at 10; variable OH 1.8 hours at 5. Output 2,000. Purchased 18,000 oz at 0.60; used 14,000 oz; actual DL 4,000 hours at 9.75; actual VOH 20,800.

Required

1. Compute material, labor and variable-overhead variances.

Model Answer

Compute the standard quantity and hours allowed for actual output, then compare actual inputs and prices/rates. For cost variances: positive = unfavorable (U); negative = favorable (F).

Standard allowed for actual output	Calculation	Allowed
Materials	2,000 x 6.00	12,000.00

Q185 · 2018-2019 · Exam Q3(b) [15 marks] (answer continued)

Standard allowed for actual output	Calculation	Allowed
Labor-hours	2,000 x 1.80	3,600.00

Variance	Calculation	Result
Materials price	18,000 x (0.60 - 0.50)	1,800.00 U
Materials quantity	0.50 x (14,000 - 12,000)	1,000.00 U
Labor rate	4,000.00 x (9.75 - 10.00)	1,000.00 F
Labor efficiency	10.00 x (4,000.00 - 3,600.00)	4,000.00 U

Variable overhead variance	Calculation	Result
Spending	20,800.00 - 4,000.00 x 5.00	800.00 U
Efficiency	5.00 x (4,000.00 - 3,600.00)	2,000.00 U

A favorable rate or price variance can offset an unfavorable usage variance. Evaluate the causes, input quality, waste and rework; the small net variance does not prove that every component is well controlled.

Q186 · 2020-2021 · Exam Q4(a) [25 marks]

Source: Ipe 2009-25.pdf, PDF pages 71, 72.

Quality Motor standard tune-up labor: 2.5 hours at 9 = 22.50. Fifty tune-ups were completed. The time record is missing; labor rate variance is 87 F and total labor variance is 93 U.

Required

1. Find actual labor-hours and actual hourly wage.

Model Answer

$$SH = 50(2.5) = 125, \quad \text{LEV} = 93 - (-87) = 180\ U\,.$$

$$9(AH - 125) = 180 \Longrightarrow AH = 145\,.$$

$$145(AR - 9) = -87 \Longrightarrow AR = 8.40\,.$$

Check actual wages = 145 x 8.40 = 1,218; standard allowed cost = 125 x 9 = 1,125; difference = 93 U.

Q187 · 2021-2022 · Exam Q2(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 80.

Xavier standard cost per unit: materials 6 ounces at 0.50; labor 0.60 hour at 30; variable MOH 0.60 hour at 10. Actual output 2,000 units; purchased 18,000 ounces costing 10,800; used 14,000 ounces; labor 1,100 hours costing 33,550; variable MOH 12,980.

Required

1. Compute materials price/quantity, labor rate/efficiency, and variable-overhead spending/efficiency variances.

Model Answer

Compute the standard quantity and hours allowed for actual output, then compare actual inputs and prices/rates. For cost variances: positive = unfavorable (U); negative = favorable (F).

Standard allowed for actual output	Calculation	Allowed
Materials	2,000 x 6.00	12,000.00
Labor-hours	2,000 x 0.60	1,200.00

Variance	Calculation	Result
Materials price	18,000 x (0.60 - 0.50)	1,800.00 U
Materials quantity	0.50 x (14,000 - 12,000)	1,000.00 U
Labor rate	1,100.00 x (30.50 - 30.00)	550.00 U
Labor efficiency	30.00 x (1,100.00 - 1,200.00)	3,000.00 F

Variable overhead variance	Calculation	Result
Spending	12,980.00 - 1,100.00 x 10.00	1,980.00 U
Efficiency	10.00 x (1,100.00 - 1,200.00)	1,000.00 F

A favorable rate or price variance can offset an unfavorable usage variance. Evaluate the causes, input quality, waste and rework; the small net variance does not prove that every component is well controlled.

Q188 · 2023-2024 · Exam Q2(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 99, 100.

Valley View Hospital completed 1,800 blood tests and 2,400 smears. It purchased 12,000 plates for 28,200; 1,500 remained unused and no beginning plates existed. Actual labor was 1,150 hours for 13,800; actual variable overhead was 7,820. Industry standards: two plates per test at 2.50; blood tests 0.30 hour, smears 0.15 hour; labor 14 per hour; variable overhead 6 per DLH.

Required

1. Compute materials price variance for purchased plates and quantity variance for used plates.

Model Answer

Actual purchase price = 28,200 / 12,000 = 2.35. Actual plates used = 12,000 - 1,500 = 10,500. Standard plates allowed = (1,800 + 2,400) x 2 = 8,400.

$$\text{MPV} = 12,000(2.35 - 2.50) = -1,800 = 1,800\ F\,.$$

$$\text{MQV} = 2.50(10,500 - 8,400) = 5,250\ U\,.$$

The purchase price is favorable, but use exceeds the standard by 2,100 plates. Do not use all plates purchased as the quantity consumed; the unused 1,500 stay in materials inventory.

Q189 · 2024-2025 · Exam Q3(a) [20 marks]

Source: Ipe 2009-25.pdf, PDF pages 110, 111.

Koontz Company produced 12,000 units in May. There were no beginning or ending materials inventories. Variable overhead is assigned using direct labor-hours.

Input	Standard per unit	Actual per unit
Materials	1.80 feet at 3.00 = 5.40	1.80 feet at 3.30 = 5.94
Labor	0.90 hours at 18.00 = 16.20	0.92 hours at 17.50 = 16.10
Variable overhead	0.90 hours at 5.00 = 4.50	0.92 hours at 4.50 = 4.14
Total unit cost	26.10	26.18

The superintendent considers the 0.08 excess to be within management's 2% acceptable limit.

Required

1. Compute materials price and quantity variances.

Q189 · 2024-2025 · Exam Q3(a) [20 marks] (question continued)

2. Compute labor rate and efficiency variances.

Model Answer

Compute the standard quantity and hours allowed for actual output, then compare actual inputs and prices/rates. For cost variances: positive = unfavorable (U); negative = favorable (F).

Standard allowed for actual output	Calculation	Allowed
Materials	12,000 x 1.80	21,600.00
Labor-hours	12,000 x 0.90	10,800.00

Variance	Calculation	Result
Materials price	21,600 x (3.30 - 3.00)	6,480.00 U
Materials quantity	3.00 x (21,600 - 21,600)	0.00 -
Labor rate	11,040.00 x (17.50 - 18.00)	5,520.00 F
Labor efficiency	18.00 x (11,040.00 - 10,800.00)	4,320.00 U

Variable overhead variance	Calculation	Result
Spending	49,680.00 - 11,040.00 x 5.00	5,520.00 F
Efficiency	5.00 x (11,040.00 - 10,800.00)	1,200.00 U

A favorable rate or price variance can offset an unfavorable usage variance. Evaluate the causes, input quality, waste and rework; the small net variance does not prove that every component is well controlled.

The requested material and labor variances are in the first table. The overhead check explains the total report: 6,480 U materials - 1,200 F net labor - 4,320 F net overhead = 960 U, or 0.08 per unit. Offset between large variances can conceal meaningful operational issues.

Financial statements, ratios and ROI

Q190 · 2011-2012 · Exam Q6(b) [30 marks]

Source: Ipe 2009-25.pdf, PDF pages 14, 15.

Modern Building Supply requests a 300,000 loan to strengthen cash and pay short-term obligations. Beginning last-year AR 350,000, inventory 720,000 and assets 3,100,000. Assume sales are credit sales for collection analysis.

Balance sheet	This year	Last year
Cash	90,000	200,000
Marketable securities	0	50,000
AR	650,000	400,000
Inventory	1,300,000	800,000
Prepaids	20,000	20,000
CA	2,060,000	1,470,000
PPE net	1,940,000	1,830,000
Total assets	4,000,000	3,300,000
CL	1,100,000	600,000
12% bonds	750,000	750,000
Total liabilities	1,850,000	1,350,000
Preferred stock: 50 par, 8%	200,000	200,000
Common stock: 10 par	500,000	500,000
RE	1,450,000	1,250,000
Equity	2,150,000	1,950,000

Income statement	This year	Last year
Sales	7,000,000	6,000,000
COGS	5,400,000	4,800,000
Gross margin	1,600,000	1,200,000
Operating expenses	970,000	710,000

Q190 · 2011-2012 · Exam Q6(b) [30 marks] (question continued)

Income statement	This year	Last year
NOI	630,000	490,000
Interest	90,000	90,000
Pretax income	540,000	400,000
Taxes: 40%	216,000	160,000
NI	324,000	240,000

Required

1. For both years find working capital, current/quick ratios, collection period, inventory turnover in days, debt/equity, TIE, gross margin %, common book value/share and ROA.

Model Answer

Ratio	This year	Last year
Working capital	960,000	870,000
Current ratio	1.8727	2.4500
Quick ratio	0.6727	1.0833
Collection days	27.38	22.81
Inventory days	70.97	57.79
Debt/equity	0.8605	0.6923
TIE	7.0000	5.4444
Gross margin	22.86%	20.00%
Common book value/share	39	35
ROA incl. after-tax interest	10.36%	9.19%

Common shares = 500,000/10 = 50,000; common book value excludes preferred equity 200,000. ROA uses NI + interest x (1 - 40%) over average total assets. Profits improve but liquidity weakens and working-capital turnover slows; the requested loan needs a repayment/cash forecast. If a course defines ROA simply as NI/average assets, those alternatives are 8.8767% and 7.50%.

Q191 · 2016-2017 · Exam Q8 [$5x7$ marks]

Source: Ipe 2009-25.pdf, PDF page 46.

Starbucks comparative data, amounts in millions. No preferred stock is outstanding.

Balance sheet	End	Beginning
Cash	113	71
Marketable securities	107	61
AR	90	76
Inventory	221	202
Other CA	63	48
Total CA	594	458
PPE net	1,136	931
Other assets	121	103
Total assets	1,851	1,492
AP	128	74
Short-term loans	62	56
Accrued payables	245	174
Other CL	10	8
Total CL	445	312
Long-term liabilities	30	32
Total liabilities	475	344
Common stock/paid capital	792	751
RE	584	397
Equity	1,376	1,148

Current income statement	Amount
Revenue	2,678
COGS	1,113
Gross margin	1,565
Store operating expense	875

Q191 · 2016-2017 · Exam Q8 [$5x7$ marks] (question continued)

Current income statement	Amount
Other operating expense	93
Depreciation/amortization	164
G&A	151
Total operating expense	1,283
NOI	282
Internet investment losses	3
Interest income	11
Interest expense	0
Pretax income	290
Income tax	108
NI	182

Required

1. Compute ROA, common ROE, leverage effect, current/quick ratios, inventory turnover and average sale period.

Model Answer

Measure	Calculation	Result
ROA	182 / [(1,851 + 1,492)/2]	10.89%
ROE	182 / [(1,376 + 1,148)/2]	14.42%
Current ratio	594 / 445	1.3348
Quick ratio	(113 + 107 + 90) / 445	0.6966
Inventory turnover	1,113 / [(221 + 202)/2]	5.2624
Average sale period	365 / turnover	69.36 days

Financial leverage is favorable: common ROE exceeds ROA. Interest expense is zero, so the usual after-tax interest addition does not change ROA. Quick assets exclude inventory and other current assets; turnover uses average inventory. Financial amounts stay in millions, and ratios are unit-free.

Q192 · 2017-2018 · Exam Q1(b) [25 marks]

Source: Ipe 2009-25.pdf, PDF page 47.

Sally Perkins evaluates Plunge Enterprises. Shares outstanding did not change across the three years.

Measure	Year 1	Year 2	Year 3
Current ratio	2	2.50	2.80
Quick ratio	1.20	0.90	0.70
AR turnover	10.40	9.50	8.60
Inventory turnover	6.80	5.70	5
Sales index	100	118	130
Dividends/share	2.50	2.50	2.50
Dividend yield	3%	4%	5%
Dividend payout	60%	50%	40%
ROA	10.4%	11.8%	13%
Common ROE	9%	14.5%	18.2%

Required

1. Is stock market price rising/falling?
2. Are EPS rising/falling?
3. Is P/E rising/falling?
4. Is leverage benefiting common shareholders?
5. Is bill-paying ability improving?
6. Are customers paying as quickly as in Year 1?
7. Are receivables increasing/decreasing/constant?
8. Is inventory increasing/decreasing/constant? Explain each conclusion.

> **Source clarification:** Inventory turnover uses COGS, which is not given; the usual growing-inventory inference is therefore stated with its assumption rather than claimed as certain.

Q192 · 2017-2018 · Exam Q1(b) [25 marks] (answer continued)

Model Answer

Derived measure	Year 1	Year 2	Year 3
Price = dividend / yield	83.33	62.50	50
EPS = dividend / payout	4.17	5	6.25
P/E = price / EPS	20	12.50	8
Collection days: 365 / turnover	35.10	38.42	42.44
Average AR index, Year 1 = 100	100	129.18	157.21

1. Price falls while EPS rises; P/E falls from 20 to 8.
2. ROE exceeds ROA in Years 2 and 3, consistent with favorable financial leverage; Year 1 shows the opposite.
3. The current ratio improves, but quick ratio deteriorates sharply. More current assets are tied in inventory; immediate bill-paying ability is not clearly improving.
4. AR turnover declines and collection time rises: customers pay more slowly.
5. Assuming the sales trend applies to credit sales, average AR rises by about 57.2% from Year 1. These turnover data address average balances, not necessarily each year-end balance.
6. Inventory turnover declines, suggesting slower inventory use. A definitive inventory-balance trend requires COGS. If COGS moves proportionately with the sales index, average inventory rises: index 100, 140.77, 176.80.

Q193 · 2020-2021 · Exam Q7(a) [18 marks]

Source: Ipe 2009-25.pdf, PDF page 74.

Commercial Services.com has sales 3,000,000; NOI 150,000; average operating assets 750,000. Consider independently: (i) current ROI; (ii) sales increase 50%, NOI increases 200%, assets unchanged; (iii) sales increase 1,000,000, assets increase 250,000, NOI increases 200,000. Compute ROI to two decimals.

Model Answer

Scenario	Sales	NOI	Assets	ROI
Current	3,000,000	150,000	750,000	20.00%

Q193 · 2020-2021 · Exam Q7(a) [18 marks] (answer continued)

Model Answer

Scenario	Sales	NOI	Assets	ROI
Founder	4,500,000	450,000	750,000	60.00%
CFO	4,000,000	350,000	1,000,000	35.00%

ROI = NOI / average operating assets. A 200% increase makes NOI three times its original amount; it does not mean the new NOI is only twice the original.

Q194 · 2021-2022 · Exam Q7(b) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 85.

McKenzie Sales reports the comparative statement below. The directors ask why income rose only 38,000 when sales rose by two million.

Line	This year	Last year
Sales	8,000,000	6,000,000
Cost of goods sold	4,984,000	3,516,000
Gross margin	3,016,000	2,484,000
Selling expenses	1,480,000	1,092,000
Administrative expenses	712,000	618,000
Total S&A expenses	2,192,000	1,710,000
Net operating income	824,000	774,000
Interest expense	96,000	84,000
Income before tax	728,000	690,000

Required

1. Express both years as common-size percentages to one decimal place.
2. Comment on the changes.

Model Answer

Divide every line by sales for its own year, rather than by combined sales.

Q194 · 2021-2022 · Exam Q7(b) [10 marks] (answer continued)

Line	This year	Last year
Sales	100.0%	100.0%
Cost of goods sold	62.3%	58.6%
Gross margin	37.7%	41.4%
Selling expenses	18.5%	18.2%
Administrative expenses	8.9%	10.3%
Total S&A expenses	27.4%	28.5%
Net operating income	10.3%	12.9%
Interest expense	1.2%	1.4%
Income before tax	9.1%	11.5%

Sales rose 33.3%, but the COGS ratio increased from 58.6% to 62.3%, reducing gross margin from 41.4% to 37.7%. Administrative and interest ratios fell and partly offset the deterioration, while the selling-expense ratio rose. Pretax margin fell from 11.5% to 9.1%; hence sales growth generated only 38,000 more pretax income.

Q195 · 2022-2023 · Exam Q1(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 87.

A martial-arts distributor supplies the following divisional performance data. Fill the missing values.

Measure	Alpha	Bravo	Charlie
Sales	?	11,500,000	?
NOI	?	920,000	210,000
Average operating assets	800,000	?	?
Margin	4%	?	7%
Turnover	5	?	?
ROI	?	20%	14%

Q195 · 2022-2023 · Exam Q1(b) [15 marks] (answer continued)

Model Answer

$$\text{Margin} = \text{NOI} / \text{Sales}, \quad \text{Turnover} = \text{Sales} / \text{Assets},$$
$$\text{ROI} = \text{Margin} \times \text{Turnover} = \text{NOI} / \text{Assets}.$$

Completed measure	Alpha	Bravo	Charlie
Sales	4,000,000	11,500,000	3,000,000
NOI	160,000	920,000	210,000
Average operating assets	800,000	4,600,000	1,500,000
Margin	4%	8%	7%
Turnover	5.00	2.50	2.00
ROI	20%	20%	14%

Alpha: sales = 5 x 800,000, then NOI = 4% x sales. Bravo: assets = 920,000 / 20%, then margin and turnover. Charlie: sales = 210,000 / 7%; assets = 210,000 / 14%.

Q196 · 2022-2023 · Exam Q7(b) [22 marks]

Source: Ipe 2009-25.pdf, PDF pages 94, 95.

Sabin Electronics requests a 500,000 long-term loan: 100,000 for cash and 400,000 for equipment. All sales are on account. It introduced product lines, raised prices and expanded sales territories. Opening balances before last year: A/R 250,000; inventory 500,000; total assets 2,420,000.

Balance sheet	This year	Last year
Cash	70,000	150,000
Marketable securities	0	18,000
Accounts receivable	580,000	300,000
Inventory	850,000	600,000
Prepaid expenses	20,000	22,000
Current assets	1,520,000	1,090,000
Plant/equipment, net	1,480,000	1,370,000

Q196 · 2022-2023 · Exam Q7(b) [22 marks] (question continued)

Balance sheet	This year	Last year
Total assets	3,000,000	2,460,000
Current liabilities	800,000	430,000
Bonds payable (12%)	600,000	600,000
Total liabilities	1,400,000	1,030,000
Common stock	750,000	750,000
Retained earnings	850,000	680,000
Equity	1,600,000	1,430,000

Income and RE statement	This year	Last year
Sales	5,000,000	4,350,000
COGS	3,875,000	3,450,000
Gross margin	1,125,000	900,000
Selling/admin	653,000	548,000
Operating income	472,000	352,000
Interest	72,000	72,000
Pretax income	400,000	280,000
Tax (30%)	120,000	84,000
Net income	280,000	196,000
Dividends	110,000	95,000
Income retained	170,000	101,000
Beginning RE	680,000	579,000
Ending RE	850,000	680,000

Industry benchmark	Value
Current ratio	2.5
Quick ratio	1.3
Collection period	18 days

Q196 · 2022-2023 · Exam Q7(b) [22 marks] (question continued)

Industry benchmark	Value
Sale period	60 days
Debt/equity	0.90
Times interest earned	6.0

Required

1. For both years compute working capital, current/quick ratios, collection/sale periods, operating cycle, asset turnover and debt/equity.
2. Compare with industry benchmarks and discuss whether the loan is likely to be approved.

Source clarification: The receivables/security digits in the scan are faint. Current A/R 580,000, current securities zero and prior securities 18,000 reconcile to the printed current-asset totals and are used consistently in the ratios.

Model Answer

Use closing balances for liquidity/debt ratios and two-year averages for turnover. Average A/R = 440,000 this year, 275,000 last year; inventory = 725,000 and 550,000; assets = 2,730,000 and 2,440,000. A 365-day year is used.

Measure	This year	Last year	Industry
Working capital	720,000.00	660,000.00	Not specified
Current ratio	1.9000	2.5349	2.50
Quick ratio	0.8125	1.0884	1.30
Collection period	32.12 days	23.07 days	18 days
Sale period	68.29 days	58.19 days	60 days
Operating cycle	100.41 days	81.26 days	78 days (sum)
Asset turnover	1.8315	1.7828	Not specified
Debt/equity	0.8750	0.7203	0.90
Times interest earned (comparison)	6.5556	4.8889	6.00

Q196 · 2022-2023 · Exam Q7(b) [22 marks] (answer continued)

Profit and interest coverage improve, but cash falls and liquidity weakens. Receivables and inventory grow faster than sales; the operating cycle lengthens beyond industry benchmarks, tying up cash. Current leverage is below 0.90, but a new 500,000 loan would raise liabilities/equity to 1,900,000 / 1,600,000 = 1.1875 before any subsequent earnings.

The bank should treat approval as conditional on improved collections, inventory control and a credible cash-flow/repayment forecast. Profitability and current interest coverage favor the application, but liquidity and post-loan leverage make an unconditional approval difficult to justify. A final bank decision cannot be known from these ratios alone.

Q197 · 2022-2023 · Exam Q7(c) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 95.

What are cash equivalents, and why include them with cash on the cash-flow statement?

Model Answer

Cash equivalents are short-term, highly liquid investments readily convertible to known amounts of cash and subject to an insignificant risk of changes in value. They normally have a maturity of three months or less from the acquisition date; examples include qualifying treasury bills and money-market holdings.

They are included with cash because they are held to meet short-term cash commitments and behave much like cash. Transfers between cash and qualifying cash equivalents are cash-management movements rather than operating, investing or financing cash flows.

Q198 · 2023-2024 · Exam Q5(a) [25 marks]

Source: Ipe 2009-25.pdf, PDF pages 103, 104.

Weller's comparative statements below are in thousands of dollars. All sales are on account; 800,000 common shares were outstanding throughout the year. Bonds bear 12%; income tax = 40%; common dividend = 0.40 per share; year-end market price = 18.

Balance sheet (thousands)	This year	Last year
Cash	1,280	1,560
Accounts receivable	12,300	9,100
Inventory	9,700	8,200

Q198 · 2023-2024 · Exam Q5(a) [25 marks] (question continued)

Balance sheet (thousands)	This year	Last year
Prepaid expenses	1,800	2,100
Total current assets	25,080	20,960
Land	6,000	6,000
Buildings/equipment, net	19,200	19,000
Total assets	50,280	45,960
Accounts payable	9,500	8,300
Accrued liabilities	600	700
Short-term notes	300	300
Total current liabilities	10,400	9,300
Bonds payable	5,000	5,000
Total liabilities	15,400	14,300
Common stock	800	800
Paid-in capital	4,200	4,200
Retained earnings	29,880	26,660
Total equity	34,880	31,660

Income/retained earnings (thousands)	This year	Last year
Sales	79,000	74,000
COGS	52,000	48,000
Gross margin	27,000	26,000
Selling expenses	8,500	8,000
Administrative expenses	12,000	11,000
Total S&A	20,500	19,000
Operating income	6,500	7,000
Interest	600	600
Pretax income	5,900	6,400
Taxes	2,360	2,560

Q198 · 2023-2024 · Exam Q5(a) [25 marks] (question continued)

Income/retained earnings (thousands)	This year	Last year
Net income	3,540	3,840
Dividends	320	320
Added to retained earnings (printed)	3,220	3,240
Beginning retained earnings (printed)	26,660	23,420
Ending retained earnings	29,880	26,660

Required

1. Compute for both years: working capital; current/acid-test ratios; A/R turnover and collection period; inventory turnover and sale period; operating cycle; total asset turnover; gross/net margin; return on assets and equity.
2. Comment on financial performance and position.

> **Source clarification:** Prior-year opening balances are missing, so the prior-year average-based ratio results are explicitly conditional. The prior-year retained-earnings reconciliation also contains an arithmetic inconsistency.

Model Answer

For this year, average receivables = 10,700; inventory = 8,950; assets = 48,120; equity = 33,270 (all in thousands). Last year's opening balance sheet is absent. For a numerical comparison, explicitly use its closing balances as proxies for its averages; exact prior-year average-based ratios cannot be determined without the missing opening balances.

$$\text{ROA} = \frac{\text{NI} + \text{Interest}(1 - \text{Tax rate})}{\text{Average total assets}}, \quad \text{ROE} = \frac{\text{NI}}{\text{Average equity}}.$$

Use net credit sales / average A/R, COGS / average inventory, sales / average assets, and 365 / turnover for the day measures. Liquidity ratios use closing balances.

Measure	This year	Last year (closing-average proxy)
Working capital (thousands)	14,680.00	11,660.00
Current ratio	2.4115	2.2538
Acid-test ratio	1.3058	1.1462
A/R turnover	7.3832	8.1319
Collection days	49.44	44.89

Q198 · 2023-2024 · Exam Q5(a) [25 marks] (answer continued)

Measure	This year	Last year (closing-average proxy)
Inventory turnover	5.8101	5.8537
Sale days	62.82	62.35
Operating cycle (days)	112.26	107.24
Total asset turnover	1.6417	1.6101
Gross margin %	34.18%	35.14%
Net profit margin %	4.48%	5.19%
ROA %	8.10%	9.14%
ROE %	10.64%	12.13%

Working capital and current/quick ratios improve, but collections slow and the operating cycle lengthens. Gross and net margins decline; ROA and ROE also fall on the stated prior-year proxy. Sales growth has not produced stronger profitability.

The printed prior-year retained-earnings addition is inconsistent: 3,840 net income minus 320 dividends equals 3,520, not 3,240. The ratio calculations retain the reported income and balance-sheet amounts rather than silently changing them.

Q199 · 2023-2024 · Exam Q6(a) [6 marks]

Source: Ipe 2009-25.pdf, PDF page 104.

What are the three major sections of a statement of cash flows, and which cash inflows/outflows belong in each?

Model Answer

Section	Cash inflows	Cash outflows
Operating	Collections from customers and other operating receipts	Payments to suppliers/employees and other operating cash costs
Investing	Sale of long-term assets or investments; collections of loan principal	Purchase of property/equipment or investments; lending principal
Financing	Issue of shares or debt; borrowing	Repayment of debt principal, repurchase of shares, cash dividends

Q199 · 2023-2024 · Exam Q6(a) [6 marks] (answer continued)

Noncash investing/financing events are disclosed separately; they are not cash flows. In the usual textbook convention, interest paid and income taxes are operating cash flows, while repayment of loan principal is financing.

Q200 · 2023-2024 · Exam Q8(c) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 107.

What is the basic purpose of examining trends in financial ratios and other data? What other comparisons can an analyst make?

Model Answer

Trend analysis tracks whether liquidity, operating efficiency, profitability and financing risk improve or deteriorate over time. Ratios connect statement amounts, making growth or weakness easier to interpret than isolated totals.

Compare several years, actual results with budgets or targets, competitors, and industry benchmarks. Common-size statements and cash-flow patterns add context. Use comparable accounting policies, seasonal periods and business models; changes in prices, acquisitions or exceptional items can distort a ratio trend.

Q201 · 2024-2025 · Exam Q1(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 109.

Joel de Paris, Inc. reports the following financial data. Investment in Buisson S.A. is stock of another company; the land is undeveloped. Dividends paid last year were 15,000.

Balance-sheet item	Beginning	Ending
Cash	140,000	120,000
Accounts receivable	450,000	530,000
Inventory	320,000	380,000
Plant and equipment, net	680,000	620,000
Investment in Buisson S.A.	250,000	280,000
Undeveloped land	180,000	170,000
Total assets	2,020,000	2,100,000

Q201 · 2024-2025 · Exam Q1(b) [15 marks] (question continued)

Balance-sheet item	Beginning	Ending
Accounts payable	360,000	310,000
Long-term debt	1,500,000	1,500,000
Stockholders' equity	160,000	290,000

Income-statement item	Amount
Sales	4,050,000
Operating expenses	3,645,000
Net operating income	405,000
Interest expense	150,000
Tax expense	110,000
Net income	145,000

Required

1. Compute margin, turnover and return on investment (ROI) for last year.

Model Answer

For managerial ROI, use net operating income and average operating assets. The stock investment and undeveloped land are nonoperating, so exclude them; do not use net income after financing and tax.

Average operating-assets calculation	Beginning	Ending
Total assets	2,020,000	2,100,000
Less: stock investment	250,000	280,000
Less: undeveloped land	180,000	170,000
Operating assets	1,590,000	1,650,000

$$\text{Average operating assets} = \frac{1,590,000 + 1,650,000}{2} = 1,620,000\,.$$

Measure	Calculation	Answer
Margin	405,000 / 4,050,000	10.00%

Q201 · 2024-2025 · Exam Q1(b) [15 marks] (answer continued)

Measure	Calculation	Answer
Turnover	4,050,000 / 1,620,000	2.50 times
ROI	405,000 / 1,620,000	25.00%

Check: margin x turnover = 10% x 2.5 = 25%. Dividends and the balance-sheet equity amount are not part of this operating ROI calculation.

Q202 · 2024-2025 · Exam Q7(b) [10 marks]

Source: Ipe 2009-25.pdf, PDF page 116.

McKenzie Sales reports the comparative statement below. The directors ask why income rose only 38,000 when sales rose by two million.

Line	This year	Last year
Sales	8,000,000	6,000,000
Cost of goods sold	4,984,000	3,516,000
Gross margin	3,016,000	2,484,000
Selling expenses	1,480,000	1,092,000
Administrative expenses	712,000	618,000
Total S&A expenses	2,192,000	1,710,000
Net operating income	824,000	774,000
Interest expense	96,000	84,000
Income before tax	728,000	690,000

Required

1. Express both years as common-size percentages to one decimal place.
2. Comment on the changes.

Model Answer

Divide every line by sales for its own year, rather than by combined sales.

Q202 · 2024-2025 · Exam Q7(b) [10 marks] (answer continued)

Line	This year	Last year
Sales	100.0%	100.0%
Cost of goods sold	62.3%	58.6%
Gross margin	37.7%	41.4%
Selling expenses	18.5%	18.2%
Administrative expenses	8.9%	10.3%
Total S&A expenses	27.4%	28.5%
Net operating income	10.3%	12.9%
Interest expense	1.2%	1.4%
Income before tax	9.1%	11.5%

Sales rose 33.3%, but the COGS ratio increased from 58.6% to 62.3%, reducing gross margin from 41.4% to 37.7%. Administrative and interest ratios fell and partly offset the deterioration, while the selling-expense ratio rose. Pretax margin fell from 11.5% to 9.1%; hence sales growth generated only 38,000 more pretax income.

Q203 · 2024-2025 · Exam Q8(a) [5 marks]

Source: Ipe 2009-25.pdf, PDF page 116.

What are cash equivalents, and why are they included with cash on a statement of cash flows?

Model Answer

Cash equivalents are short-term, highly liquid investments readily convertible to known amounts of cash and subject to an insignificant risk of changes in value. They normally have a maturity of three months or less from the acquisition date; examples include qualifying treasury bills and money-market holdings.

They are included with cash because they are held to meet short-term cash commitments and behave much like cash. Transfers between cash and qualifying cash equivalents are cash-management movements rather than operating, investing or financing cash flows.

Q204 · 2024-2025 · Exam Q8(b) [20 marks]

Source: Ipe 2009-25.pdf, PDF page 117.

Castile Products' balance sheet and income statement follow. Beginning accounts receivable = 25,000; beginning inventory = 60,000. All sales were on account.

Balance-sheet item	Amount
Cash	6,500
Accounts receivable	35,000
Merchandise inventory	70,000
Prepaid expenses	3,500
Total current assets	115,000
Property/equipment, net	185,000
Total assets	300,000
Current liabilities	50,000
Bonds payable (10%)	80,000
Total liabilities	130,000
Common stock (5 par)	30,000
Retained earnings	140,000
Total equity	170,000

Income-statement item	Amount
Sales	420,000
COGS	292,500
Gross margin	127,500
Selling/admin	89,500
Operating income	38,000
Interest	8,000
Pretax income	30,000
Tax (30%)	9,000
Net income	21,000

Q204 · 2024-2025 · Exam Q8(b) [20 marks] (question continued)

Required

1. Compute working capital, current ratio, acid-test ratio, debt-to-equity ratio, times interest earned, average collection period, average sale period and operating cycle.

Model Answer

Use ending balances for liquidity/debt ratios. Use average receivables and average inventory for turnover measures; this answer uses a 365-day year.

Measure	Calculation	Answer
Working capital	115,000 - 50,000	65,000
Current ratio	115,000 / 50,000	2.30
Acid-test ratio	(6,500 + 35,000) / 50,000	0.83
Debt-to-equity	130,000 / 170,000	0.7647
Times interest earned	38,000 / 8,000	4.75 times
Average receivables	(25,000 + 35,000) / 2	30,000
Receivables turnover	420,000 / 30,000	14.00 times
Collection period	365 / 14	26.07 days
Average inventory	(60,000 + 70,000) / 2	65,000
Inventory turnover	292,500 / 65,000	4.50 times
Average sale period	365 / 4.5	81.11 days
Operating cycle	Collection + sale period	107.18 days

Quick assets exclude inventory and prepayments. Debt-to-equity uses total liabilities, not only interest-bearing debt. Times interest earned uses earnings before interest and tax, which here equal operating income.

Capital budgeting and investment appraisal

Q205 · 2020-2021 · Exam Q8 [35 marks]

Source: Ipe 2009-25.pdf, PDF page 75.

Tiger Computers considers a machine costing 900,000 plus installation/software 650,000. The old machine can be sold now for 70,000. Annual labor/material savings: 240,000 / 96,000. Maintenance increases 4,250 per month; overhaul 90,000 at Year 6. Life 10 years; terminal scrap 210,000; required return 18%. All amounts are Singapore dollars; ignore income taxes.

Required

1. Find net annual savings.
2. Compute incremental NPV and recommend.

Model Answer

$$\text{Annual net saving} = 240,000 + 96,000 - 12(4,250) = 285,000\,.$$

Incremental cash flow	Timing	Present value at 18%
Machine + installation - old sale	Now	-1,480,000
Annual net savings	Years 1-10	1,280,814.59
Overhaul	Year 6	-33,338.84
Terminal scrap	Year 10	40,123.54
NPV		-192,400.71

Reject on financial grounds: NPV is negative at the required 18% return. Installation/software are initial cash costs; old-machine proceeds reduce the initial investment. Depreciation is not a cash flow in this no-tax analysis. Full-precision discount factors are used; rounded exam tables may give a slightly different total.

Q206 · 2021-2022 · Exam Q3(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 81.

Explain how discounted payback, NPV and IRR support capital-budgeting decisions, with numerical examples.

Q206 · 2021-2022 · Exam Q3(b) [15 marks] (answer continued)

Model Answer

Example: invest 10,000 now for four annual cash inflows of 4,000; required return 10%. These amounts illustrate the methods rather than being extra exam data.

Year	Cash flow	PV at 10%	Cumulative PV inflows
0	-10,000	-10,000	-
1	4,000	3,636.36	3,636.36
2	4,000	3,305.79	6,942.15
3	4,000	3,005.26	9,947.41
4	4,000	2,732.05	12,679.46

Discounted payback is the time to recover the initial investment from discounted inflows. After Year 3, 52.59 remains; conventional within-year interpolation gives 3 + 52.59 / 2,732.05 = 3.019 years. With strictly year-end lump receipts, recovery occurs at the end of Year 4. Compare with a stated payback cutoff; this method ignores later benefits.

$$\mathrm{NPV} = -10,000 + \sum_{t=1}^{4} \frac{4,000}{1.10^{t}} = 2,679.46\,.$$

Accept a positive-NPV independent project: it earns more than the required return.

$$0 = -10,000 + \sum_{t=1}^{4} \frac{4,000}{(1+r)^{t}}\,.$$

The IRR is approximately 21.8623%, exceeding 10%, so the conventional independent project is acceptable. For mutually exclusive projects, prioritize NPV at the appropriate required return; unconventional cash flows can produce multiple or no useful IRRs.

Q207 · 2023-2024 · Exam Q4(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 102.

Evaluate this five-year project at a 10% discount rate.

Year	Benefits	Costs
0	0	100,000

Q207 · 2023-2024 · Exam Q4(b) [15 marks] (question continued)

Year	Benefits	Costs
1	20,000	10,000
2	25,000	8,000
3	30,000	6,000
4	15,000	5,000
5	10,000	4,000

Required

1. Compute the benefit-cost ratio and judge financial viability.

Model Answer

Discount benefits and costs separately; include the year-zero cost without discounting.

Year	Discount factor	PV benefits	PV costs
0	1.000000	0.00	100,000.00
1	0.909091	18,181.82	9,090.91
2	0.826446	20,661.16	6,611.57
3	0.751315	22,539.44	4,507.89
4	0.683013	10,245.20	3,415.07
5	0.620921	6,209.21	2,483.69

Evaluation	Calculation	Result
PV benefits	Sum of discounted benefits	77,836.83
PV costs	Initial cost + discounted annual costs	126,109.12
BCR	PV benefits / PV costs	0.6172
NPV	PV benefits - PV costs	(48,272.29)

Reject on these cash flows: BCR is below 1 and NPV is negative. Adding undiscounted annual benefits would ignore the time value of money.

Q208 · 2024-2025 · Exam Q2(b) [15 marks]

Source: Ipe 2009-25.pdf, PDF page 110.

An investment costs 10,000 today and generates 4,000 per year for the next four years. Compute the internal rate of return (IRR), showing the calculations.

Model Answer

Assume each annual receipt occurs at year-end. IRR is the discount rate making NPV zero; the required four-year annuity factor is 10,000 / 4,000 = 2.5.

$$0 = -10,000 + 4,000 \sum_{t=1}^{4} \frac{1}{(1+r)^t}.$$

$$\frac{1-(1+r)^{-4}}{r} = 2.5\,.$$

Trial rate	Annuity factor	PV of receipts	NPV
20%	2.588735	10,354.94	354.94
22%	2.493641	9,974.56	(25.44)

The solution lies between 20% and 22%. Linear interpolation between the two NPV trials gives an approximation; solving the equation more accurately gives:

$$\text{IRR} = 21.862270\% \approx 21.9\%\,.$$

Substituting this rate makes the present value of the four receipts 10,000, so NPV is zero apart from rounding.

Sources and Coverage

Question source: Ipe 2009-25.pdf, 117 PDF pages. The collection covers IPE 105, Principles of Cost and Management Accounting. Available exam sessions are listed below.

2009-2010: 16 · 2011-2012: 19 · 2012-2013: 13 · 2013-2014: 17 · 2014-2015: 14 · 2016-2017: 12 · 2017-2018: 15 · 2018-2019: 13 · January 2020 Term: 8 · 2020-2021: 11 · 2021-2022: 17 · 2022-2023: 16 · 2023-2024: 19 · 2024-2025: 18.

Count convention: A case with several linked requirements is retained as one entry. Independently numbered theory or case parts receive their own entries. Repeated questions remain under their original exam tags.

Layout: The design follows your approved topic-wise workbooks, with every question followed by its answer and no ruled practice space.

Numerical conventions: Unrounded rates are used for cost assignments; displayed amounts normally round to two decimal places. U means unfavorable and F means favorable. Where required, whole units are rounded upward to meet a target.

Financial ratios: A 365-day year is used unless a question specifies otherwise. Turnover/return ratios use averages when opening and closing balances are supplied; an explicitly labeled proxy is used only where the question permits assumptions and missing opening data prevents a unique prior-year result.

Sources and assumptions: The source-page references are printed with every question. A source clarification appears beside any case with missing data, inconsistent totals, or conflicting period labels. Historical company descriptions are treated as part of the exam prompt rather than current facts.